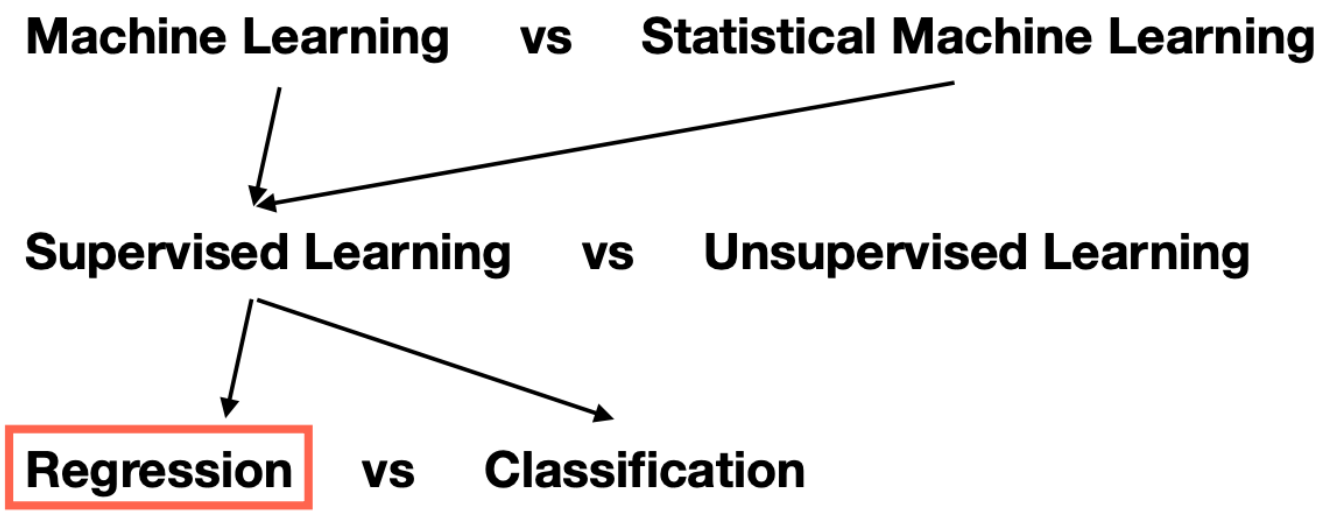
Logistic models

PSTAT 131/231

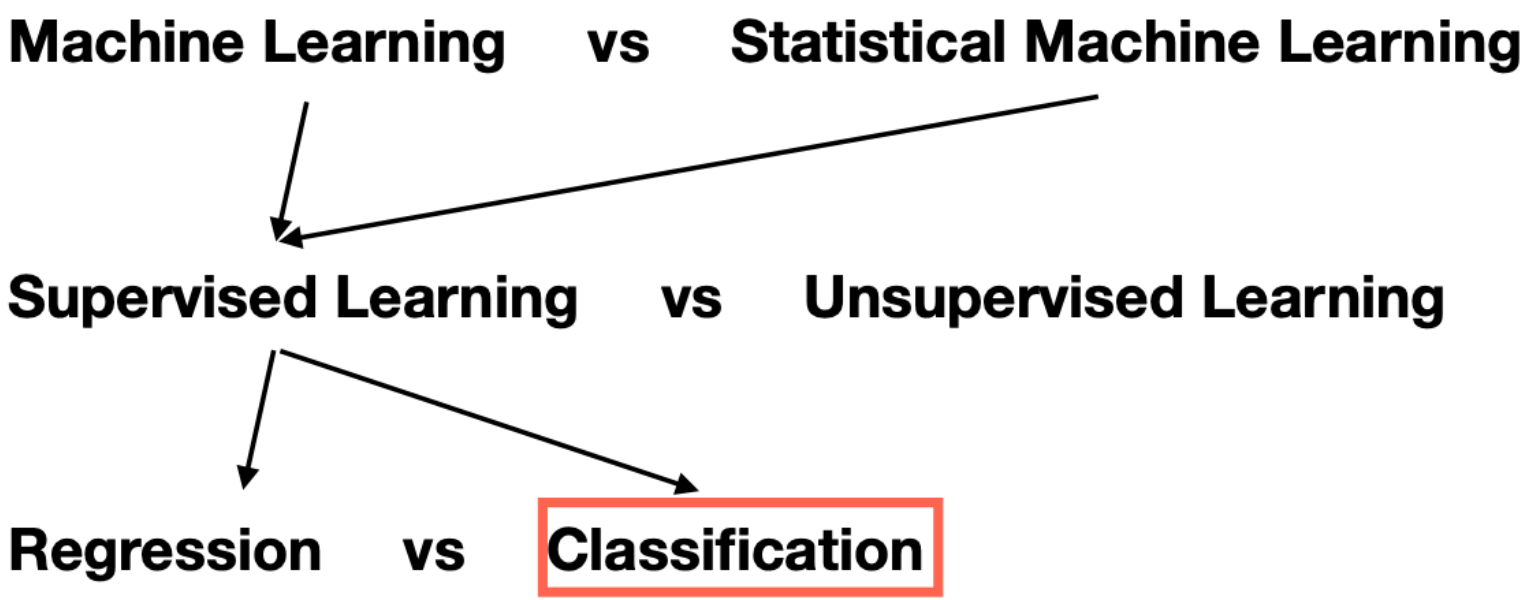
Week 3



Linear Regression

- Coefficient estimates
- Assessing accuracy
- Practical considerations
- Other perspectives…

This lecture…



A linear method in classification: Logistic Regression

Classification in general

Qualitative response

Predicting a qualitative response for an observation

= classifying that observation, i.e., assign the observation to a class

Classification methods (classifiers): K-NN, logistic regression, discriminant analysis, and many more…

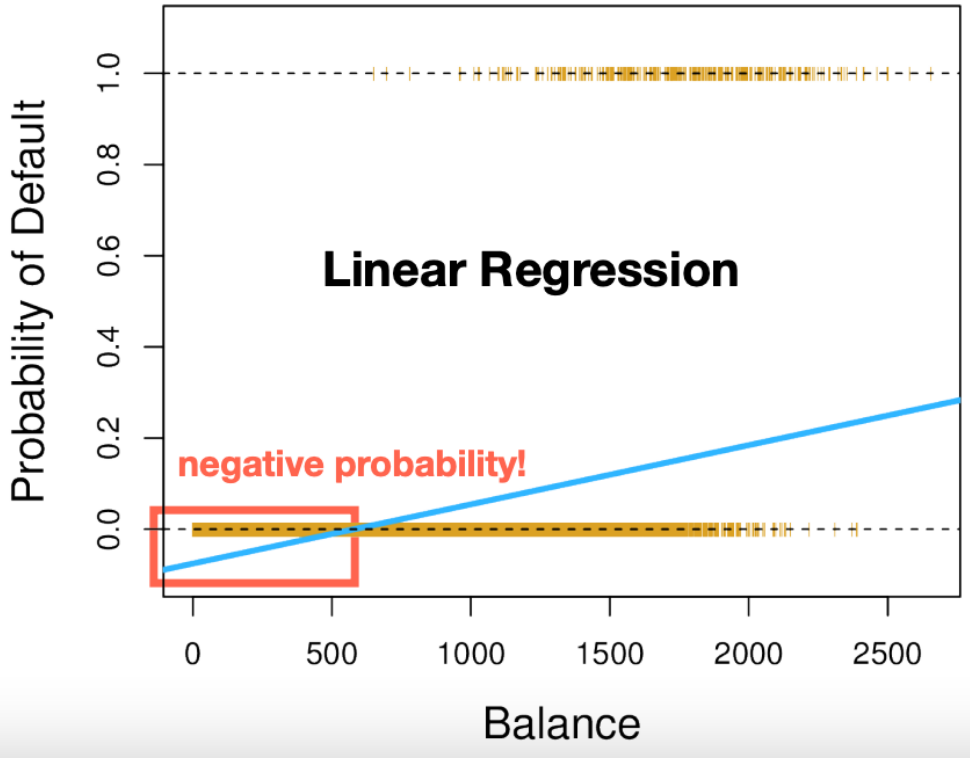
Related to Regression: often we predict the probability (which is quantitative) of each class of a qualitative response

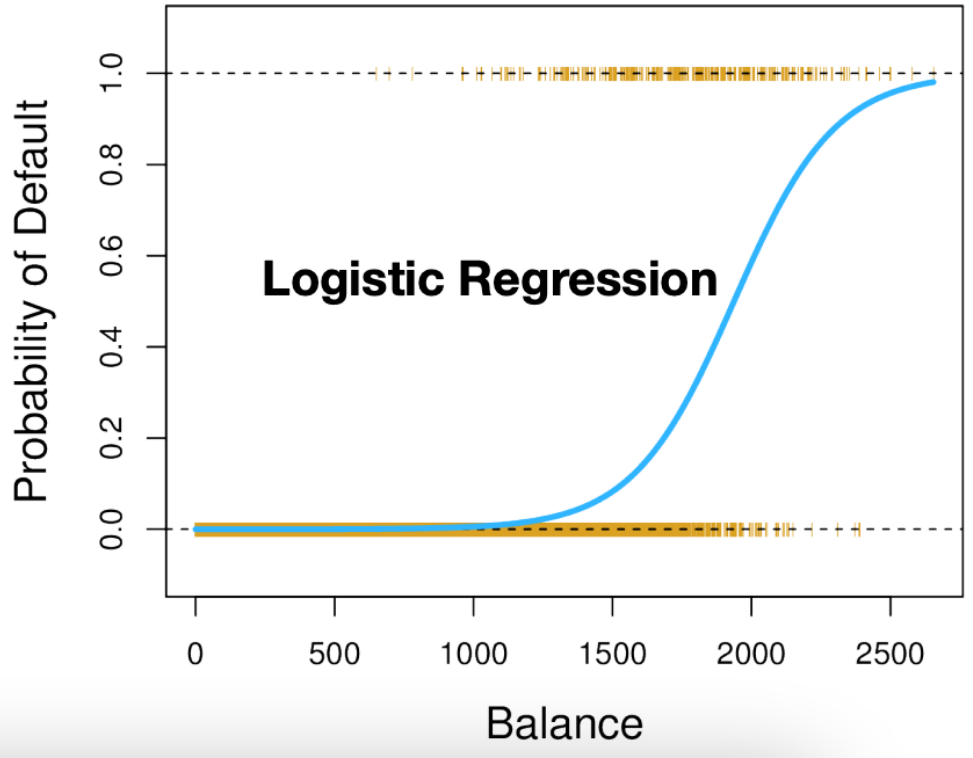
Why not Linear Regression?

You said we can predict class probability, then why not just…

Suppose $Y \in \{0,1\}$, we model

$$\Pr(Y = 1 \mid X) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$
$$\Pr(Y = 0 \mid X) = 1 - \Pr(Y = 1 \mid X)$$





Why not Linear Regression?

We know how to code qualitative predictors, why can't we code the dummy response similarly?

For example, suppose $Y \in \{\text{stroke, drug overdose, seizure}\}$, then

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if drug overdose} \\ 3 & \text{if seizure} \end{cases}$$

This implicitly implies an ordering on the response classes

difference between stroke and drug overdoes = between drug overdose and seizure

What about

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if seizure} \\ 3 & \text{if drug overdose} \end{cases}$$

What about $Y \in \{\text{dog, cat, fish}\}$

Logistic regression

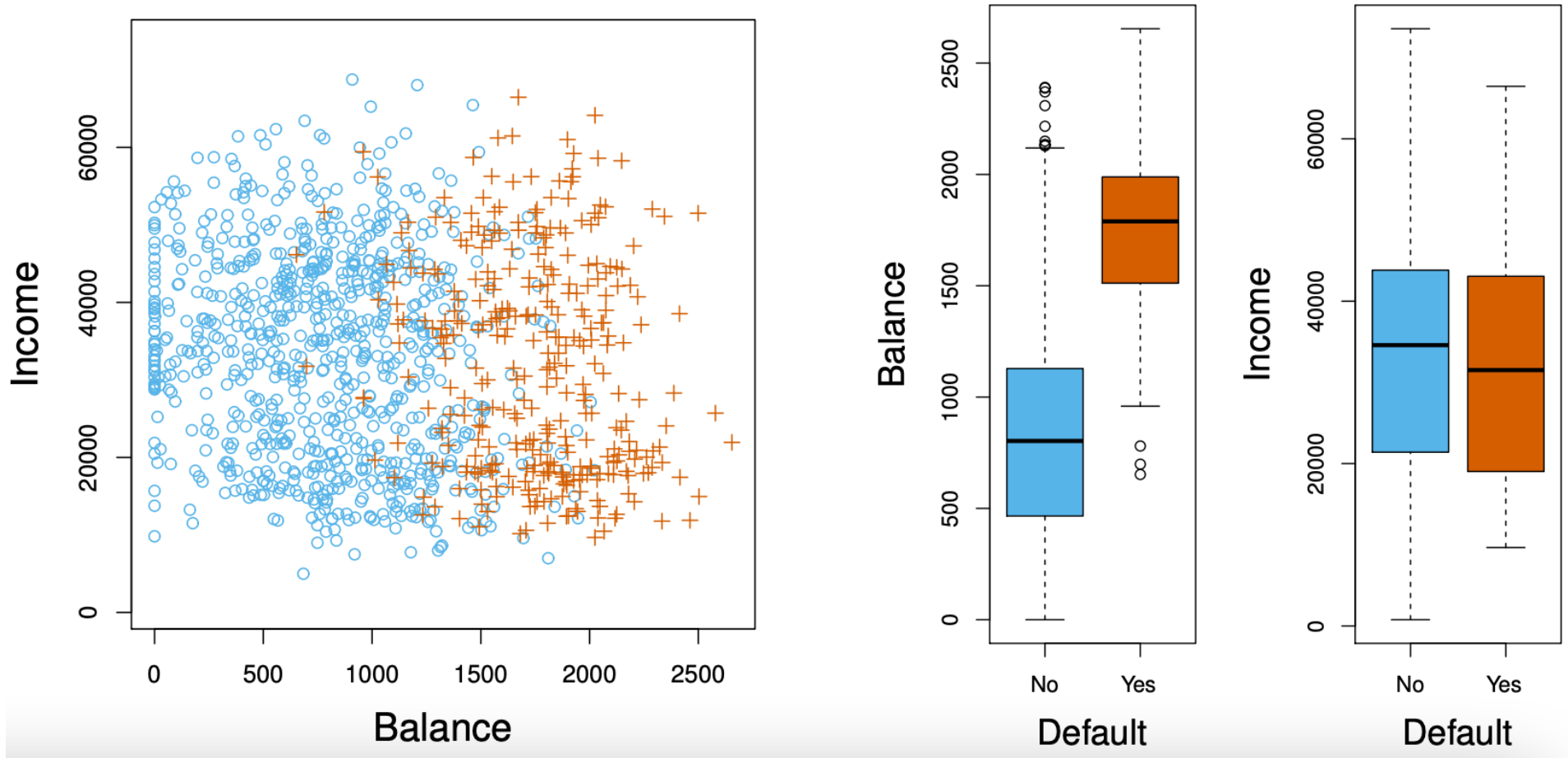
We will focus on binary classification problem

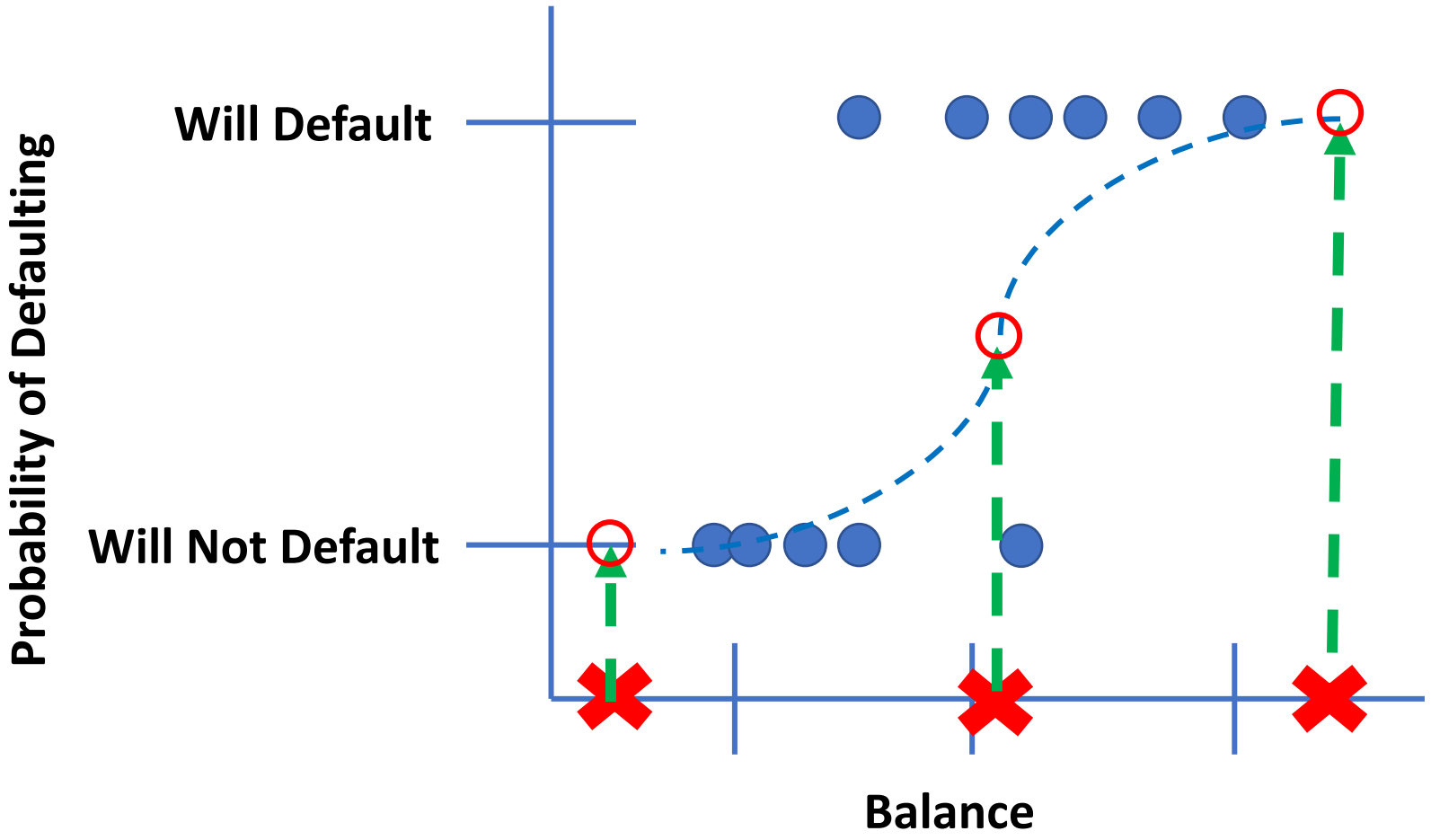
Multiple-class logistic regression exists

A data example

Y: **whether an individual will default on his/her credit card payment**

$X_1 =$ **annual income** $X_2 =$ **monthly credit card balance** $X_3 =$ **is he/she a student**




Will Default
Will Not Default
Probability of Defaulting
Balance

A simple logistic model

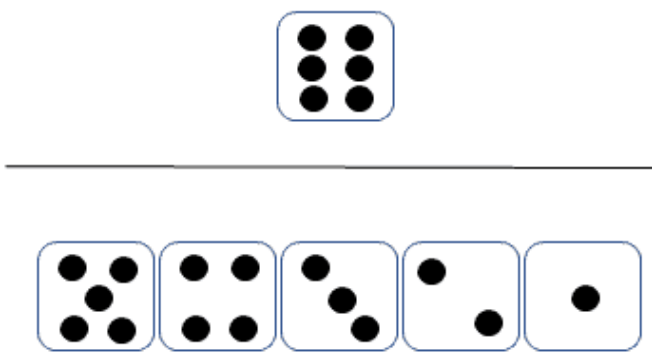
To model $\Pr(\textbf{default} = \text{TRUE} \mid \textbf{balance})$

$$\Pr(Y = 1 \mid X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

Logistic function

$\in [0,1]$

which is equivalent to…



Odds

$$\frac{\Pr(Y = 1 \mid X)}{1 - \Pr(Y = 1 \mid X)} = e^{\beta_0+\beta_1 X}$$

$\in [0, +\infty)$ **Rolling a 6-sided dice: Odds of rolling a 3 is 1/5**

A simple logistic model

$$\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)} = e^{\beta_0+\beta_1 X}$$

taking logarithm of both sides...

$$\log\left(\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)}\right) = \beta_0+\beta_1 X$$

Logit, or log-odds

$\in (-\infty, +\infty)$

logit is linear in X

Interpretation of coefficient

$$\log\left(\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)}\right)=\beta_0+\beta_1 X$$

β_1: the (additive) change of logit by increasing X by 1-unit

$$\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)}=e^{\beta_0+\beta_1 X}$$

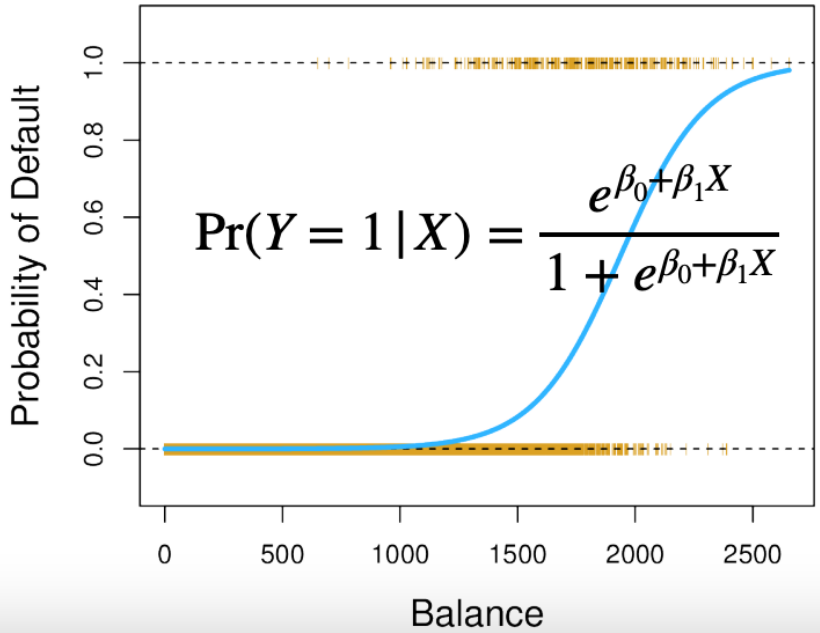
increasing X by 1-unit means multiply the odds by e^{β_1}

Interpretation of coefficient

$$\log\left(\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)}\right)=\beta_0+\beta_1 X$$

β_1: the (additive) change of logit by increasing X by 1-unit

β_1: is NOT the change in $\Pr(Y=1\,|\,X)$ associated with a 1-unit increase in X

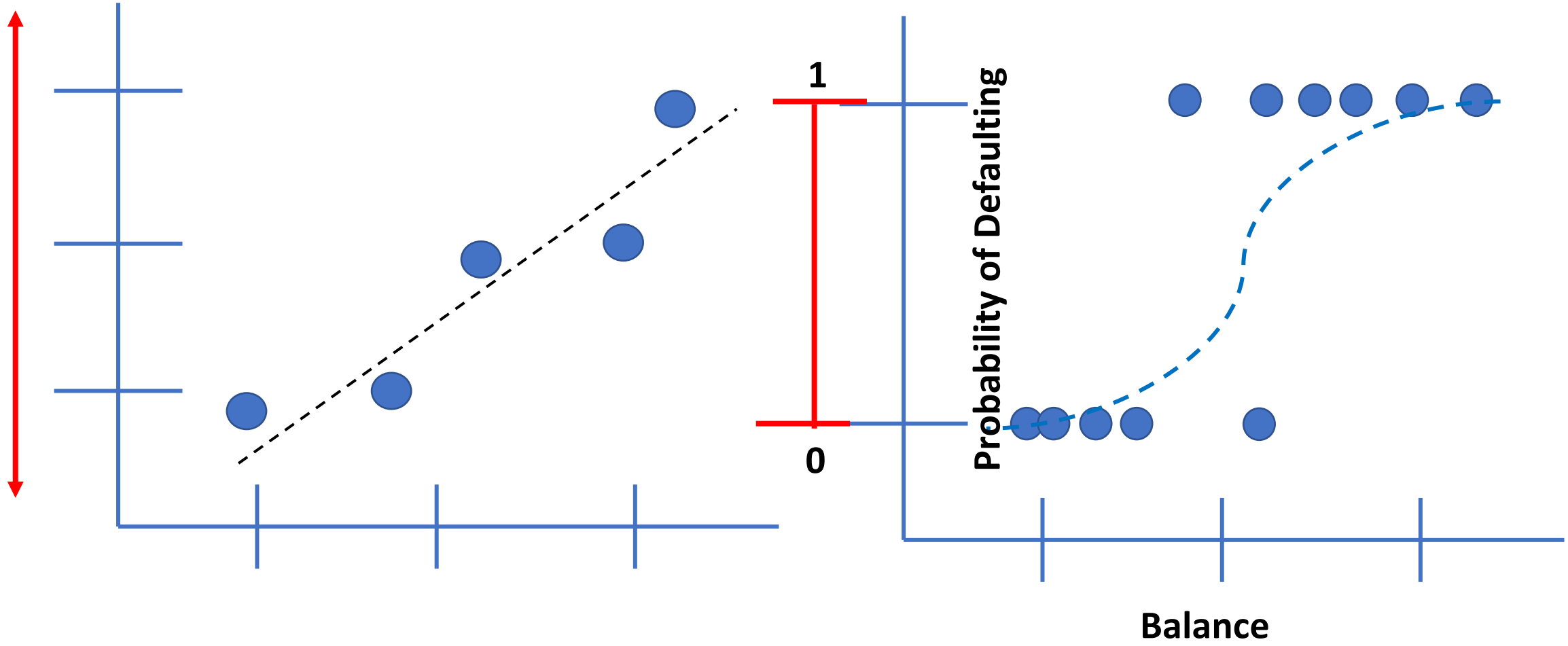


If $\beta_1 > 0$ then $X \nearrow \implies \Pr(Y=1\,|\,X) \nearrow$

If $\beta_1 < 0$ then $X \nearrow \implies \Pr(Y=1\,|\,X) \searrow$

The rate of change in $\Pr(Y=1\,|\,X)$ per unit change in X depend on the current value of X

not linear!

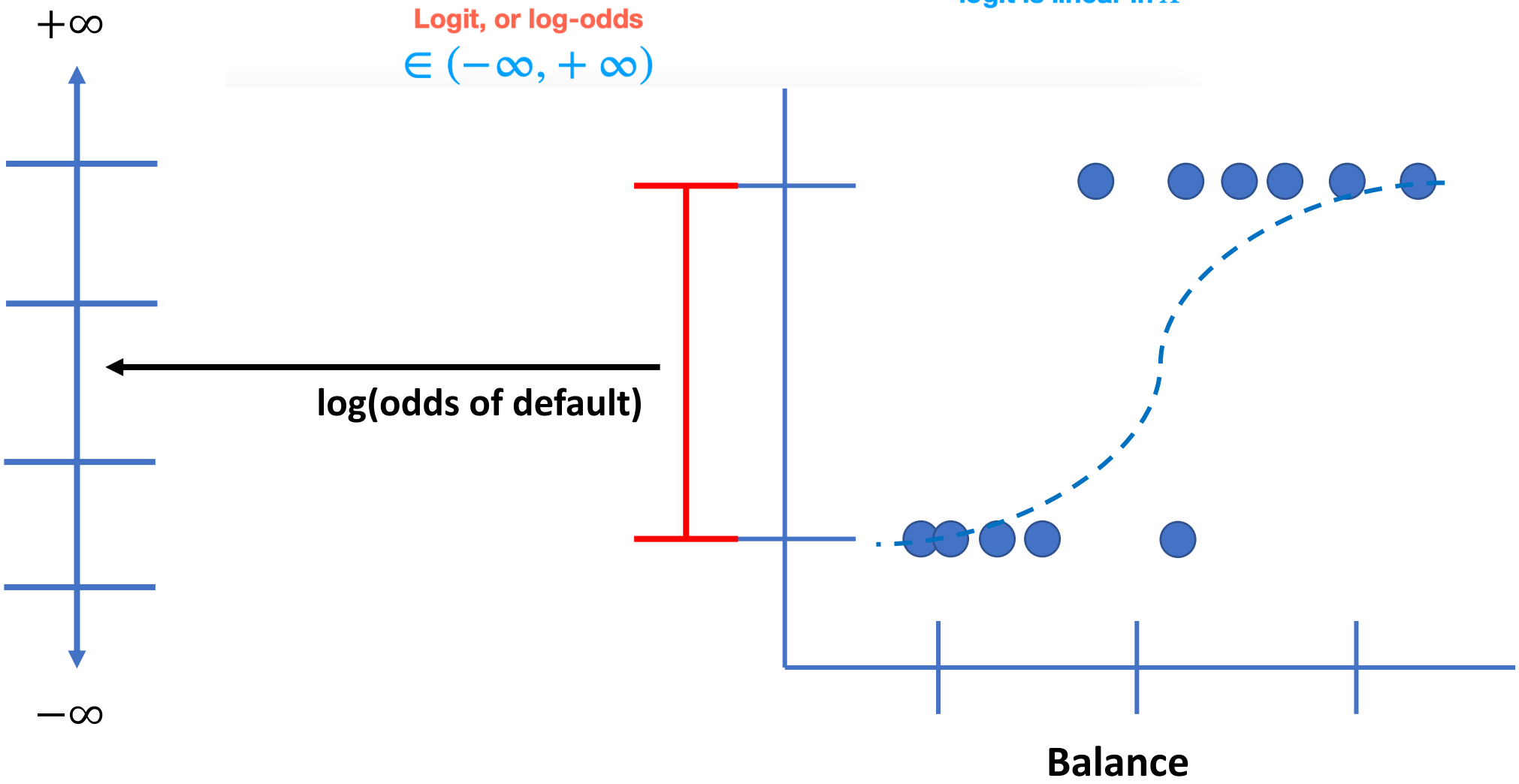

1
0
Probability of Defaulting
Balance

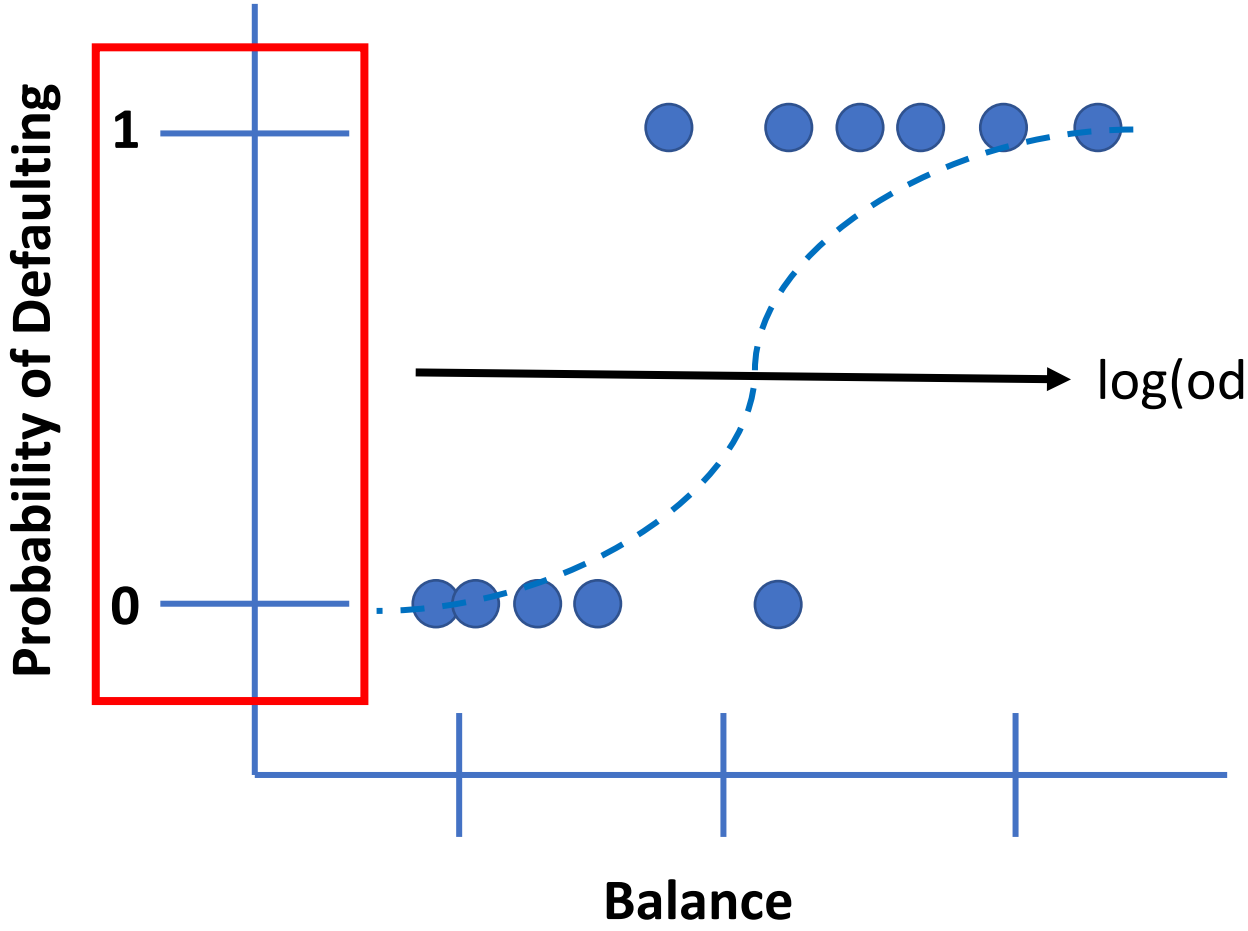
$$\log\left(\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)}\right)=\beta_0+\beta_1 X$$

logit is linear in X

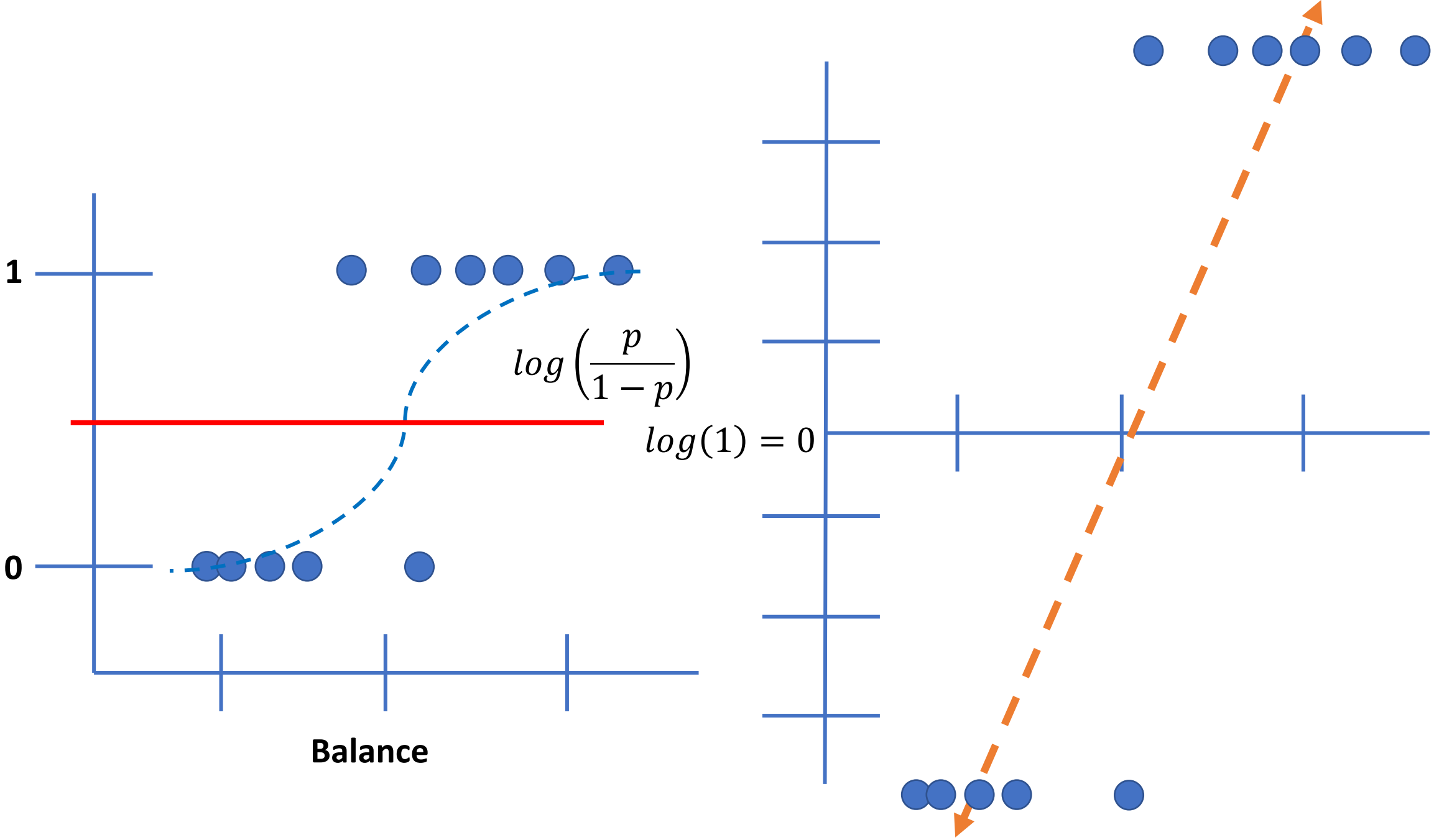
Logit, or log-odds

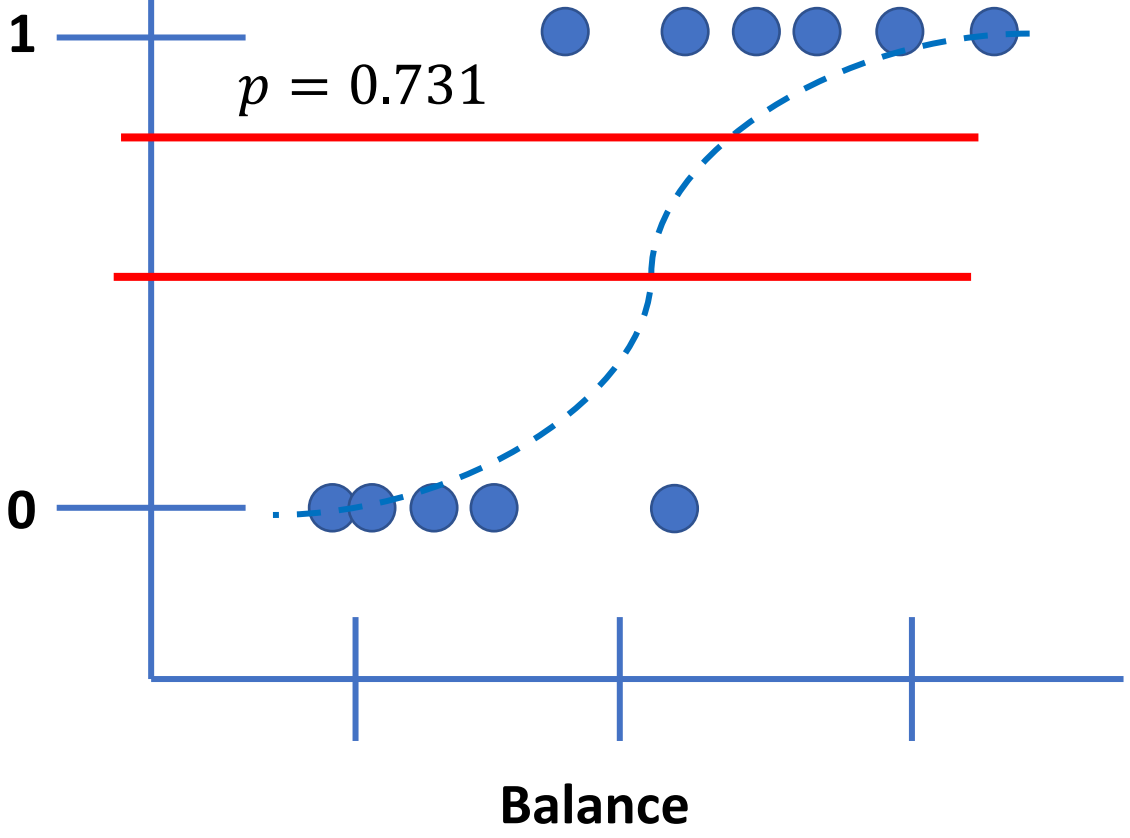
$\in(-\infty,+\infty)$

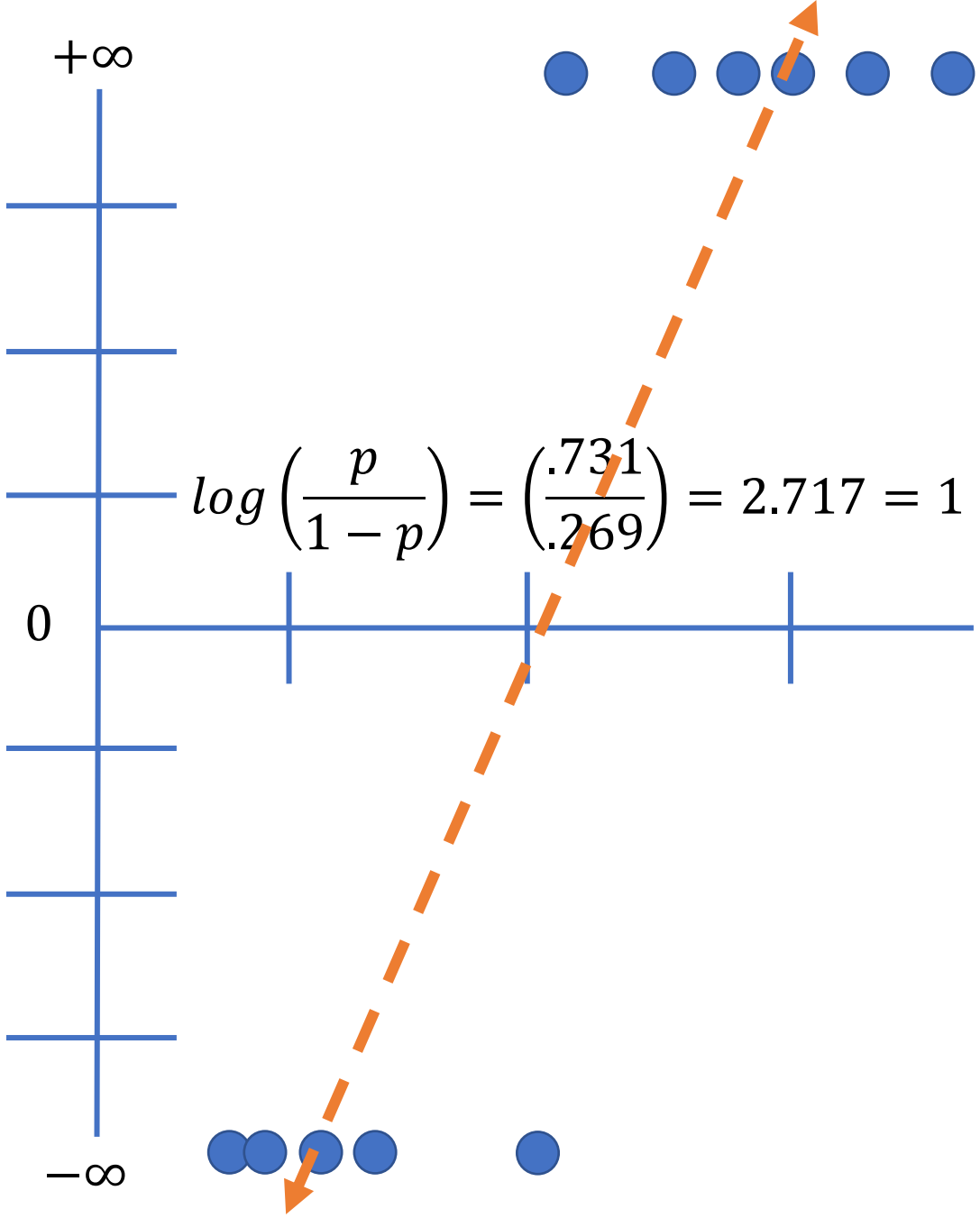


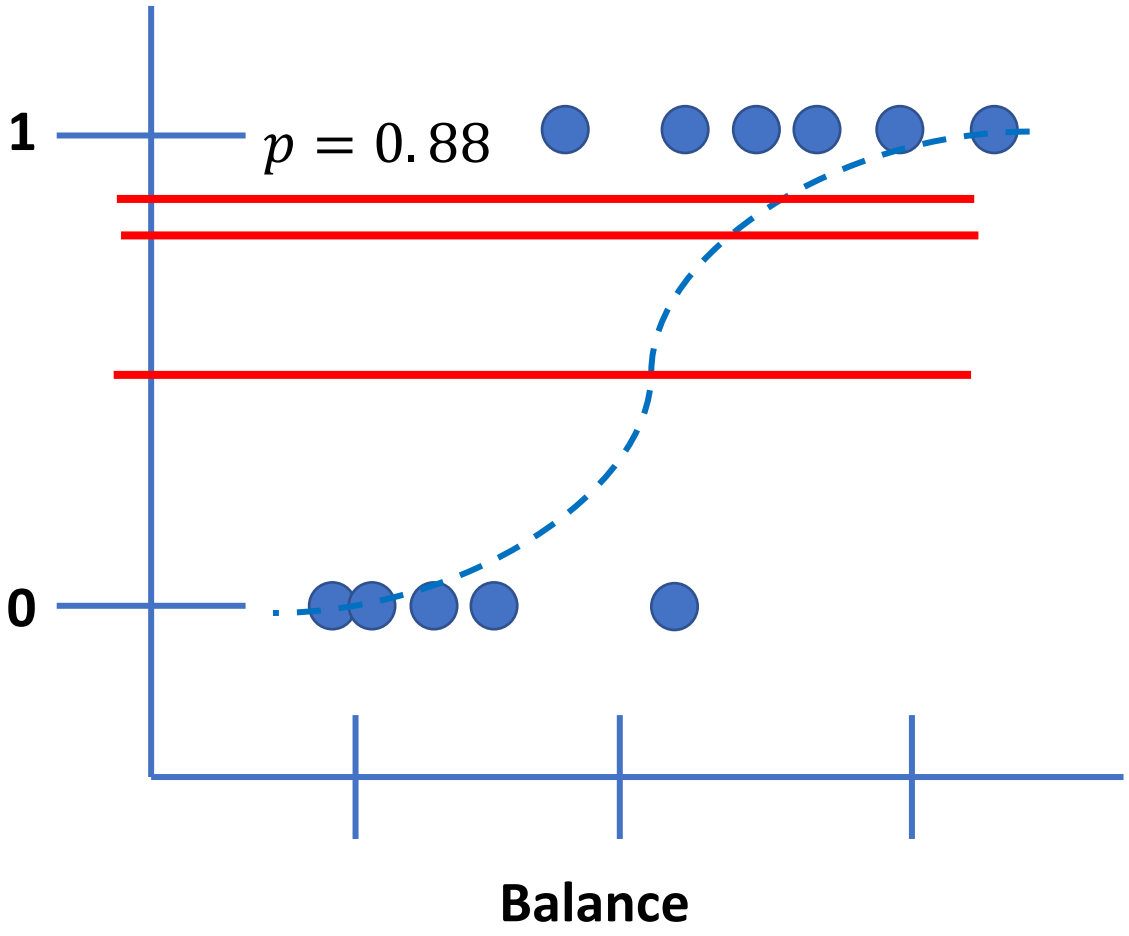


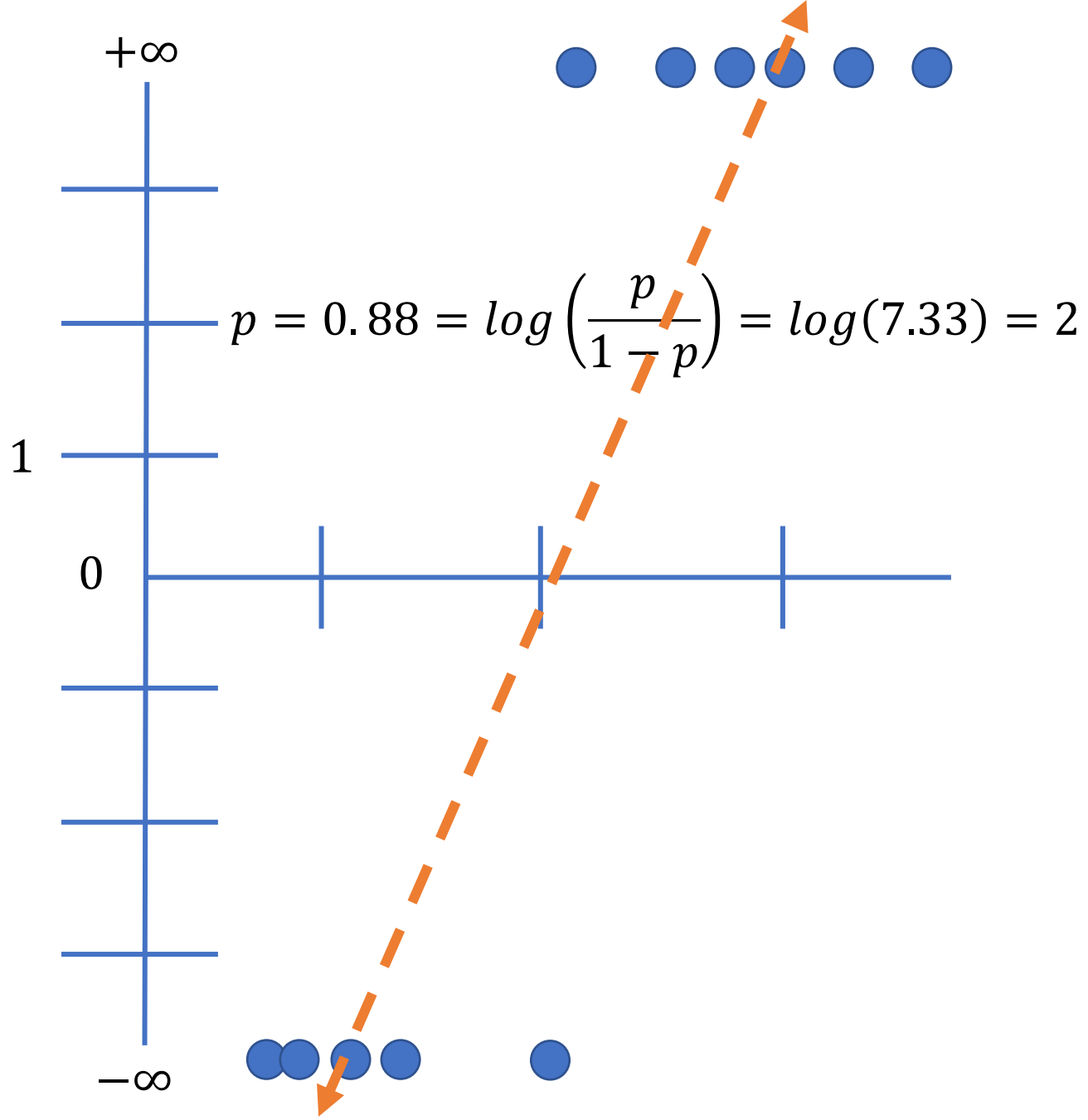
log(odds of default) $= log\left(\frac{p}{1-p}\right)$

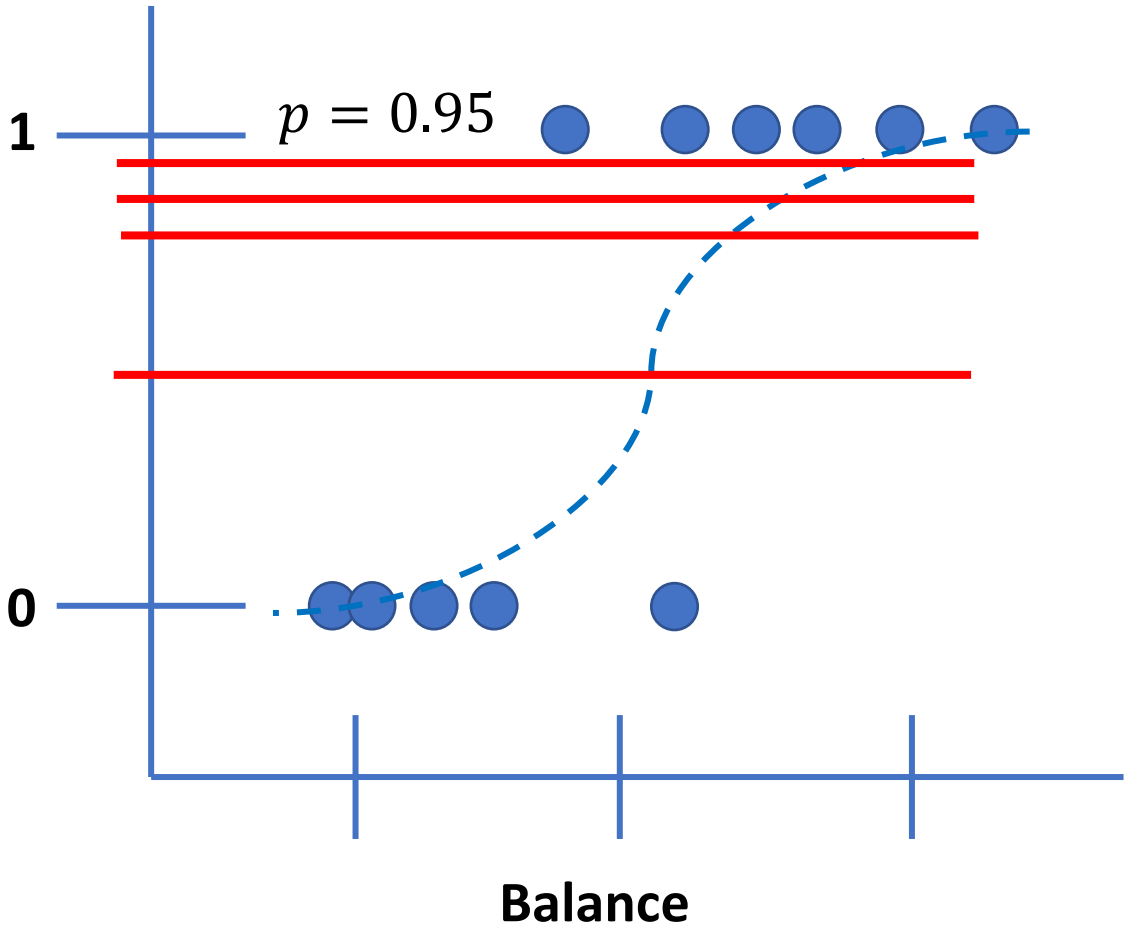

1
0
$log\left(\frac{p}{1-p}\right)$
$log(1) = 0$
Balance

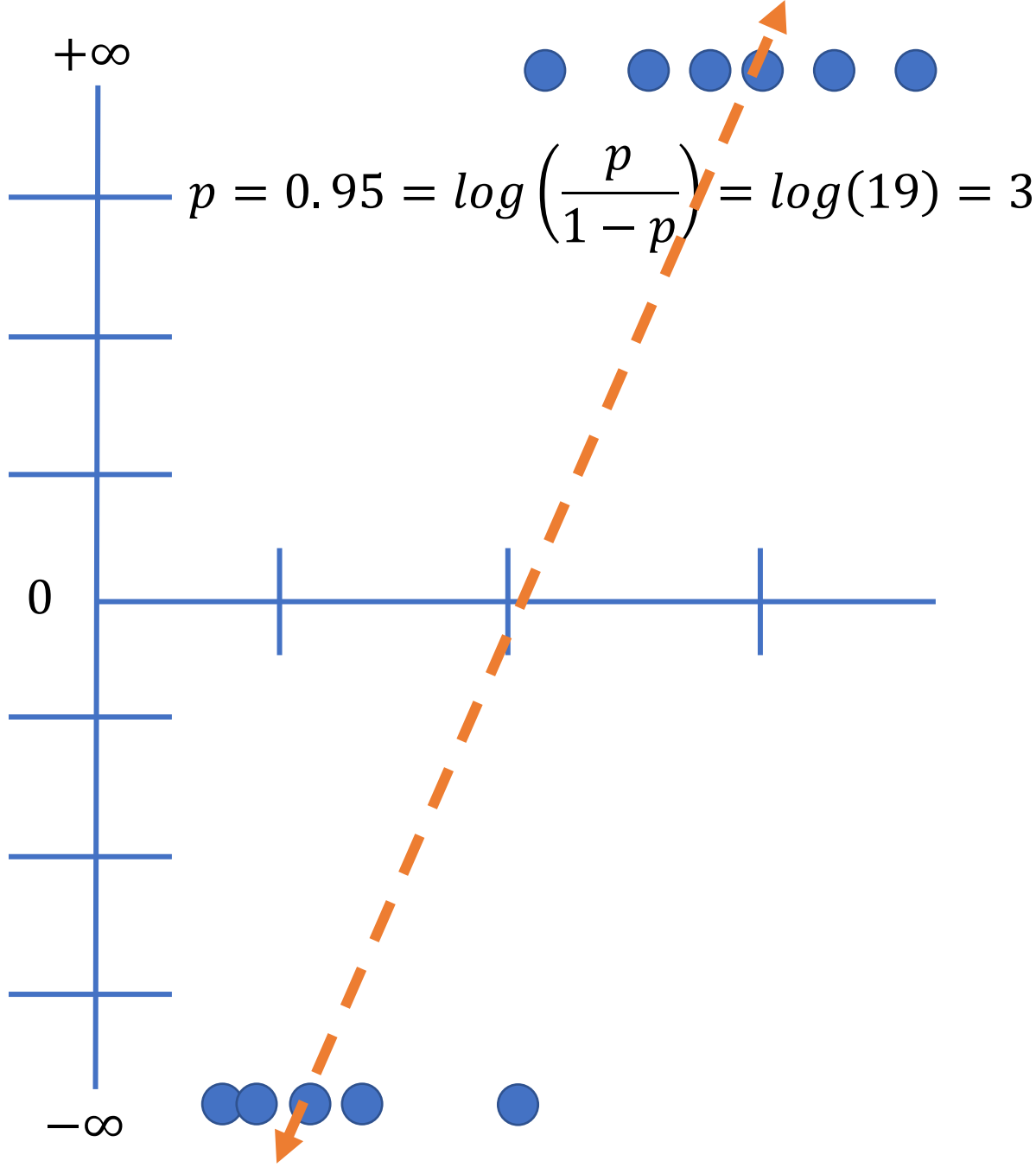

1
$p = 0.731$
0
Balance

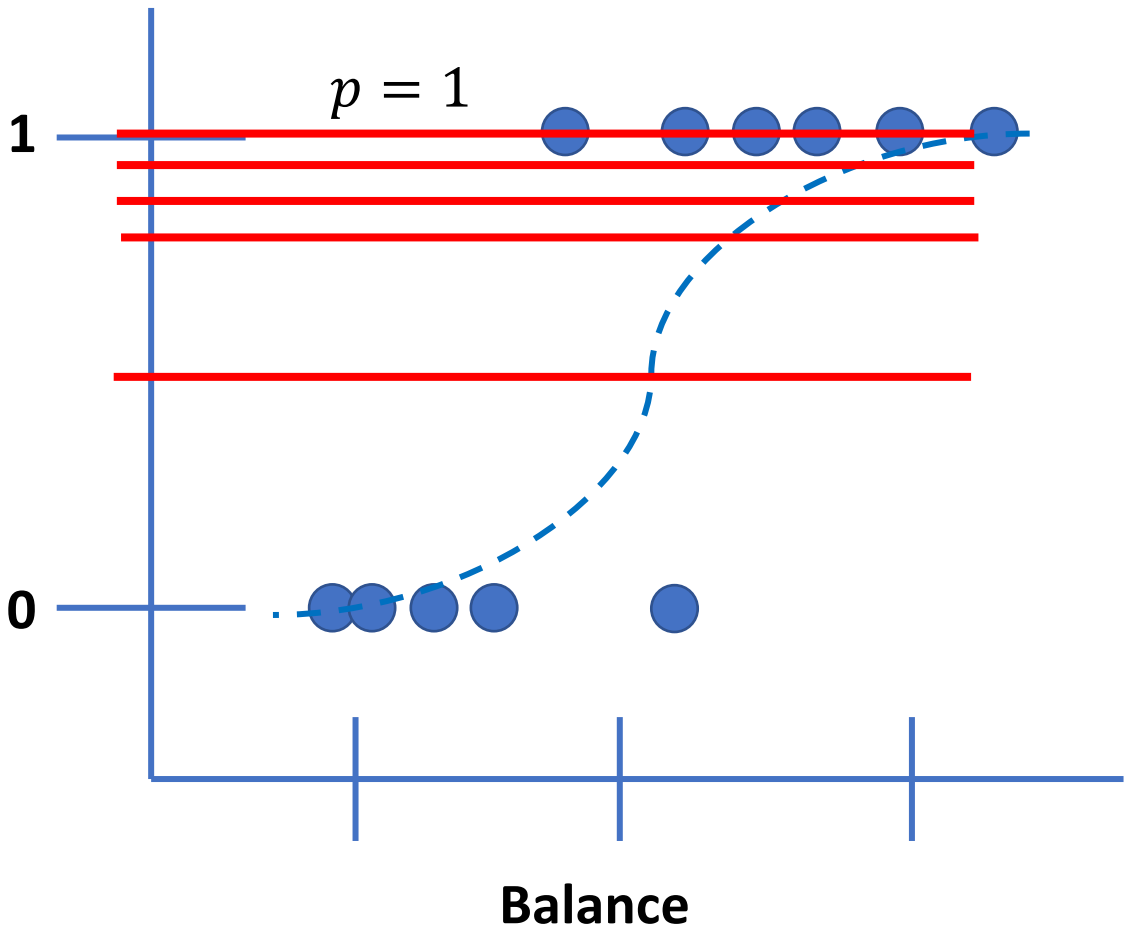

$+\infty$
$log\left(\frac{p}{1-p}\right) = \left(\frac{.731}{.269}\right) = 2.717 = 1$
0
$-\infty$

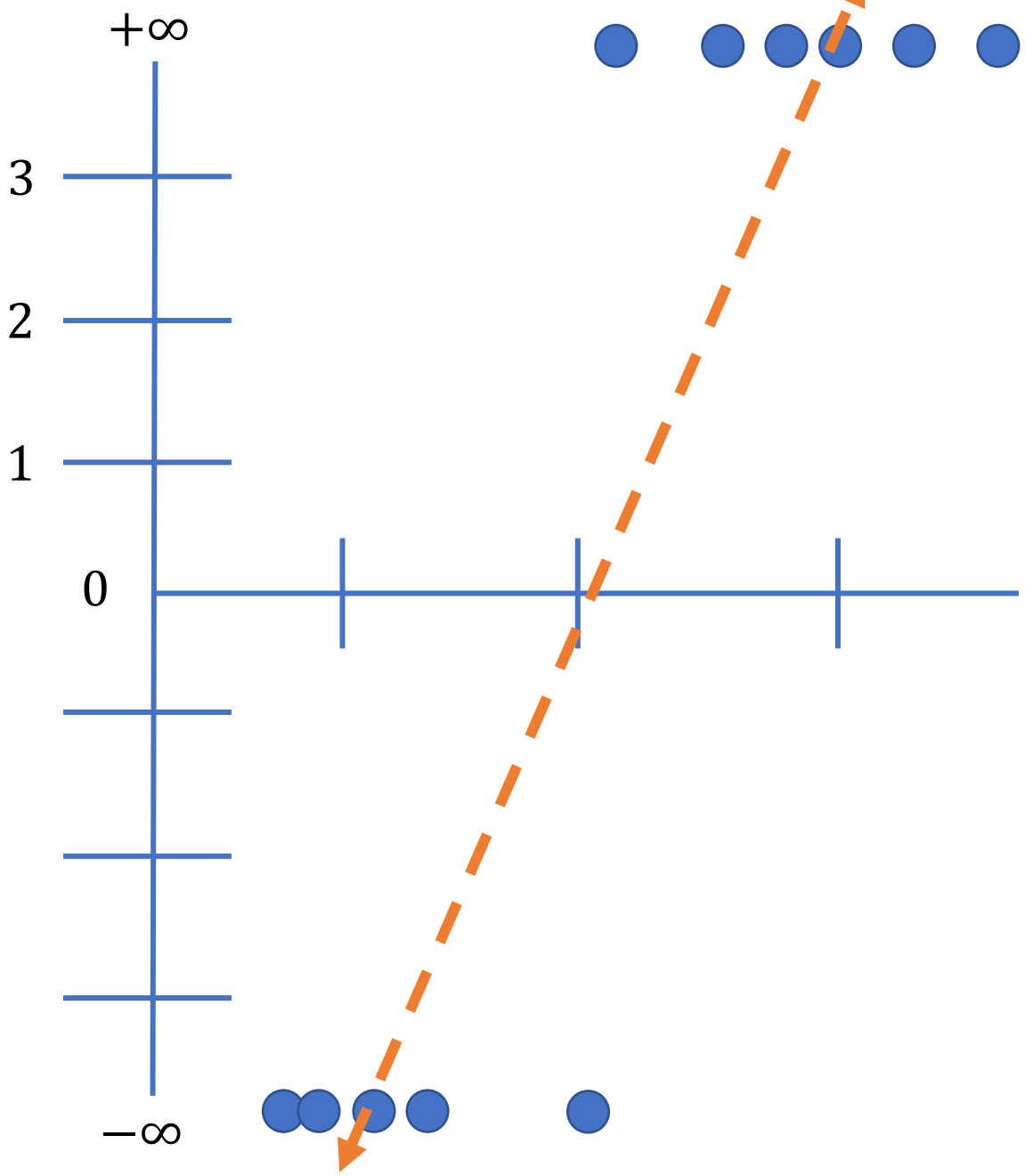

1
$p = 0.88$
0
Balance


$+\infty$
$p = 0.88 = log\left(\frac{p}{1-p}\right) = log(7.33) = 2$
1
0
$-\infty$


1
$p = 0.95$
0
Balance


$+\infty$
2
1
0
$-\infty$
$p = 0.95 = log\left(\frac{p}{1-p}\right) = log(19) = 3$


$p = 1$
1
0
Balance

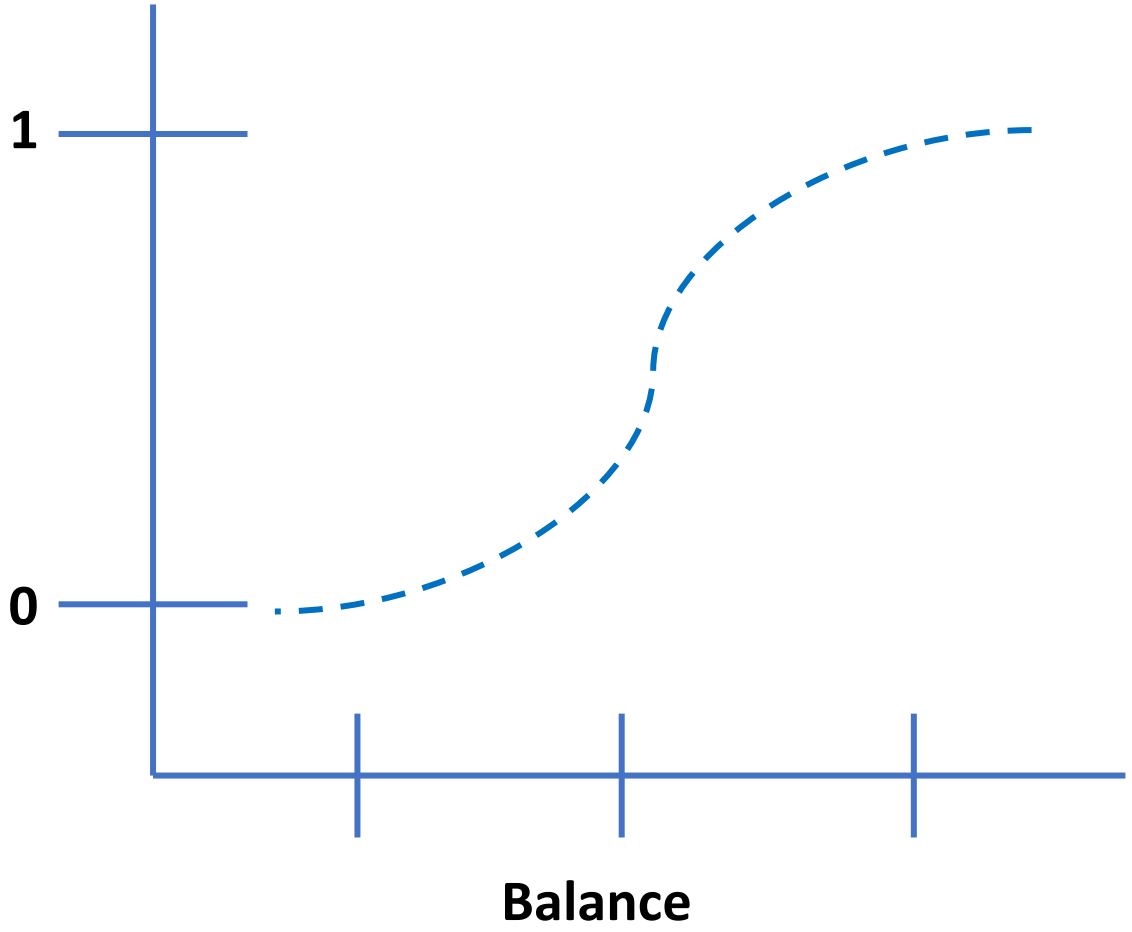

$+\infty$
3
2
1
0
$-\infty$

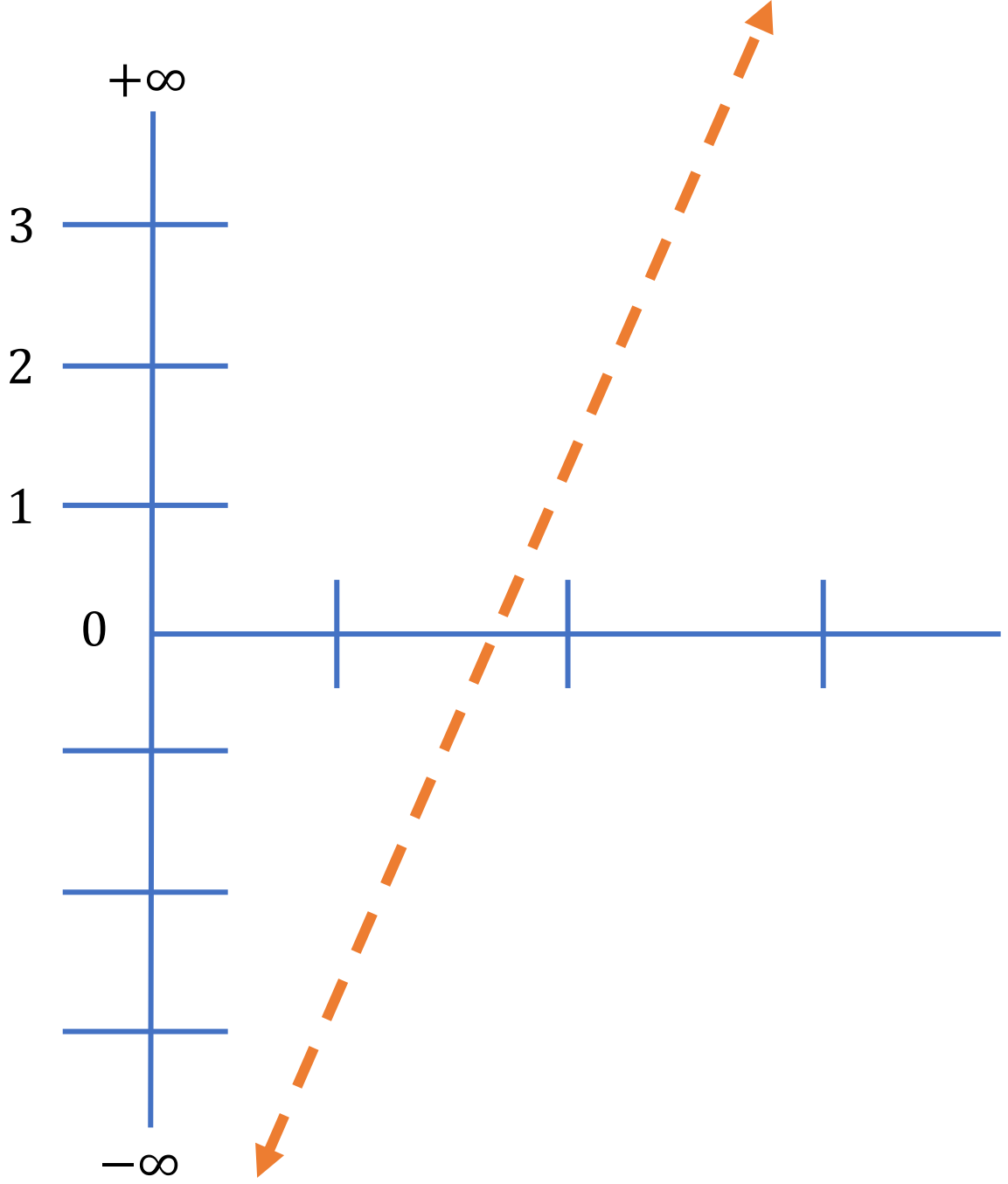
$$\log\left(\frac{\Pr(Y=1\,|\,X)}{1-\Pr(Y=1\,|\,X)}\right)=\beta_0+\beta_1 X$$

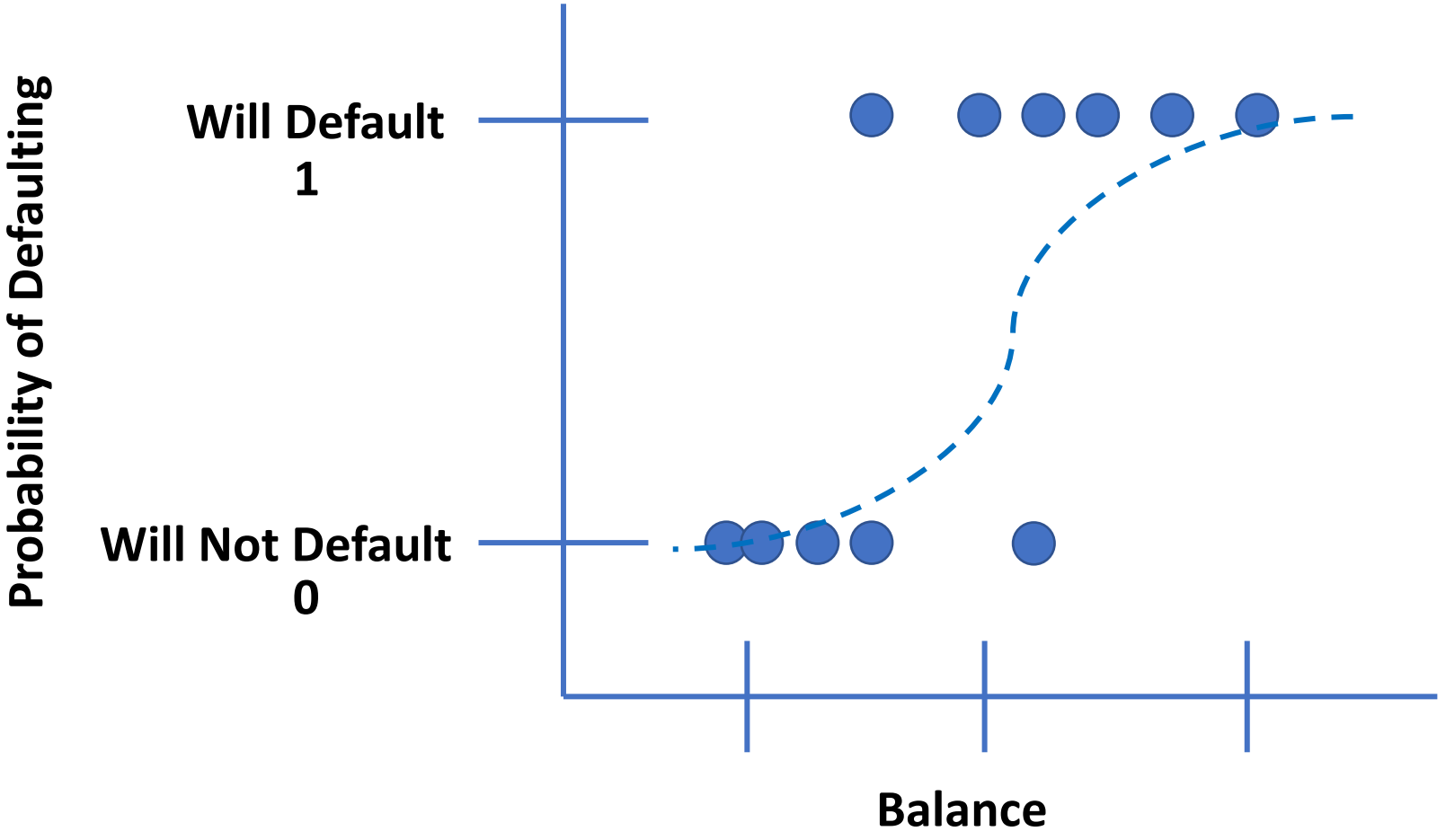
logit is linear in X

Logit, or log-odds

$\in(-\infty,+\infty)$







Default predicted by **Student** + **Balance** + **Income** ?

Multiple logistic regression

$$\Pr(Y = 1 \mid X_1, \ldots, X_p) = \frac{e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}{1 + e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}} \in [0,1]$$

which is equivalent to…

$$\frac{\Pr(Y = 1 \mid X_1, \ldots, X_p)}{1 - \Pr(Y = 1 \mid X_1, \ldots, X_p)} = e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p} \in [0, +\infty)$$

taking logarithm of both sides…

$$\log\left(\frac{\Pr(Y = 1 \mid X_1, \ldots, X_p)}{1 - \Pr(Y = 1 \mid X_1, \ldots, X_p)}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p \in (-\infty, +\infty)$$

Interpretation of coefficient

$$\log\left(\frac{\Pr(Y=1\,|\,X_1,\ldots,X_p)}{1-\Pr(Y=1\,|\,X_1,\ldots,X_p)}\right)=\beta_0+\beta_1X_1+\ldots+\beta_pX_p$$

β_j: the (additive) change of logit by increasing X_j by 1-unit, conditional on the other predictors

For any j, conditional on the other predictors $X_1,\ldots,X_{j-1},X_j\ldots,X_p$

If $\beta_j>0$ then $X_j\nearrow \implies \Pr(Y=1\,|\,X_1,\ldots,X_p)\nearrow$

If $\beta_j<0$ then $X_j\nearrow \implies \Pr(Y=1\,|\,X_1,\ldots,X_p)\searrow$

The rate of change in $\Pr(Y=1\,|\,X_1,\ldots,X_p)$ per unit change in X_j depend on the current value of X_j

not linear!

Estimating coefficients

Maximum Likelihood Estimation (MLE)

```
> glm.fit <- glm(default ~ student + balance + income, data = Default, family = binomial)
> summary(glm.fit)

Call:
glm(formula = default ~ student + balance + income, family = binomial,
    data = Default)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.4691  -0.1418  -0.0557  -0.0203   3.7383

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.087e+01  4.923e-01 -22.080  < 2e-16 ***
studentYes  -6.468e-01  2.363e-01  -2.738  0.00619 **
balance      5.737e-03  2.319e-04  24.738  < 2e-16 ***
income       3.033e-06  8.203e-06   0.370  0.71152
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2920.6  on 9999  degrees of freedom
Residual deviance: 1571.5  on 9996  degrees of freedom
AIC: 1579.5

Number of Fisher Scoring iterations: 8
```

What is wrong?

A simple logistic regression of **default** on **student**

	Coefficient	Std. error	Z-statistic	P-value
`Intercept`	−3.5041	0.0707	−49.55	<0.0001
`student[Yes]`	0.4049	0.1150	3.52	0.0004

A multiple logistic regression of **default** on **balance**, **income**, and **student**

	Coefficient	Std. error	Z-statistic	P-value
`Intercept`	−10.8690	0.4923	−22.08	<0.0001
`balance`	0.0057	0.0002	24.74	<0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	−0.6468	0.2362	−2.74	0.0062

Contradicting signs!

What is wrong?

	Coefficient
Intercept	−3.5041
student[Yes]	0.4049

A student is **riskier** than a non-student, if **no other information about the student is available**

	Coefficient
Intercept	−10.8690
balance	0.0057
income	0.0030
student[Yes]	−0.6468

A student is **less risky** than a non-student with the **same credit card balance**

student and **balance** are correlated

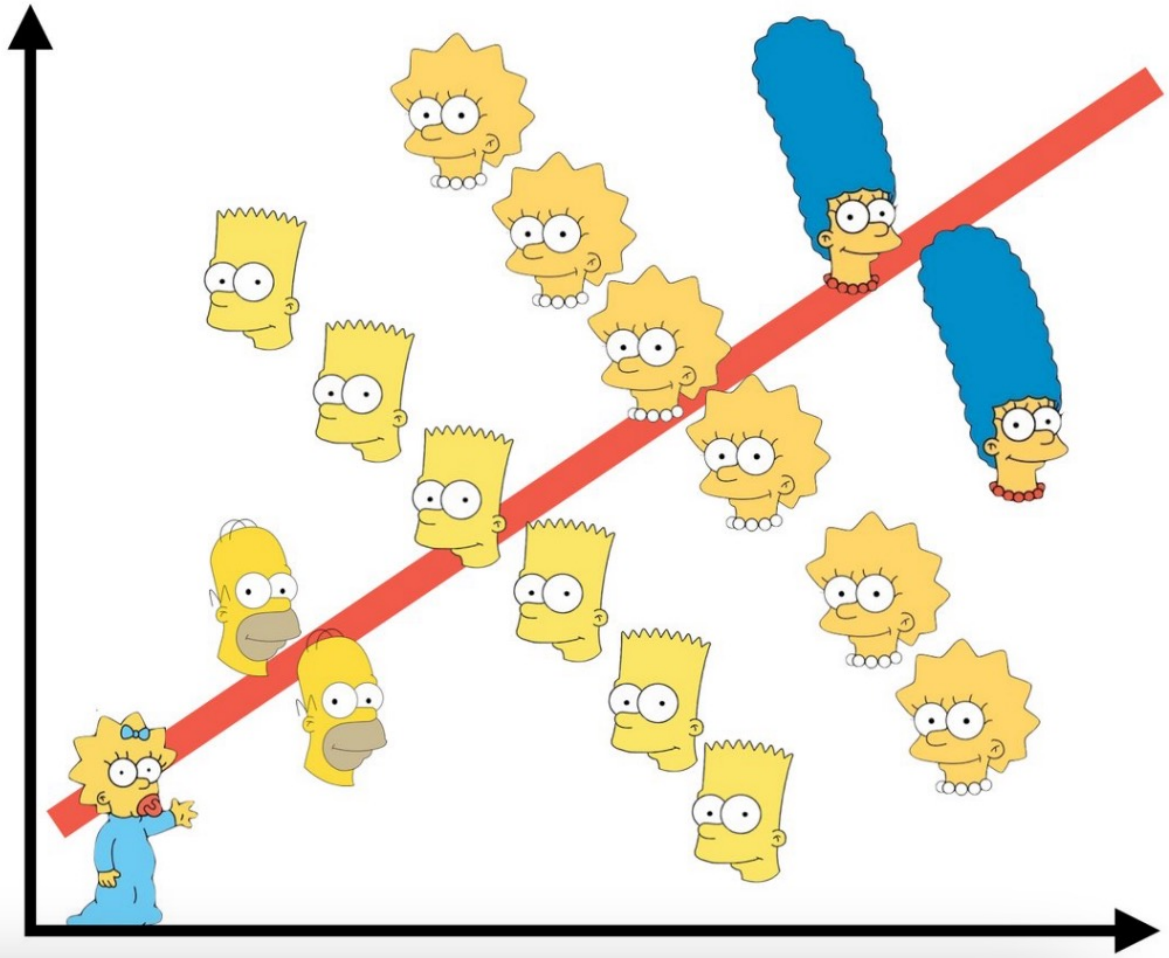
student are more likely to have large credit card balances, **which tends to be associated with high default rates**

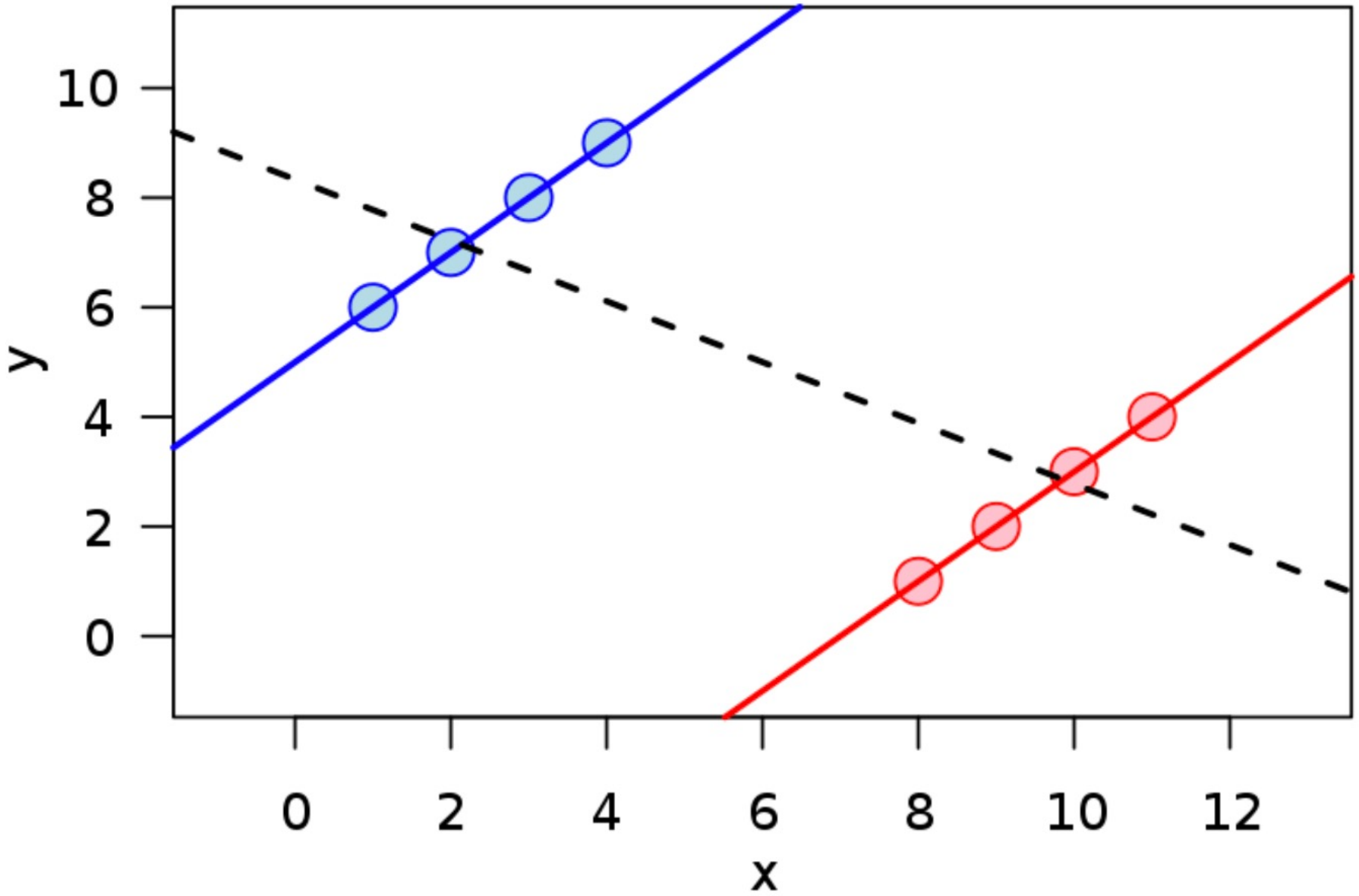
Confounding

Pay extra attention when performing regressions involving only a single predictor when other predictors may also be relevant

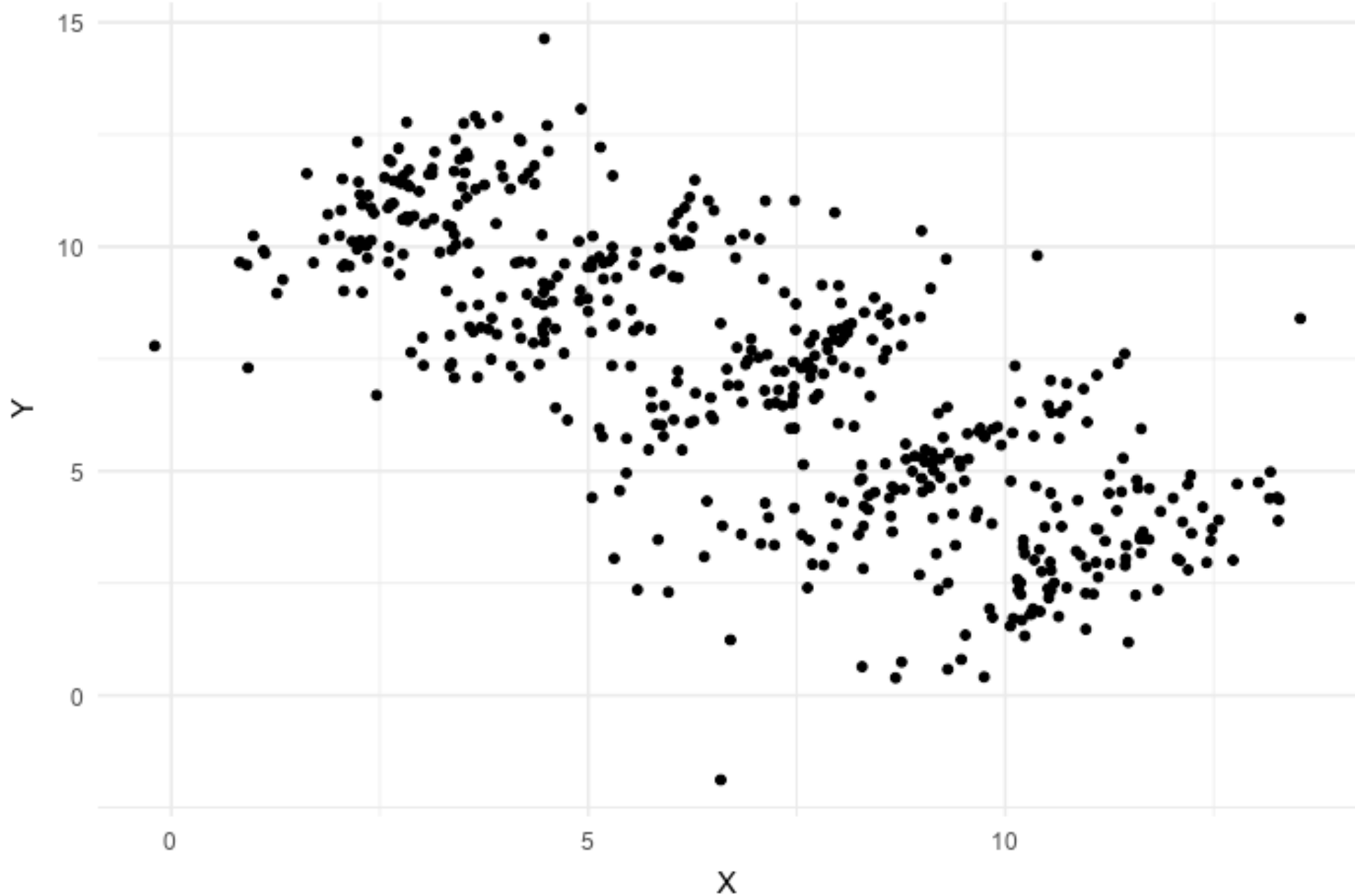
Confounding

Simpson's paradox




10
8
6
4
2
0
y
0
2
4
6
8
10
12
x

Korrelation:



Prediction of probability

	Coefficient	Std. error	Z-statistic	P-value
`Intercept`	−10.8690	0.4923	−22.08	<0.0001
`balance`	0.0057	0.0002	24.74	<0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	−0.6468	0.2362	−2.74	0.0062

$$\Pr(Y = 1 \mid X_1, \ldots, X_p) = \frac{e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}{1 + e^{\beta_0+\beta_1 X_1+\ldots+\beta_p X_p}}$$

$$\widehat{\Pr}(\text{default} = \text{TRUE} \mid \text{balance} = 1500, \text{income} = 40000, \text{student} = \text{FALSE})$$

$$= \frac{e^{-10.8690+0.0057\times1500+0.0030\times40000-0.6468\times0}}{1 + e^{-10.8690+0.0057\times1500+0.0030\times40000-0.6468\times0}} = 0.105$$

From predicted probability to predicted class

$$\widehat{\Pr}(\text{default} = \text{TRUE} \mid \text{balance} = 1500, \text{income} = 40000, \text{student} = \text{FALSE}) = 0.105$$

We can predict class (**default = TRUE or default = FALSE**)
by **thresholding the predicted probability**

For example, one might predict **default = TRUE** for any individual
for whom the **predicted probability > 0.5**

Or, if the credit card company is super **conservative** in predicting individuals who are at risk for default, one might predict **default = TRUE** for any individual for **whom the predicted probability > 0.1**

Classification performance

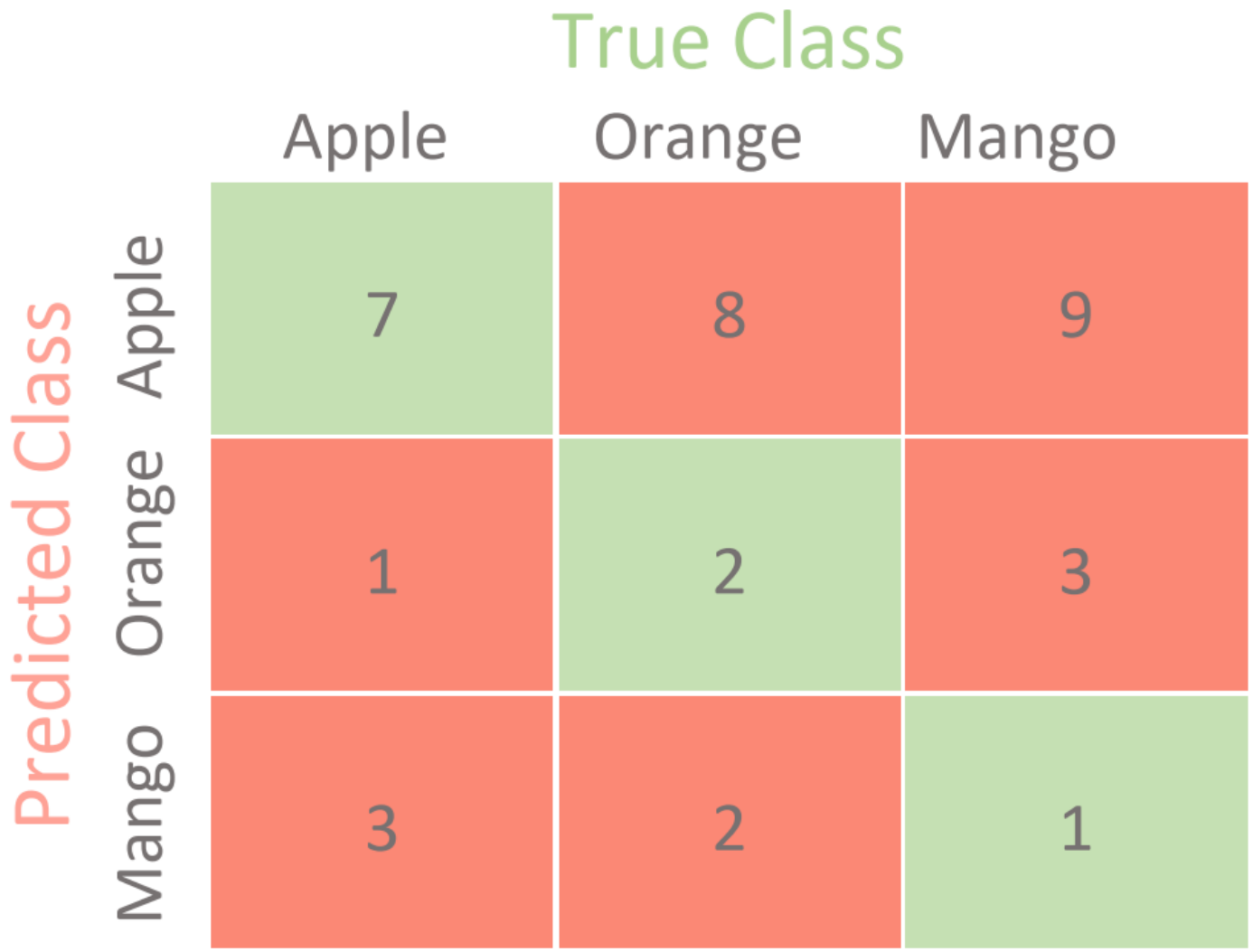
For a binary classifier, use **confusion matrix**

		Predicted Class	
		$\hat{Y} = 0$	$\hat{Y} = 1$
True Class	$Y = 0$	True Negative (**TN**)	False Positive (**FP**)
	$Y = 1$	False Negative (**FN**)	True Positive (**TP**)

True positive rate (TPR) = TP / (TP + FN), a.k.a. 1 - Type II error, power, sensitivity, recall

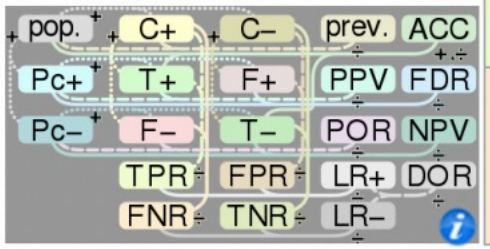
False positive rate (FPR) = FP / (FP + TN), a.k.a. Type I error, 1 - Specificity

Confusion matrices (multiclass)

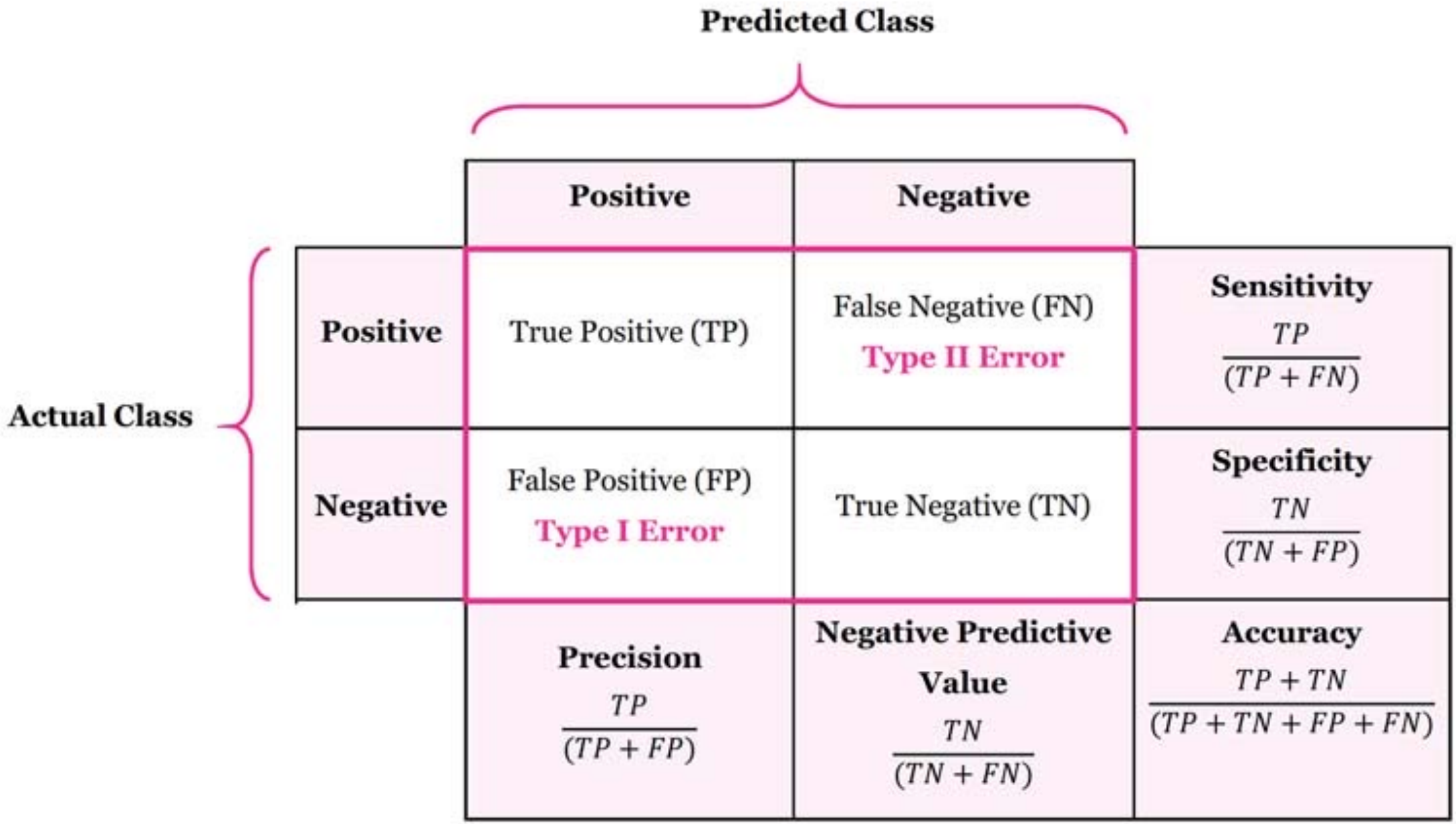


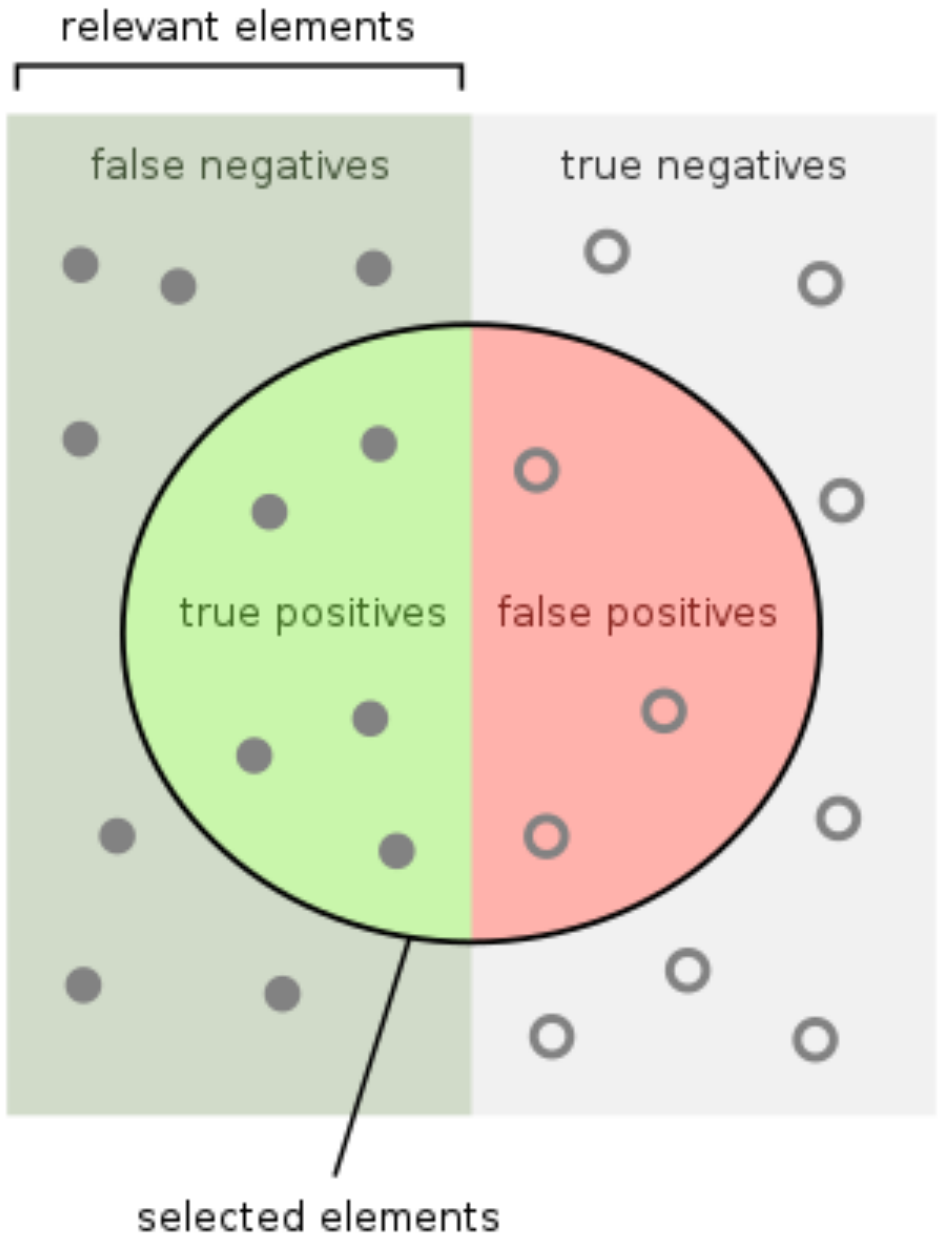
Many definitions and synonyms

		True condition			
	Total population	Condition positive	Condition negative	Prevalence $= \frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) $= \frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**	**False positive** (Type I error)	Positive predictive value (PPV), Precision $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative** (Type II error)	**True negative**	False omission rate (FOR) $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR−}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	True negative rate (TNR), Specificity (SPC) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) $= \frac{\text{FNR}}{\text{TNR}}$	F_1 score $= \frac{2}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}}$



https://en.wikipedia.org/wiki/Receiver_operating_characteristic


Predicted Class
Positive
Negative
Actual Class
Positive
True Positive (TP)
False Negative (FN)
Type II Error
Sensitivity
$\frac{TP}{(TP+FN)}$
Negative
False Positive (FP)
Type I Error
True Negative (TN)
Specificity
$\frac{TN}{(TN+FP)}$
Precision
$\frac{TP}{(TP+FP)}$
Negative Predictive Value
$\frac{TN}{(TN+FN)}$
Accuracy
$\frac{TP+TN}{(TP+TN+FP+FN)}$



How many relevant items are selected? e.g. How many sick people are correctly identified as having the condition.

$$\text{Sensitivity} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

How many negative selected elements are truly negative? e.g. How many healthy people are identified as not having the condition.

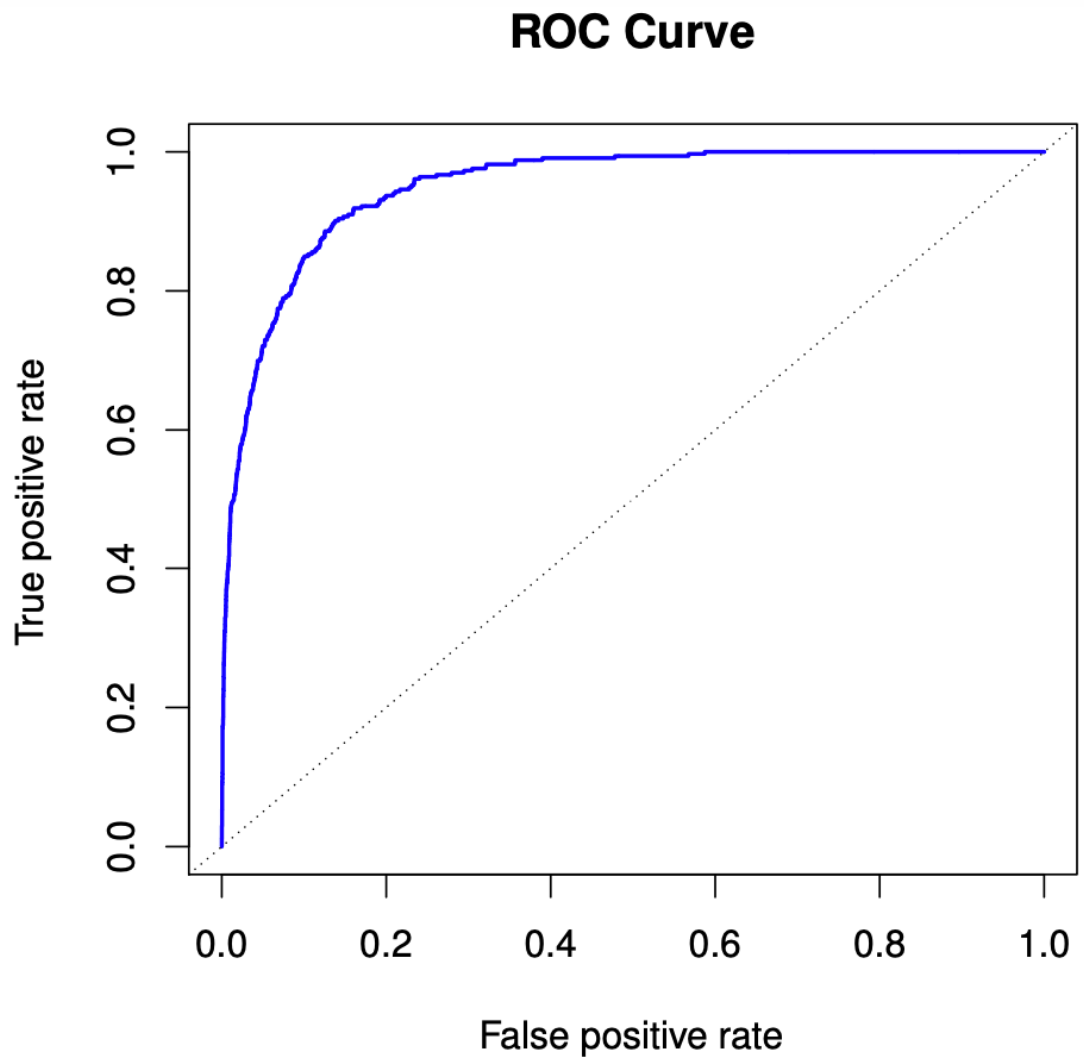
$$\text{Specificity} = \frac{\text{true negatives}}{\text{true negatives} + \text{false positives}}$$

ROC

A popular graphic for displaying the
2 types of error, for **all threshold values**

Useful in **comparing** different classifiers,
e.g., logistic regression v.s. K-NN

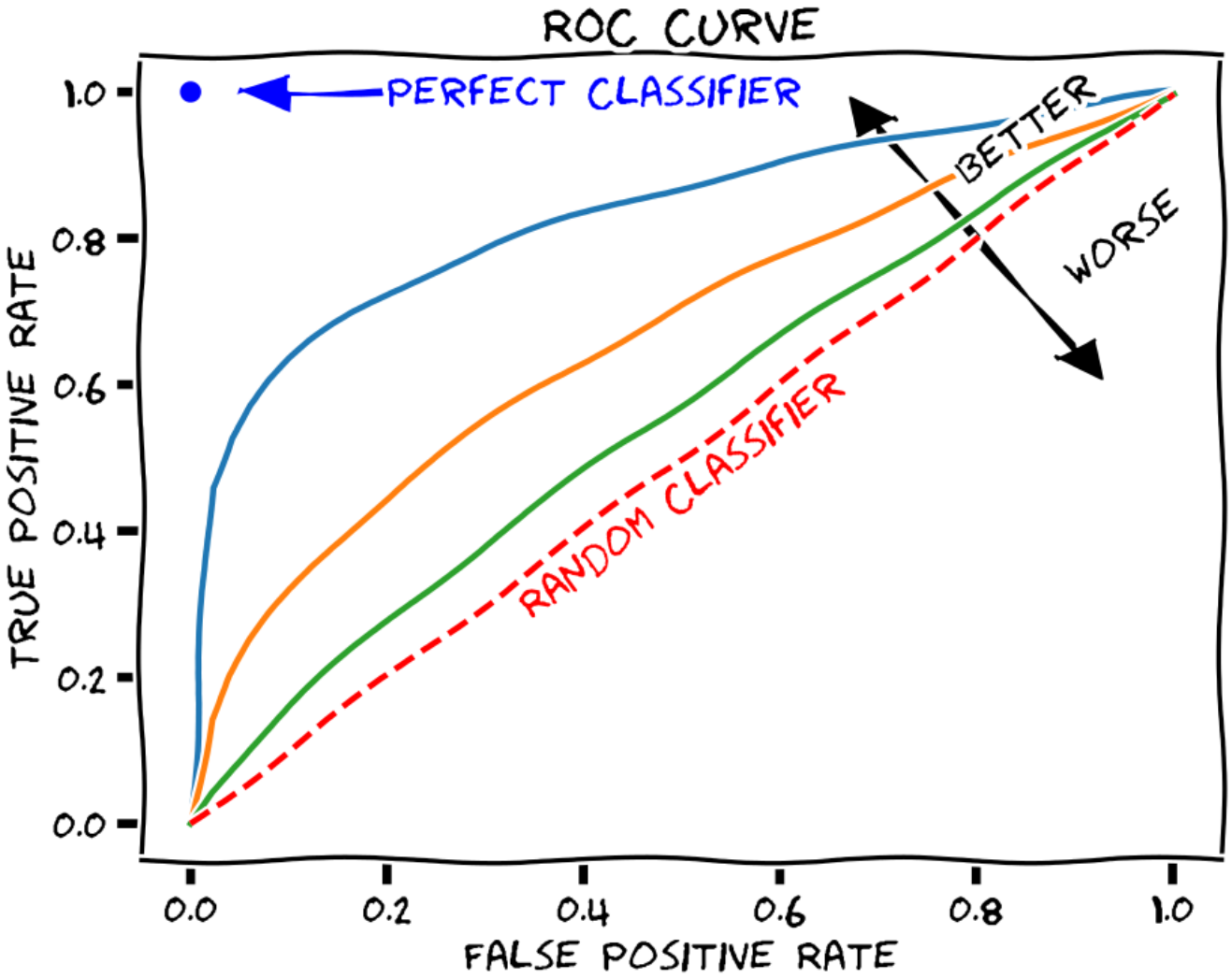
Methods perform differently in different
parts of the curve (i.e., different thresholds)

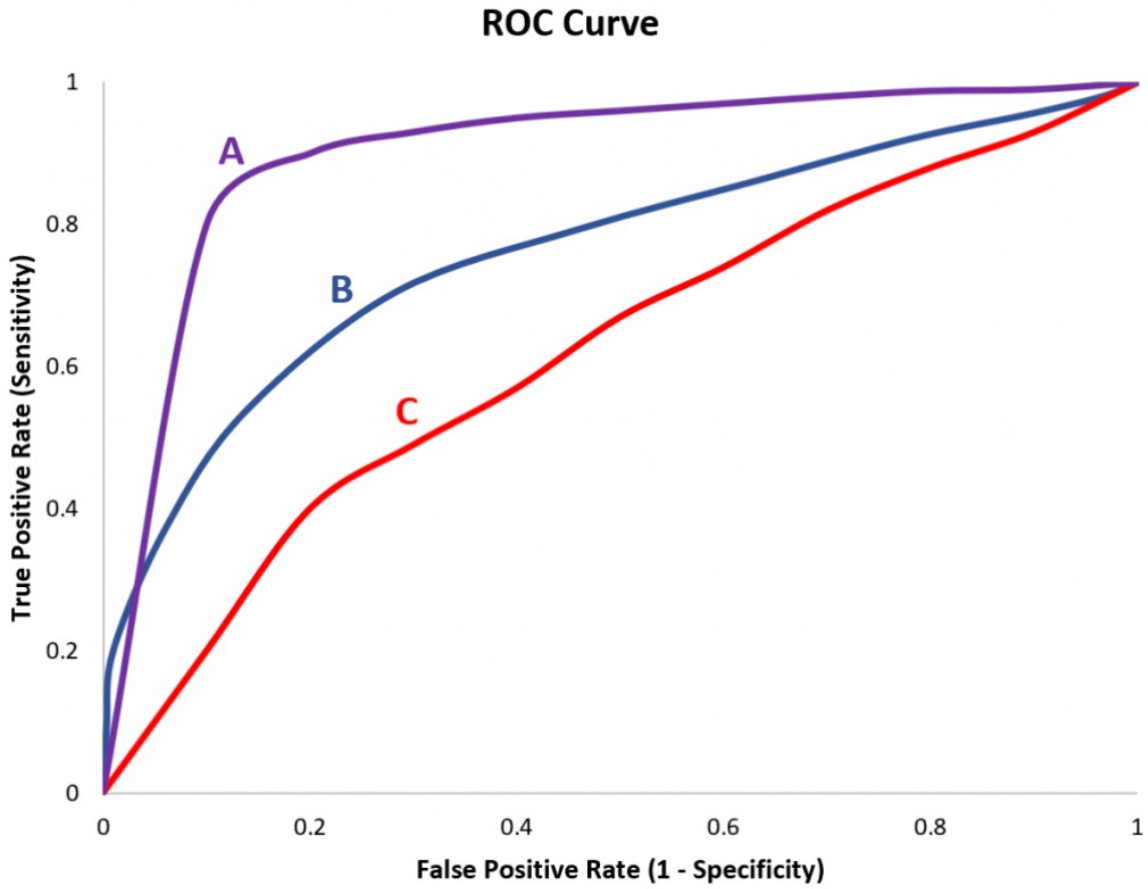


Area under curve (AUC) measures the overall performance of a classifier

Area under curve (AUC) summarized over all possible thresholds

Higher AUC, the better





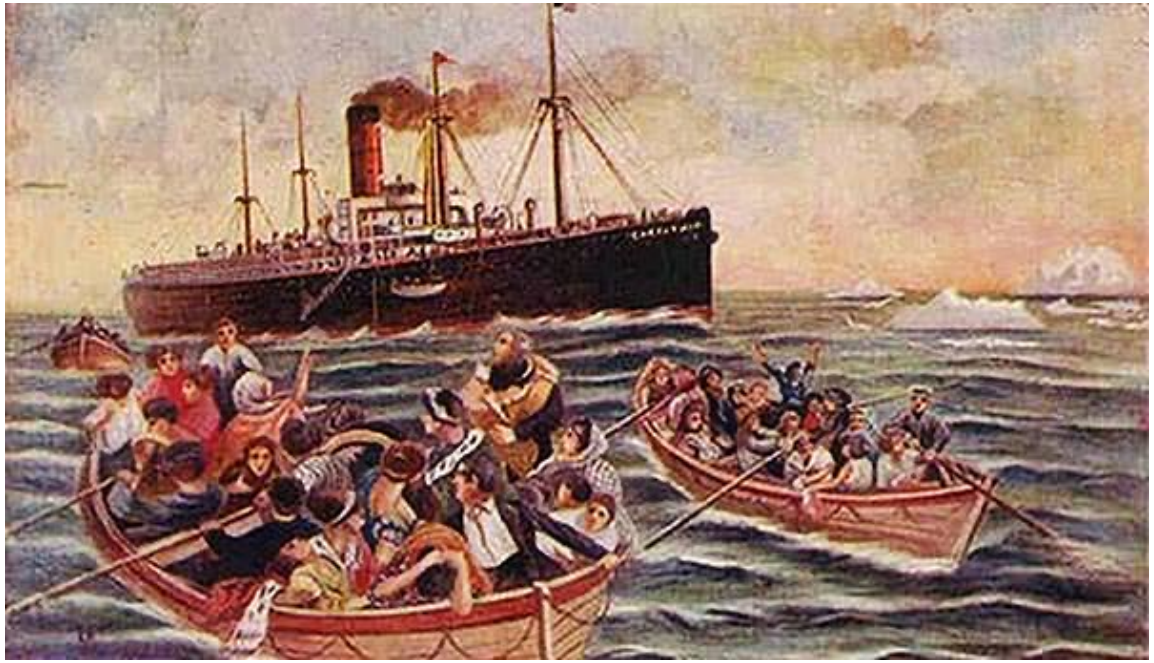
- **Model A:** AUC = 0.923
- **Model B:** AUC = 0.794
- **Model C:** AUC = 0.588

Detour on ROC curves

Lecture Slides: Logistic Regression

ROC Curves: Deep Dive





In summary

Logistic regression

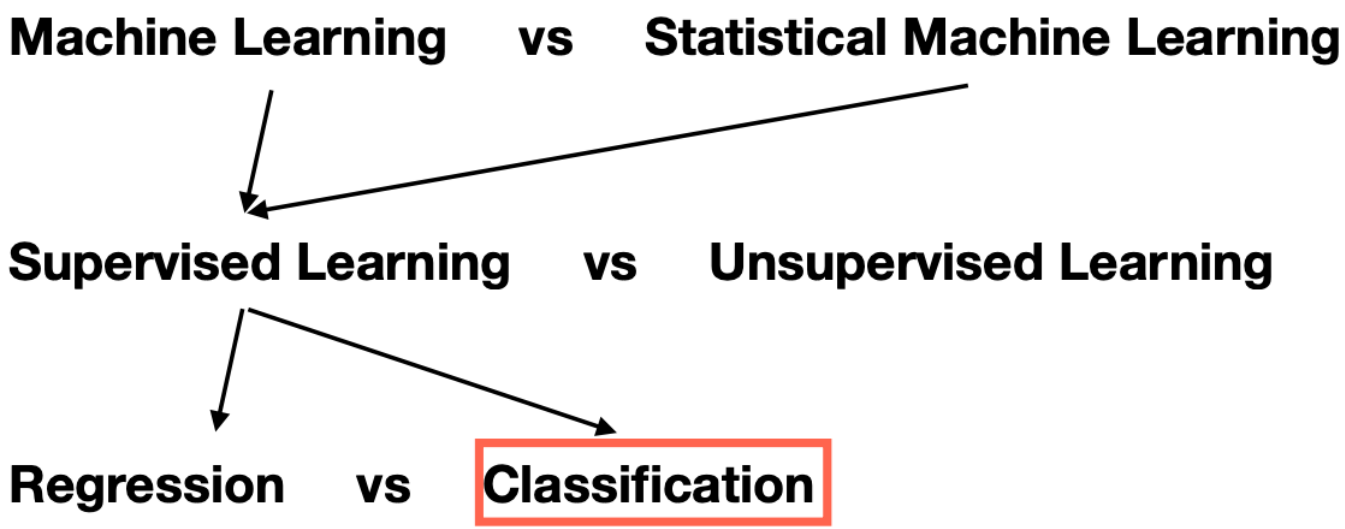
- **Model**
- **Coefficient interpretation**
- **Confounding**

Classification performance

- **TPR/FPR**
- **ROC curve and AUC**

Next…

Discriminant Analysis



Another linear method in classification:
Linear Discriminant Analysis (LDA)

related methods, and their comparison

Why bother considering LDA?

Reason 1: coefficient estimates in logistic regression can be unstable

Consider this example:

$$Y = \begin{cases} 0 & \text{if} \quad X \leq 50 \\ 1 & \text{if} \quad X > 50 \end{cases}$$

Perfectly separable!

```
> x <- seq(100)
> y <- c(rep(0, 50), rep(1, 50))
> fit <- glm(y~x, family = "binomial")
Warning messages:
1: glm.fit: algorithm did not converge
2: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

LDA does not suffer from this problem!

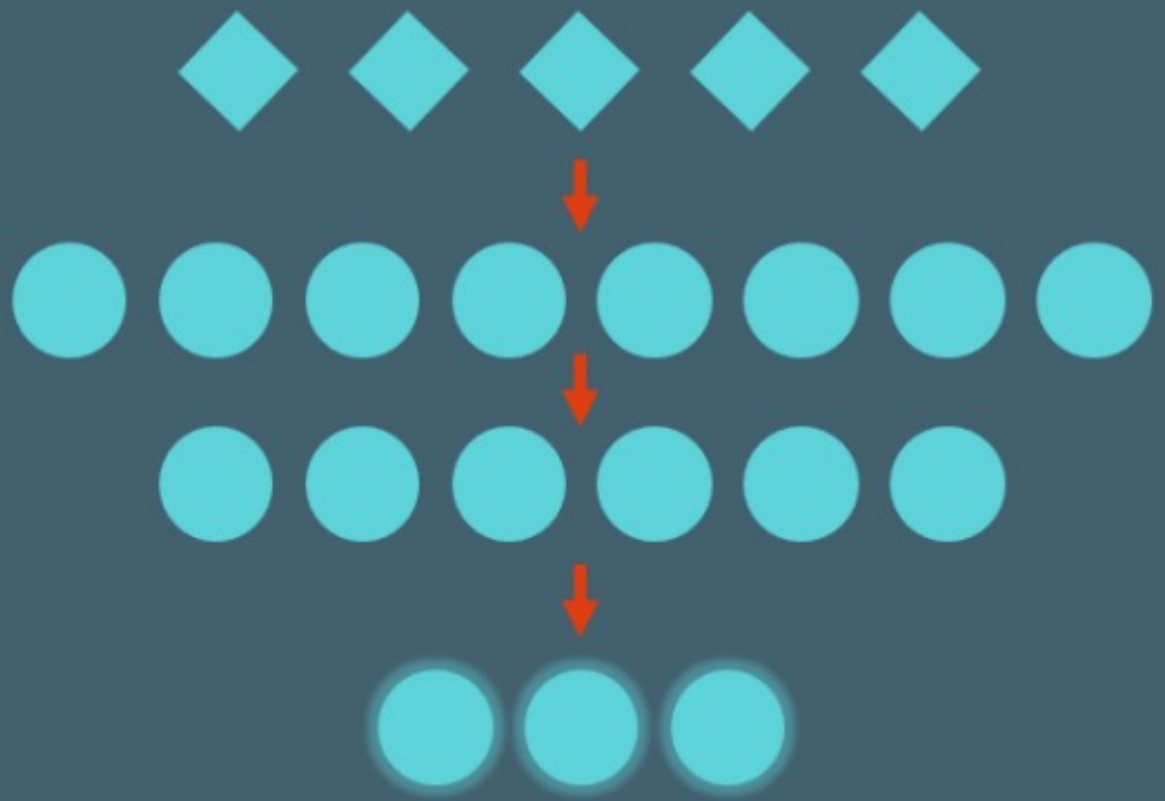
Why bother considering LDA?

Reason 1: coefficient estimates in logistic regression can be unstable

Reason 2: when n is small & distribution of X is close to normal

Reason 3: multi-class classification problem

feature extraction



feature selection



Motivating LDA

Back in Lecture 3, the Bayes classifier is the "best" we can do

$$\text{argmax}_{k=1,\ldots,K} \left\{ \Pr\left(Y = k \,|\, X_1, \ldots, X_p \right) \right\}$$

Logistic regression ($K = 2$ case)

$$\Pr(Y = 1 \,|\, X_1, \ldots, X_p) = \frac{e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}$$

LDA: uses Bayes theorem & makes assumptions on $X_1, \ldots, X_p \,|\, Y$

Bayes theorem

Prior probability $\pi_k = \Pr\left(Y = k\right)$

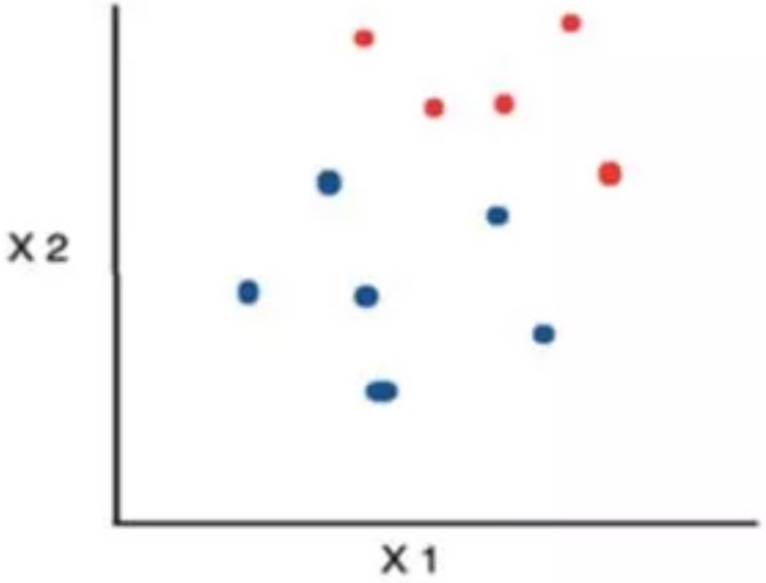
Density function **of** $(X_1, \ldots, X_p) \,|\, Y = k$

$$f_k(x_1, \ldots, x_p) = \Pr\left(X_1 = x_1, \ldots, X_p = x_p \,|\, Y = k\right)$$

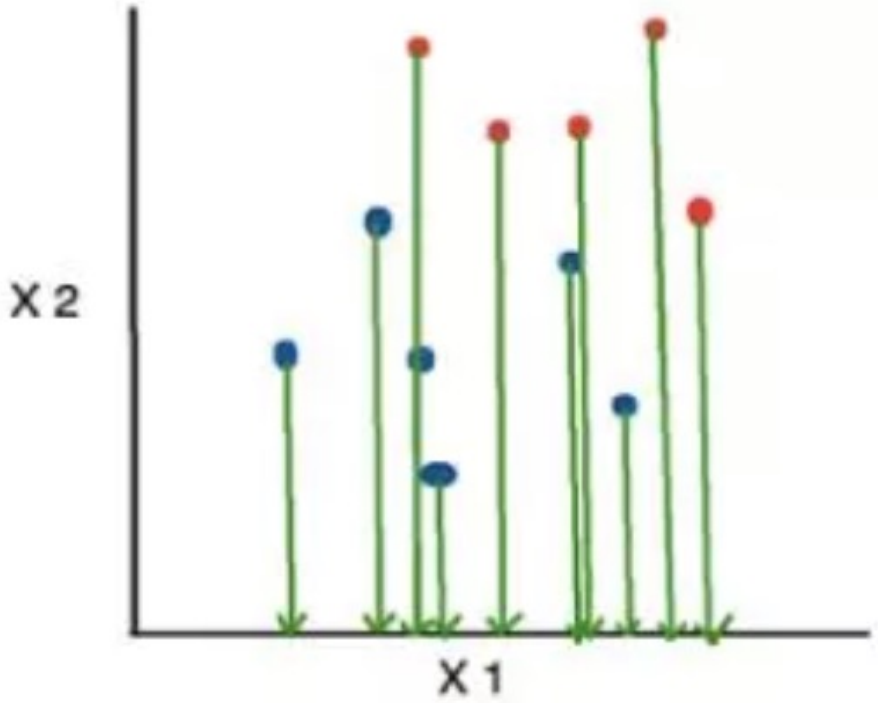
Bayes theorem:

$$\Pr\left(Y = k \,|\, X_1 = x_1, \ldots, X_p = x_p\right) = \frac{\Pr\left(Y = k, X_1 = x_1, \ldots, X_p = x_p\right)}{\Pr\left(X_1 = x_1, \ldots, X_p = x_p\right)}$$

$$= \frac{\Pr\left(Y = k, X_1 = x_1, \ldots, X_p = x_p\right)}{\sum_{j=1}^{K} \Pr\left(Y = j, X_1 = x_1, \ldots, X_p = x_p\right)} = \frac{\pi_k f_k(x_1, \ldots, x_p)}{\sum_{j=1}^{K} \pi_j f_j(x_1, \ldots, x_p)}$$



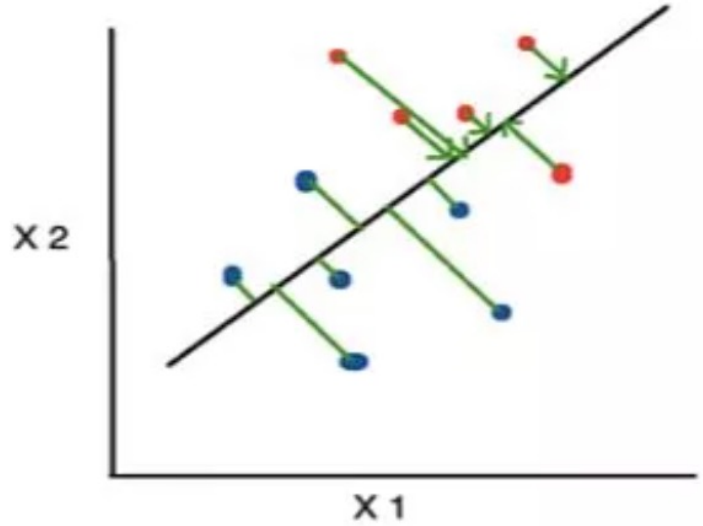
2D



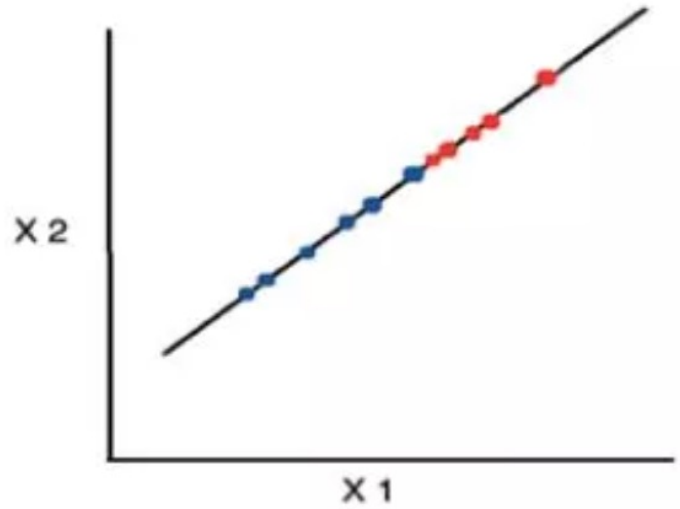
2D

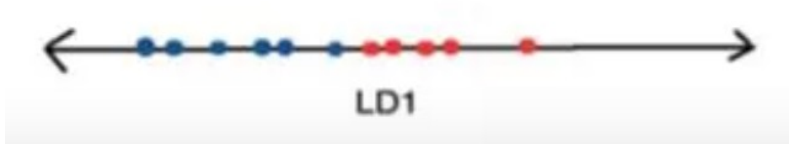


1D



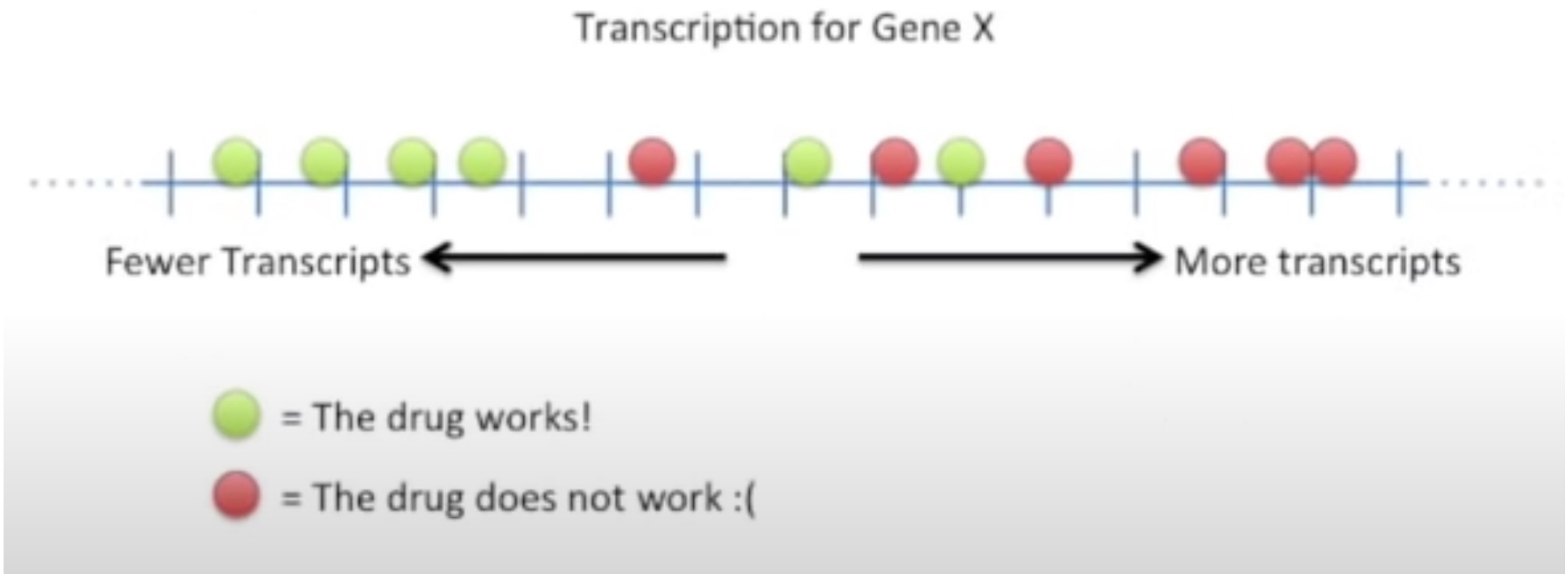
2D





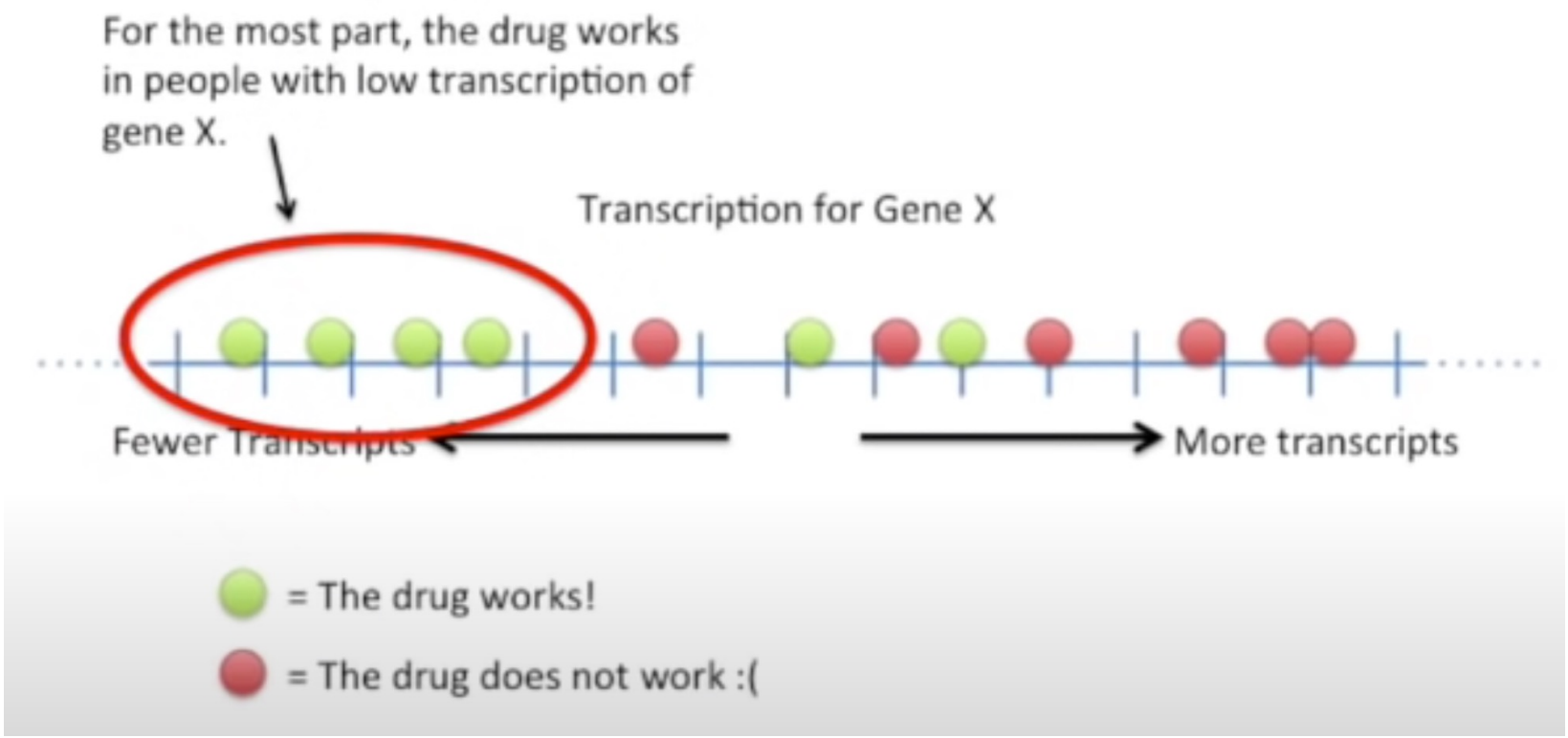
1D

LDA: A motivating example

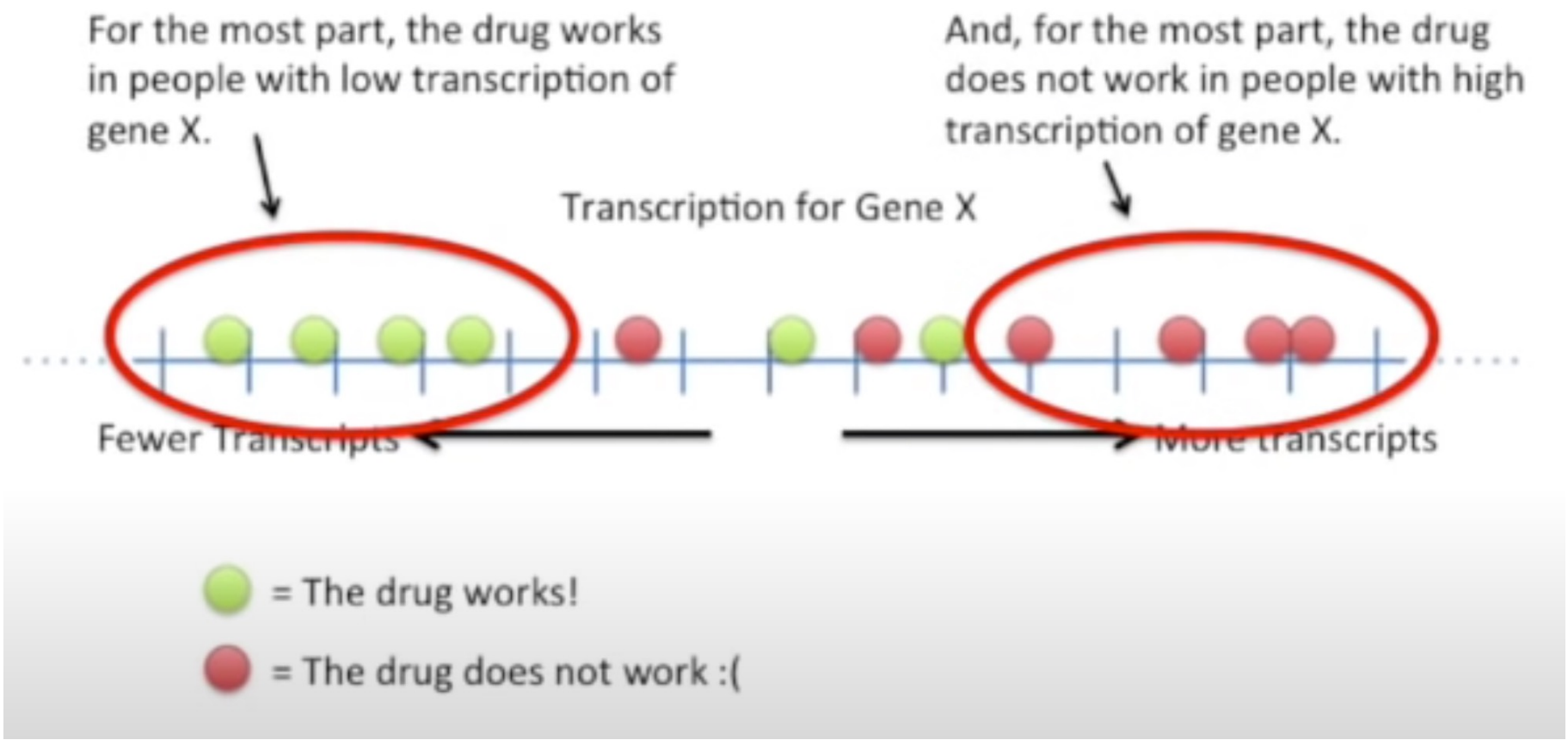


https://youtu.be/azXCzI57Yfc

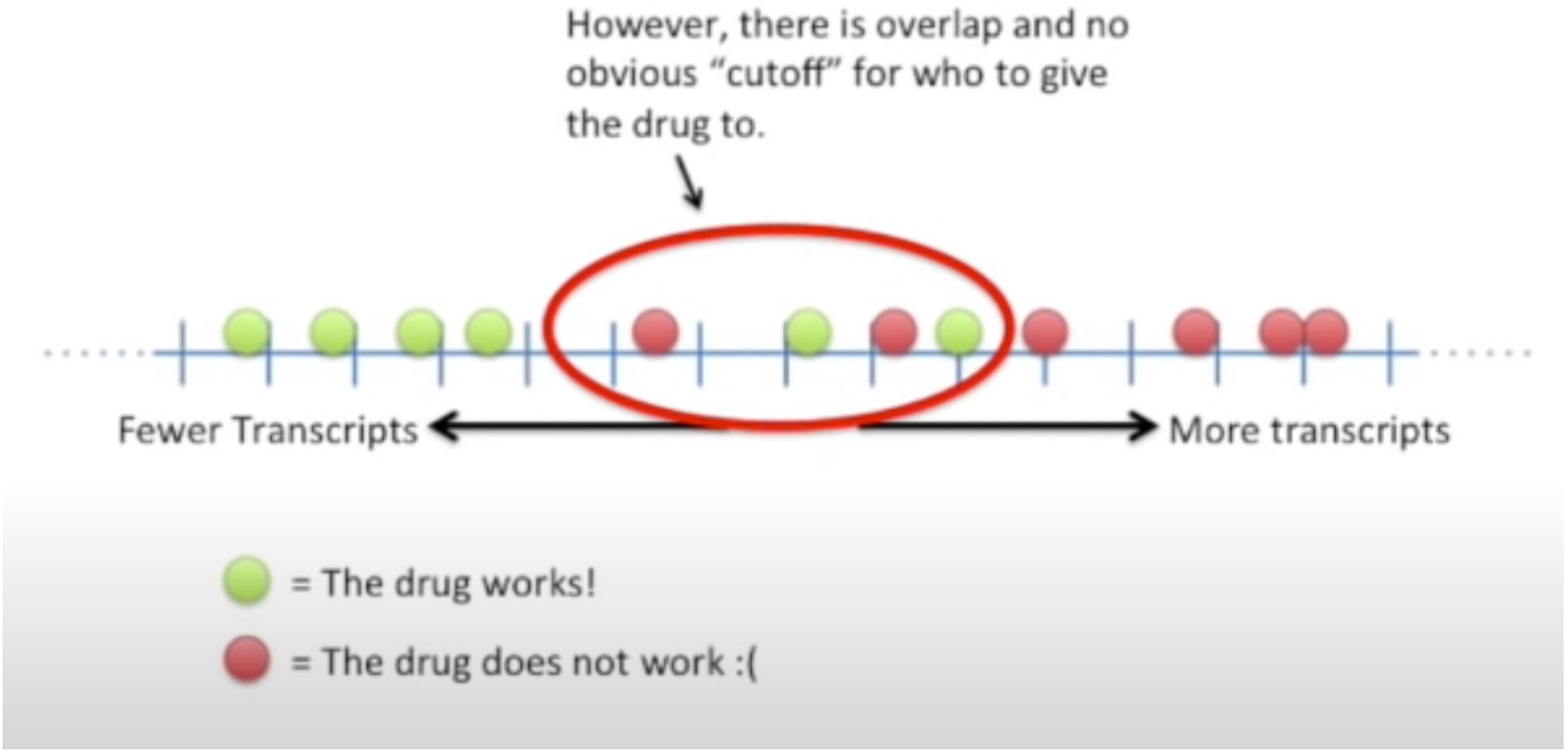
LDA: A motivating example



LDA: A motivating example

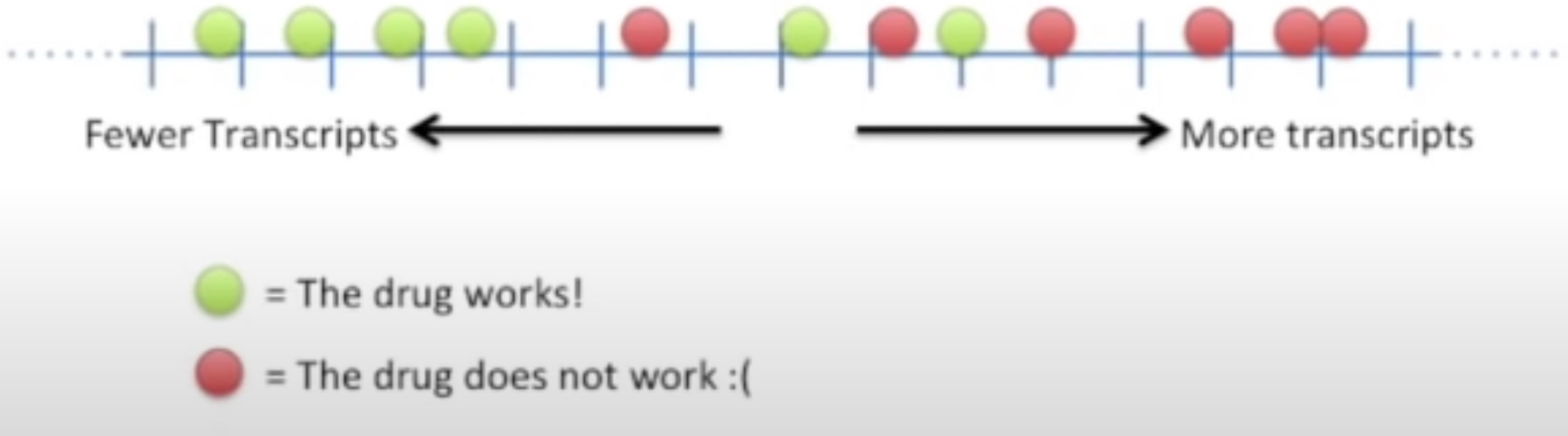


LDA: A motivating example



LDA: A motivating example

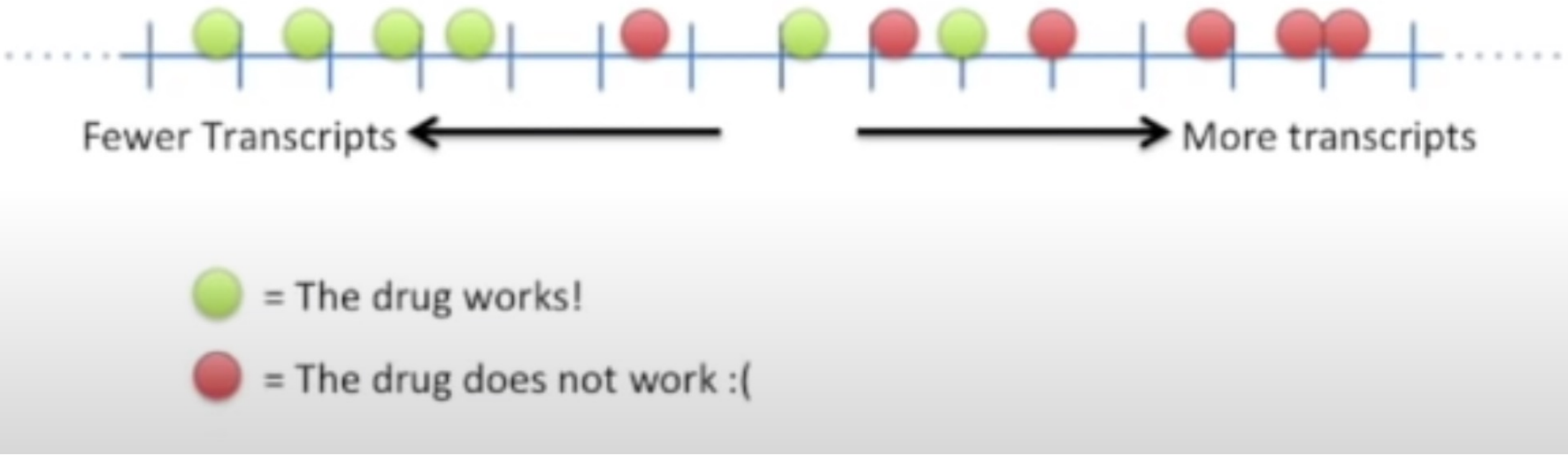
In summary, Gene X does an OK job at telling us who should take the drug (and who shouldn't).



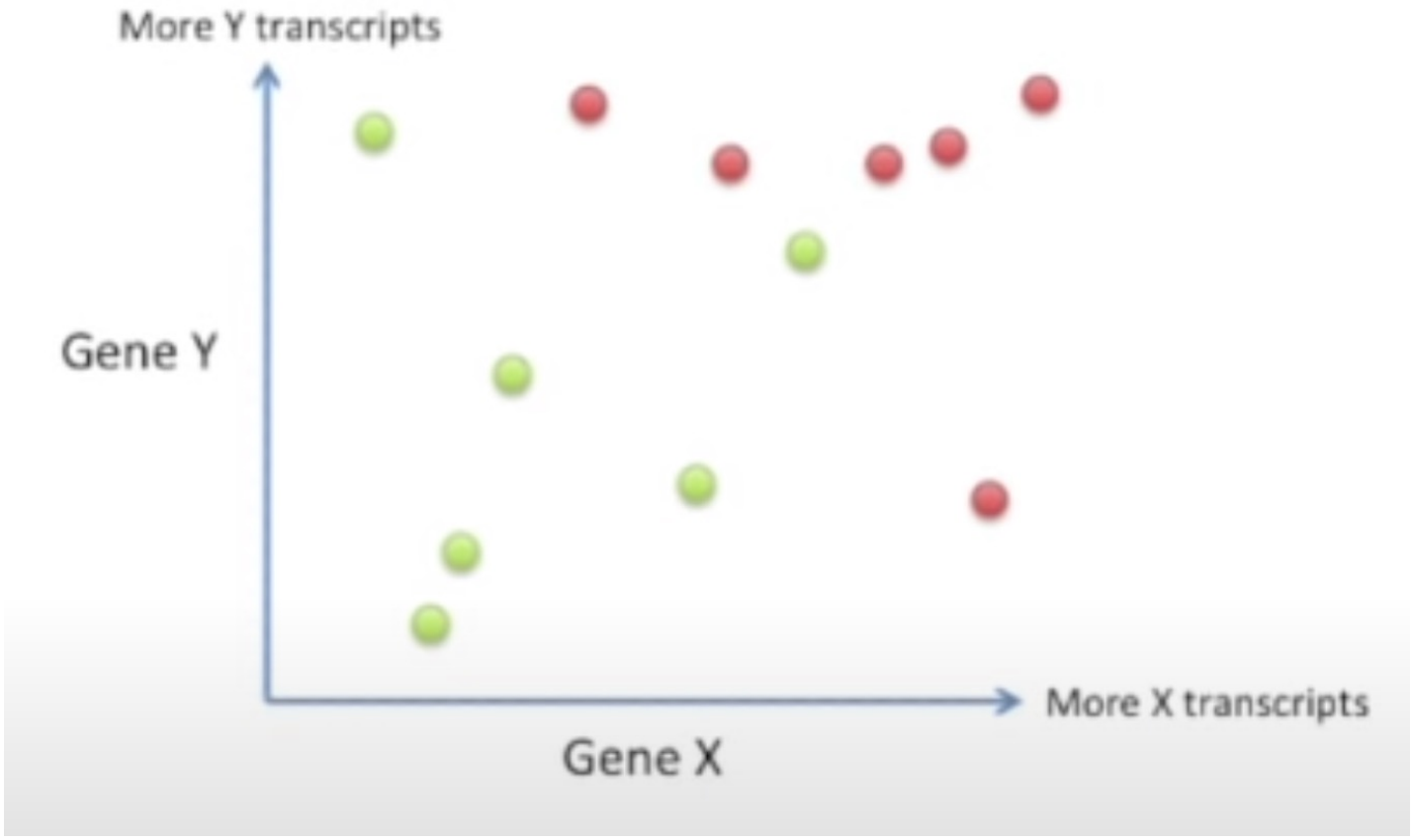
LDA: A motivating example

In summary, Gene X does an OK job at telling us who should take the drug (and who shouldn't).

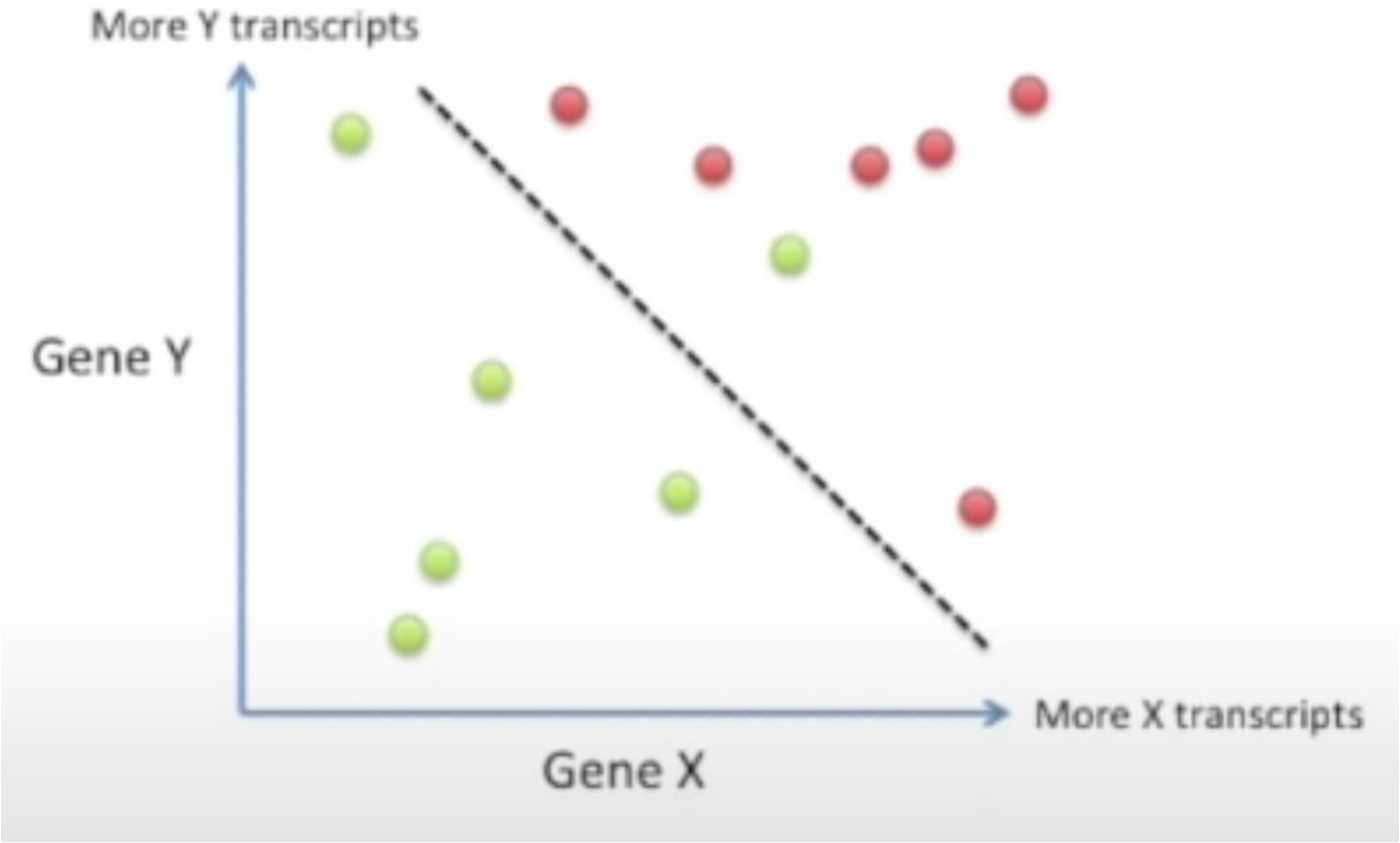
Can we do better? What if we used more than one gene?



LDA: A motivating example



LDA: A motivating example



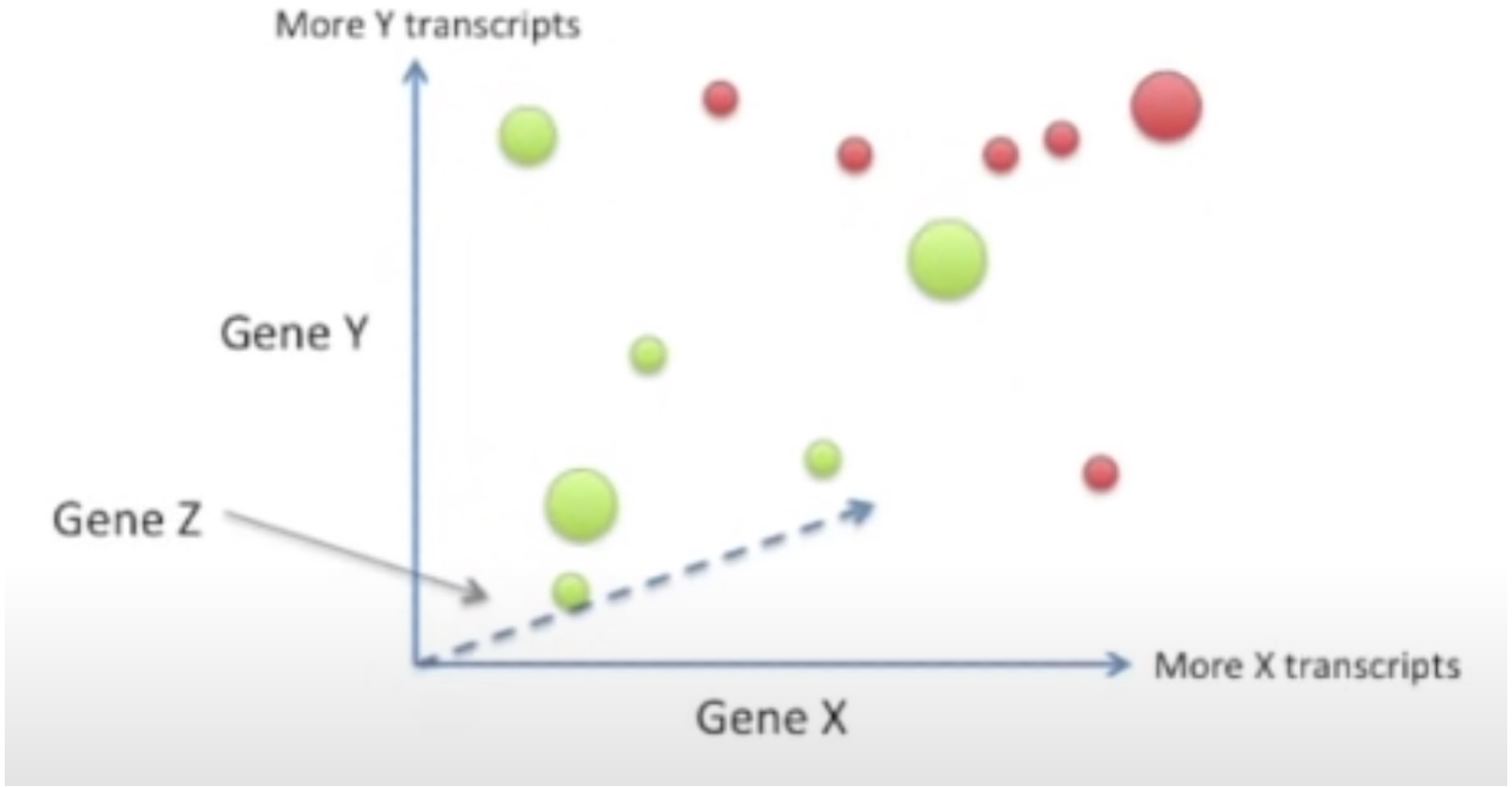
LDA: A motivating example



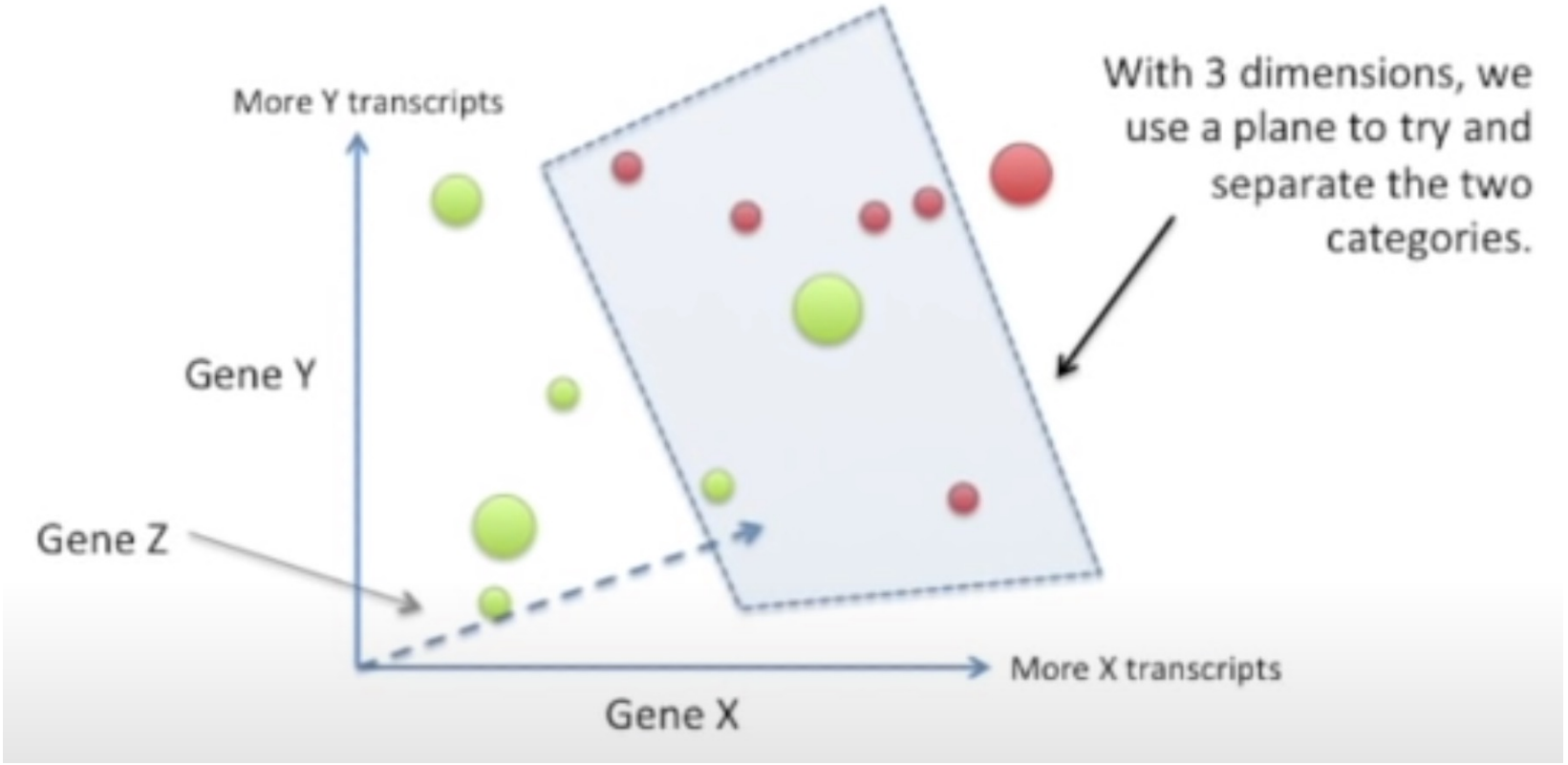
LDA: A motivating example



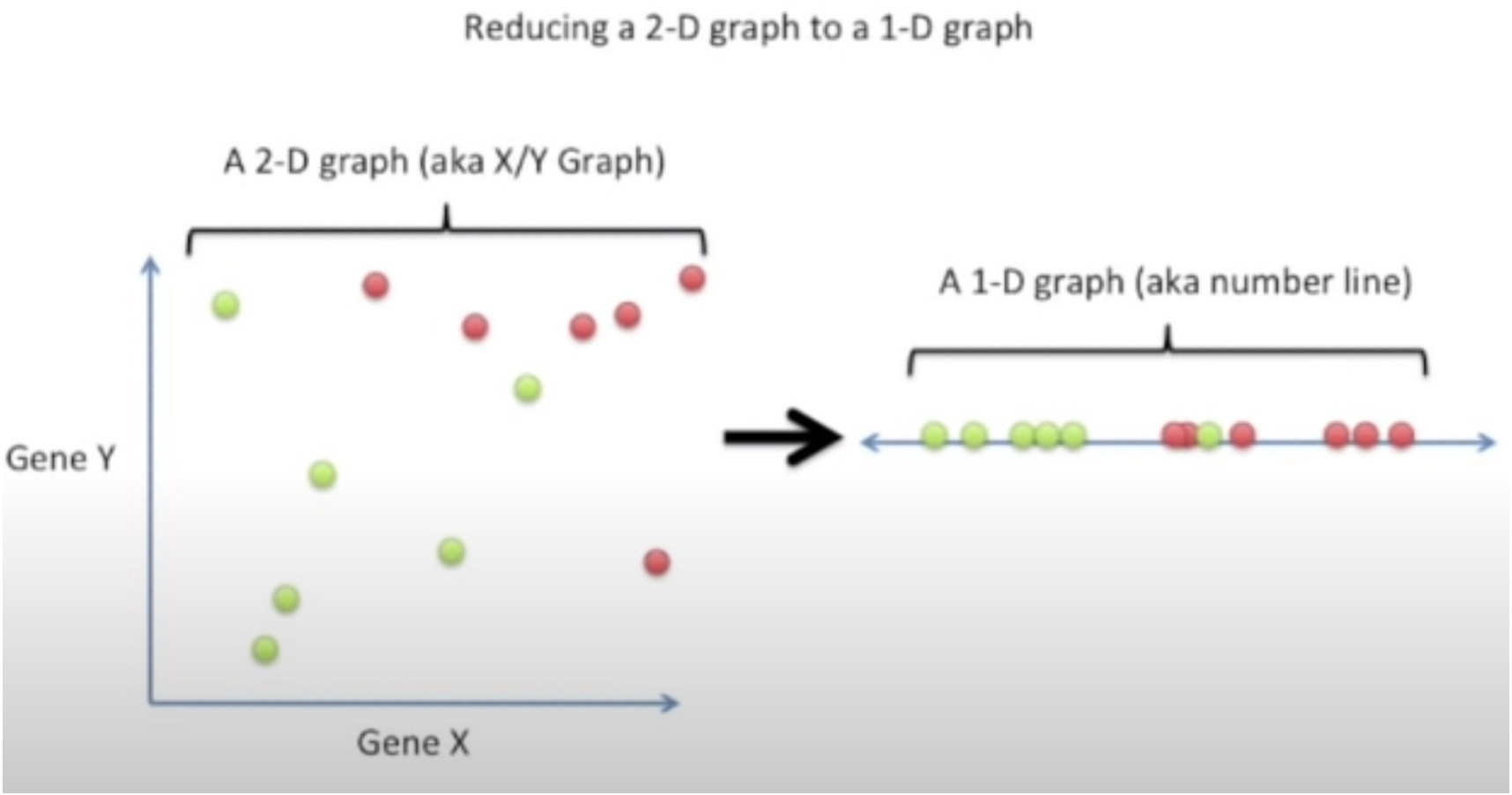
LDA: A motivating example



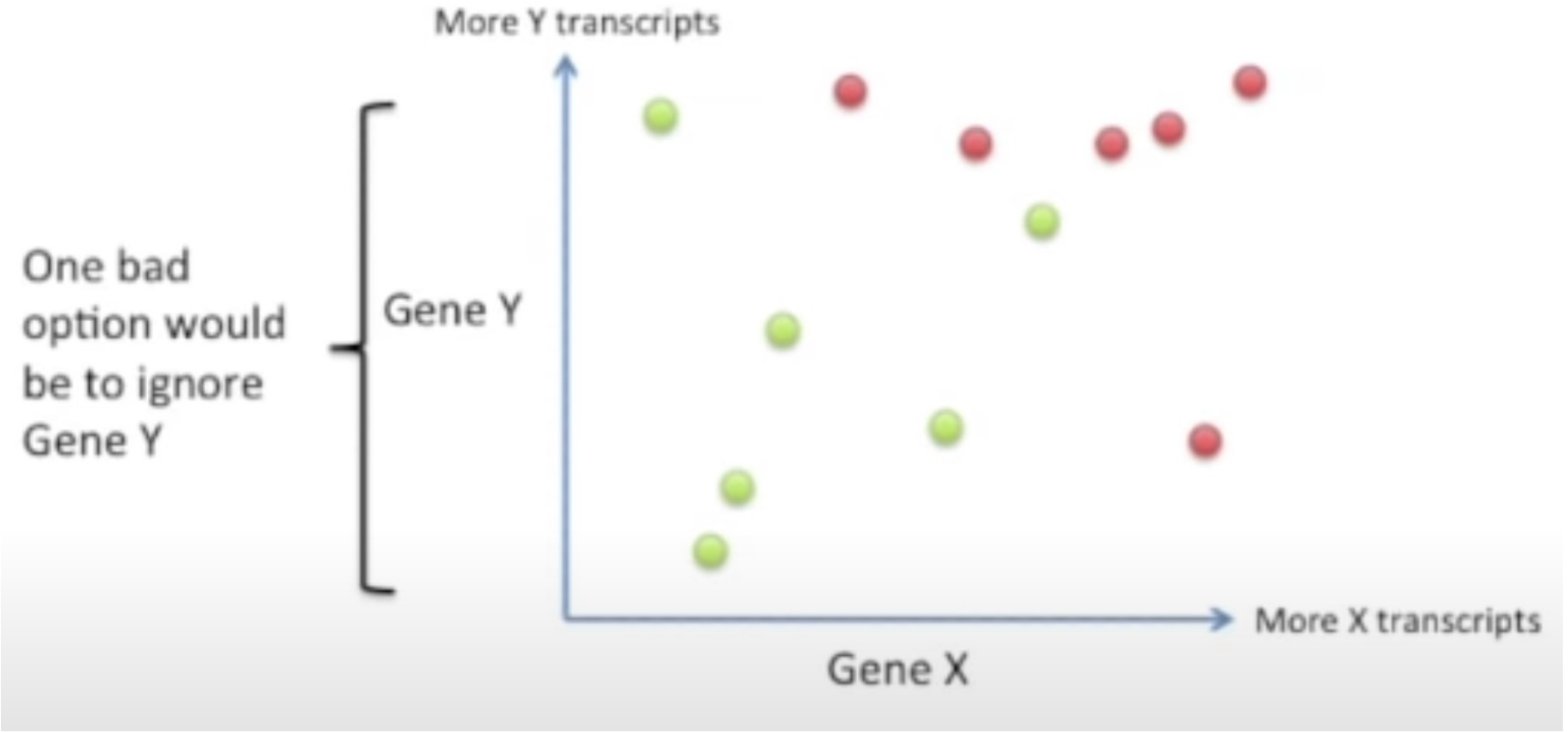
LDA: A motivating example



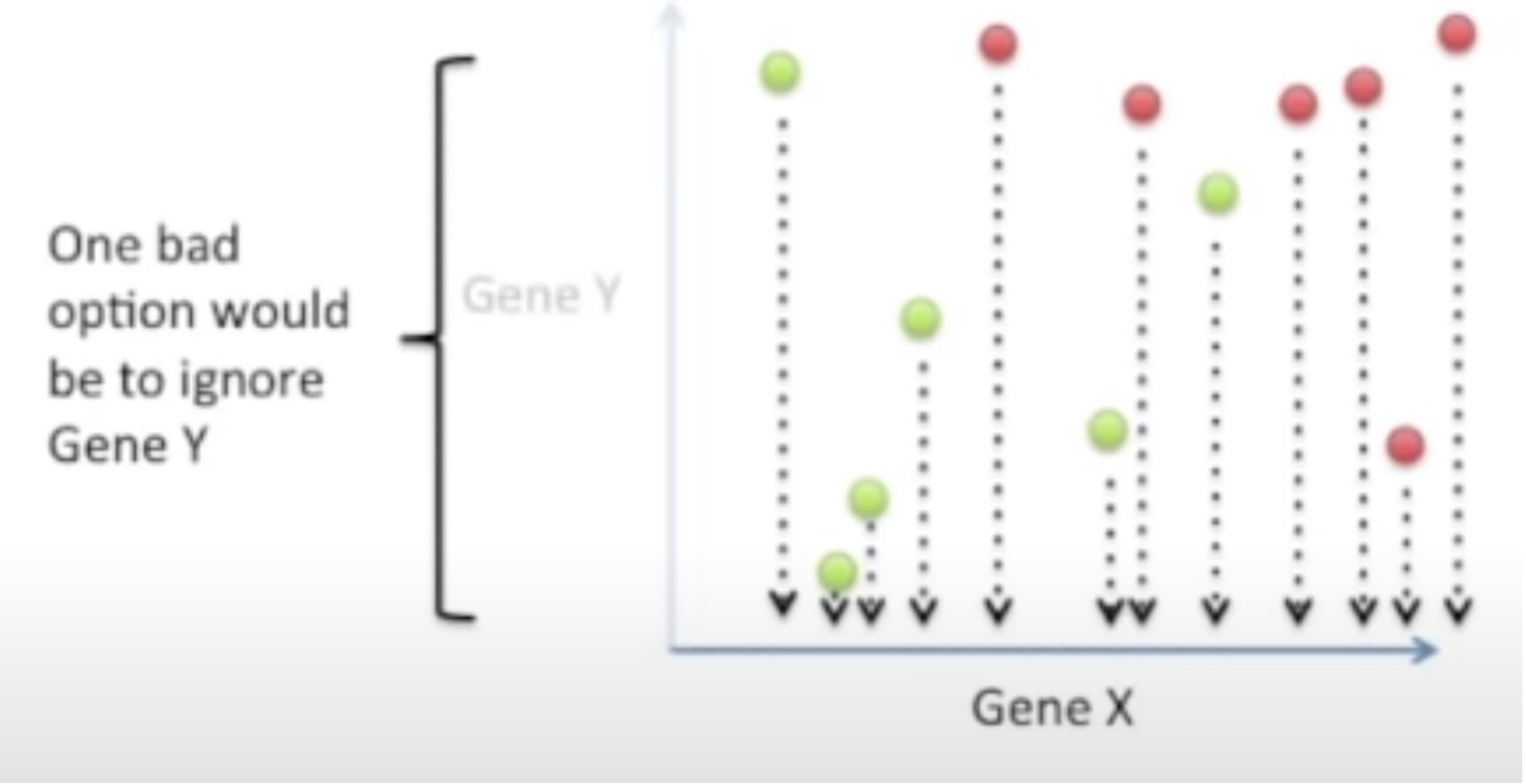
LDA: A motivating example



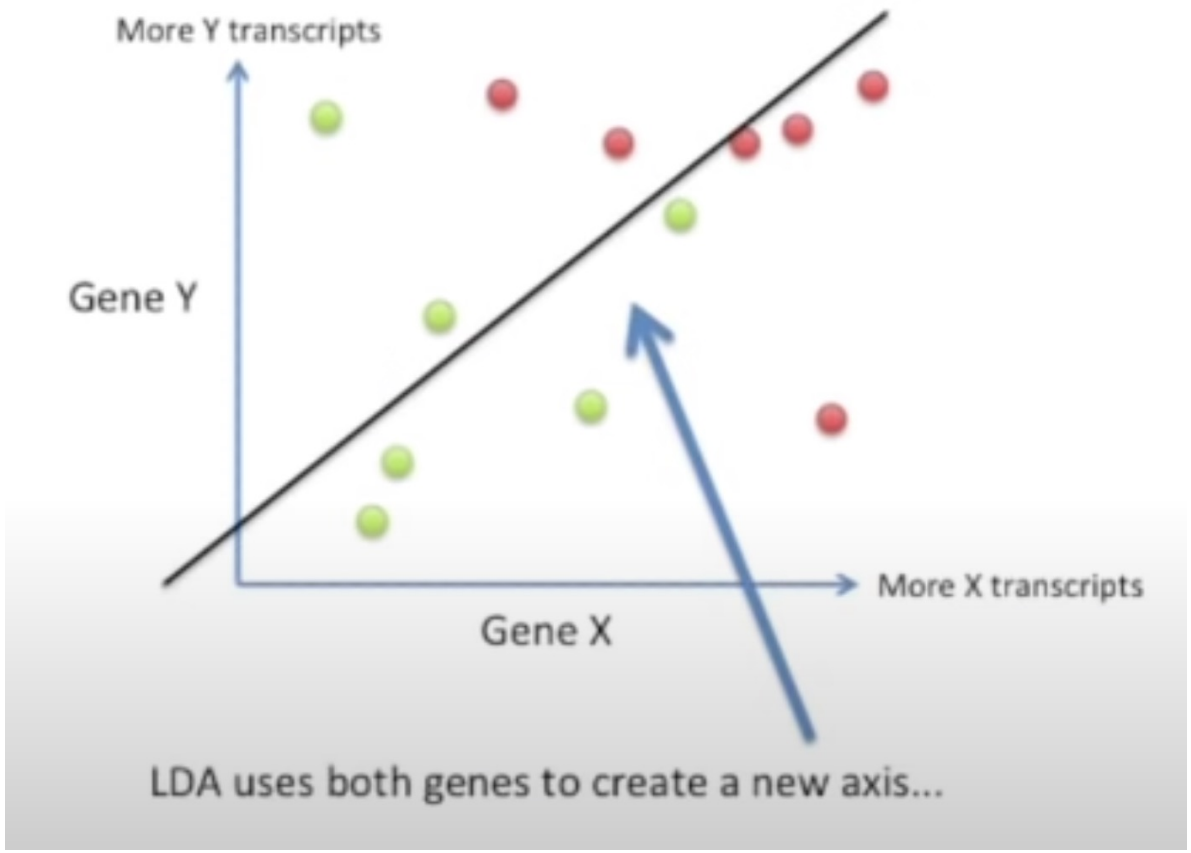
LDA: A motivating example

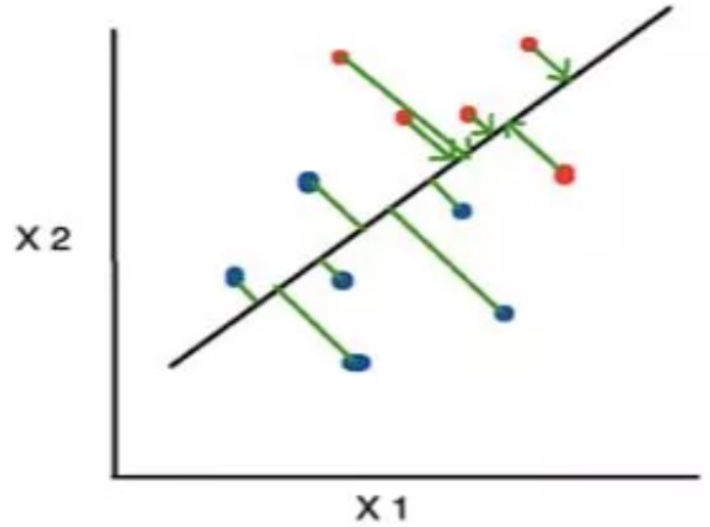


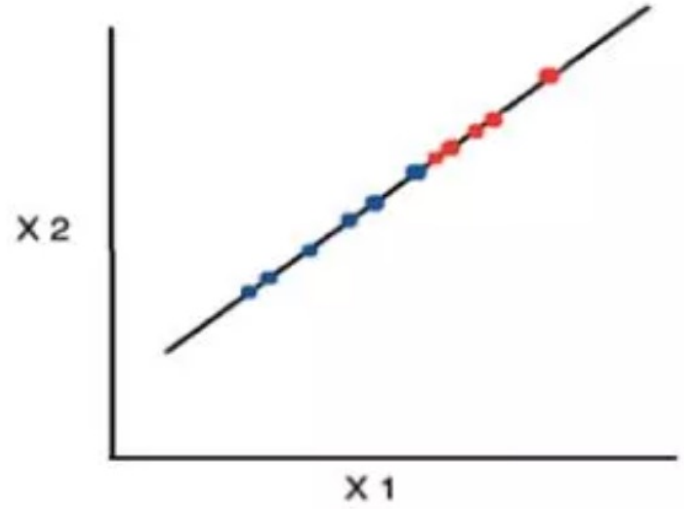
LDA: A motivating example

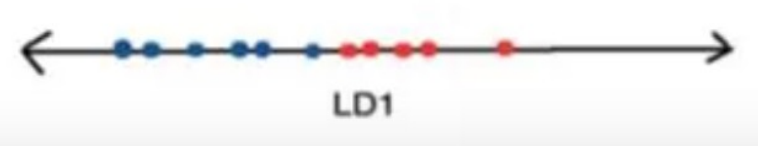


LDA: A motivating example









Discriminant Analysis: key idea

$\pi_k = \Pr(Y = k)$ density function

$$\Pr(Y = k \mid X_1 = x_1, \ldots, X_p = x_p) = \frac{\pi_k f_k(x_1, \ldots, x_p)}{\sum_{j=1}^{K} \pi_j f_j(x_1, \ldots, x_p)}$$

what we really care about

$$f_k(x_1, \ldots, x_p) = \Pr(X_1 = x_1, \ldots, X_p = x_p \mid Y = k)$$

Discriminant Analysis in one sentence:

assume $f_k(x_1, \ldots, x_p)$ is the normal distribution density

$\leftrightarrow (X_1, \ldots, X_p) \mid Y$ is normally distributed

LDA: $p = 1$ case

We only have one predictor X, which we assume $X \mid Y = k \sim N(\mu_k, \sigma^2)$

different mean, same variance

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)$$

$$\Pr\left(Y = k \mid X = x\right) = \frac{\pi_k f_k(x)}{\sum_{j=1}^{K} \pi_j f_j(x)}$$

$$= \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{j=1}^{K} \pi_j \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_j)^2\right)}$$

same for all k

LDA: $p = 1$ case

We only have one predictor X, which we assume $X \,|\, Y = k \sim N(\mu_k, \sigma^2)$

different mean, same variance

$$\Pr\left(Y = k \,|\, X = x\right) \propto \pi_k \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)$$

The Bayes classifier will assign an observation $X = x$ to the class for which $\Pr\left(Y = k \,|\, X = x\right)$ is largest…

equivalent to assigning the observation to the class for which

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{\sigma^2} + \log(\pi_k)$$ **is largest**

LDA: $p = 1$ case

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{\sigma^2} + \log(\pi_k)$$

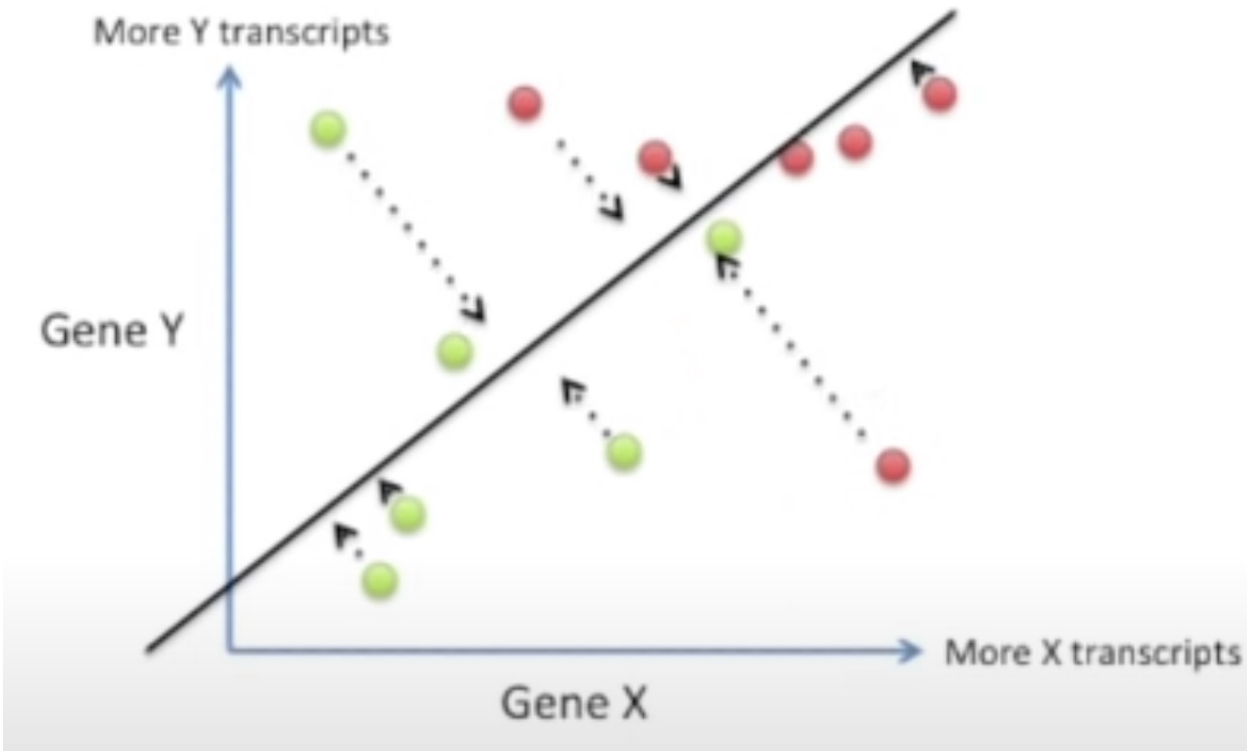
For the observation $X = x$, obtain

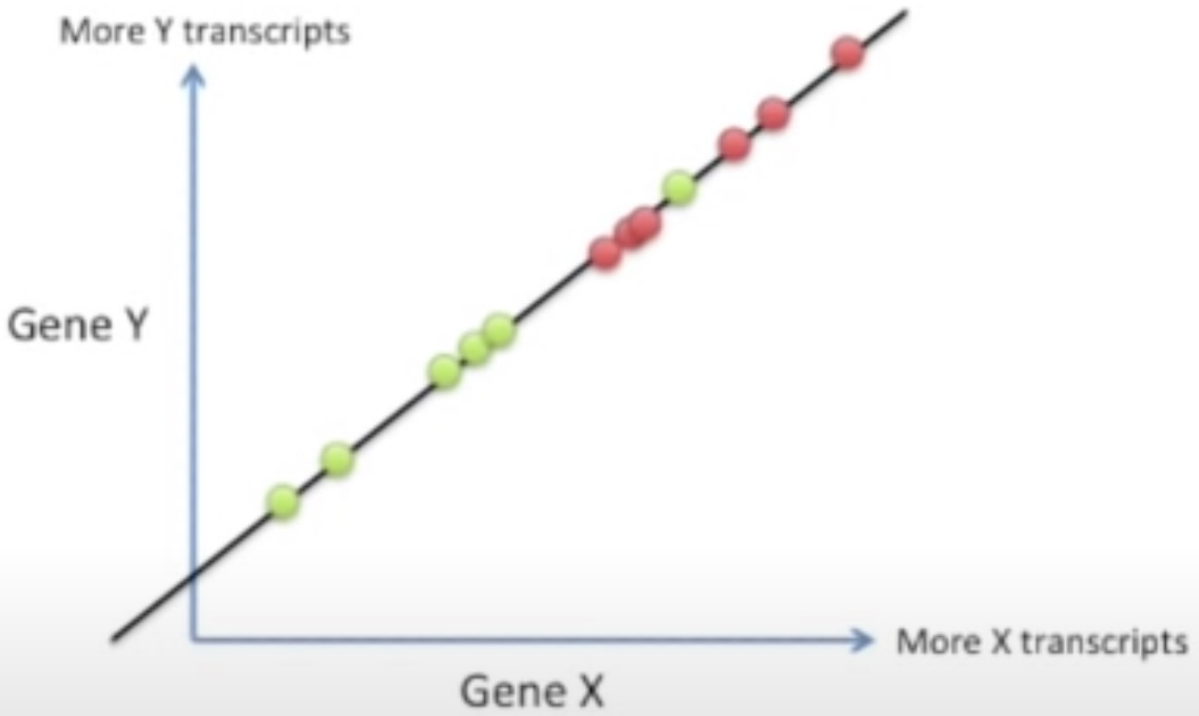
$$\delta_1(x), \delta_2(x), \ldots, \delta_K(x)$$

The Bayes classifier assigns the observation to class j if $\delta_j(x)$ is the largest

The value of $\mu_1, \mu_2, \ldots, \mu_K, \pi_1, \ldots, \pi_K, \sigma^2$ are unknown!

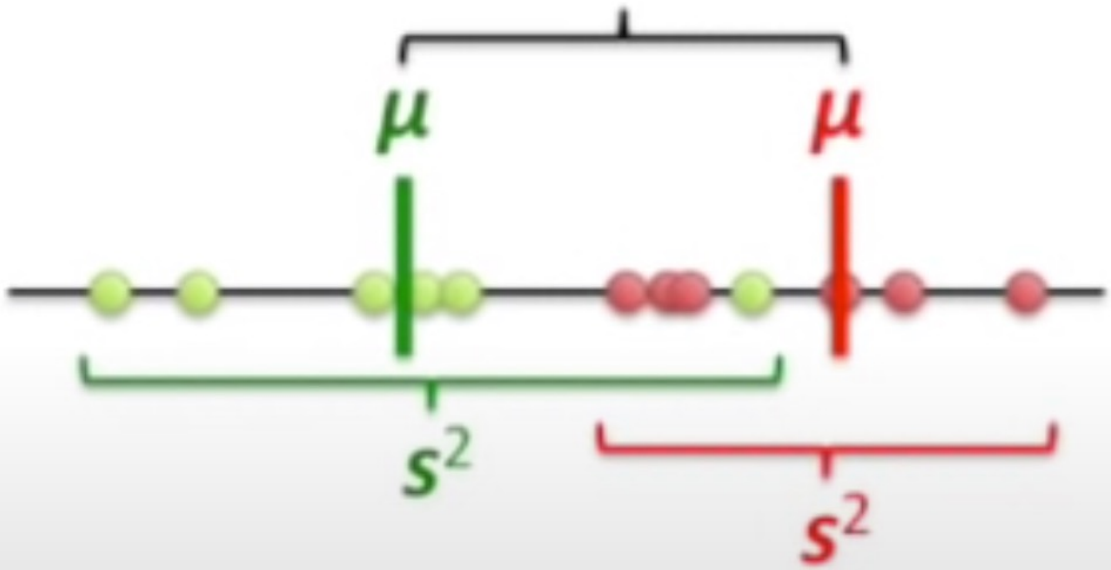
estimate them!


More Y transcripts
Gene Y
More X transcripts
Gene X


More Y transcripts
Gene Y
More X transcripts
Gene X

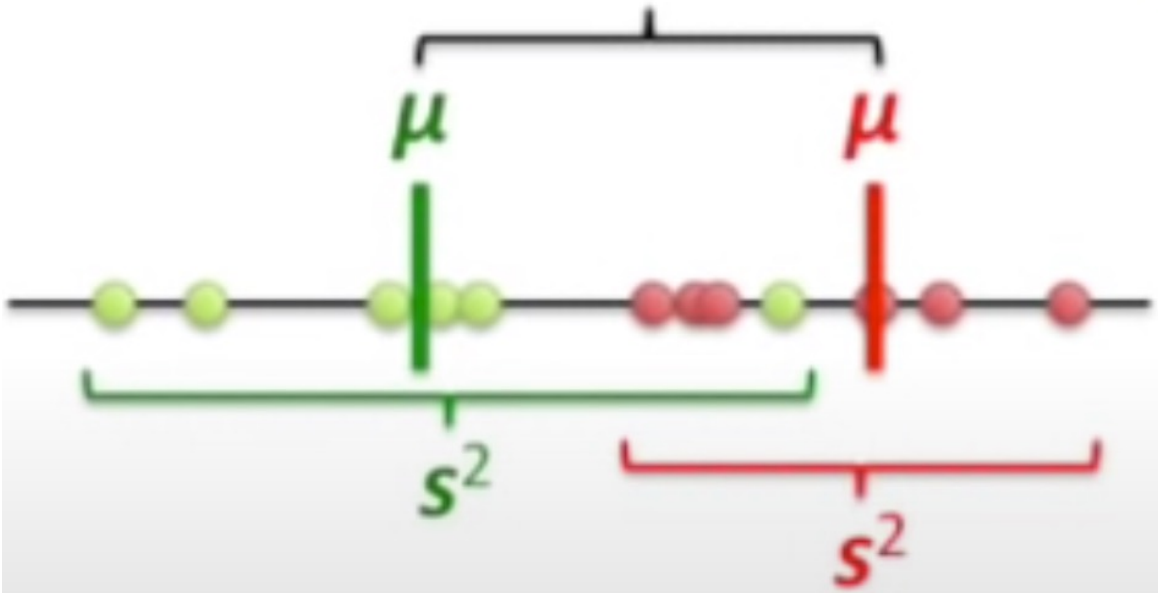
$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{\sigma^2} + \log(\pi_k)$$

1) Maximize the distance between means.



2) Minimize the variation (which LDA calls "scatter" and is represented by s^2) within each category.

1) Maximize the distance between means.

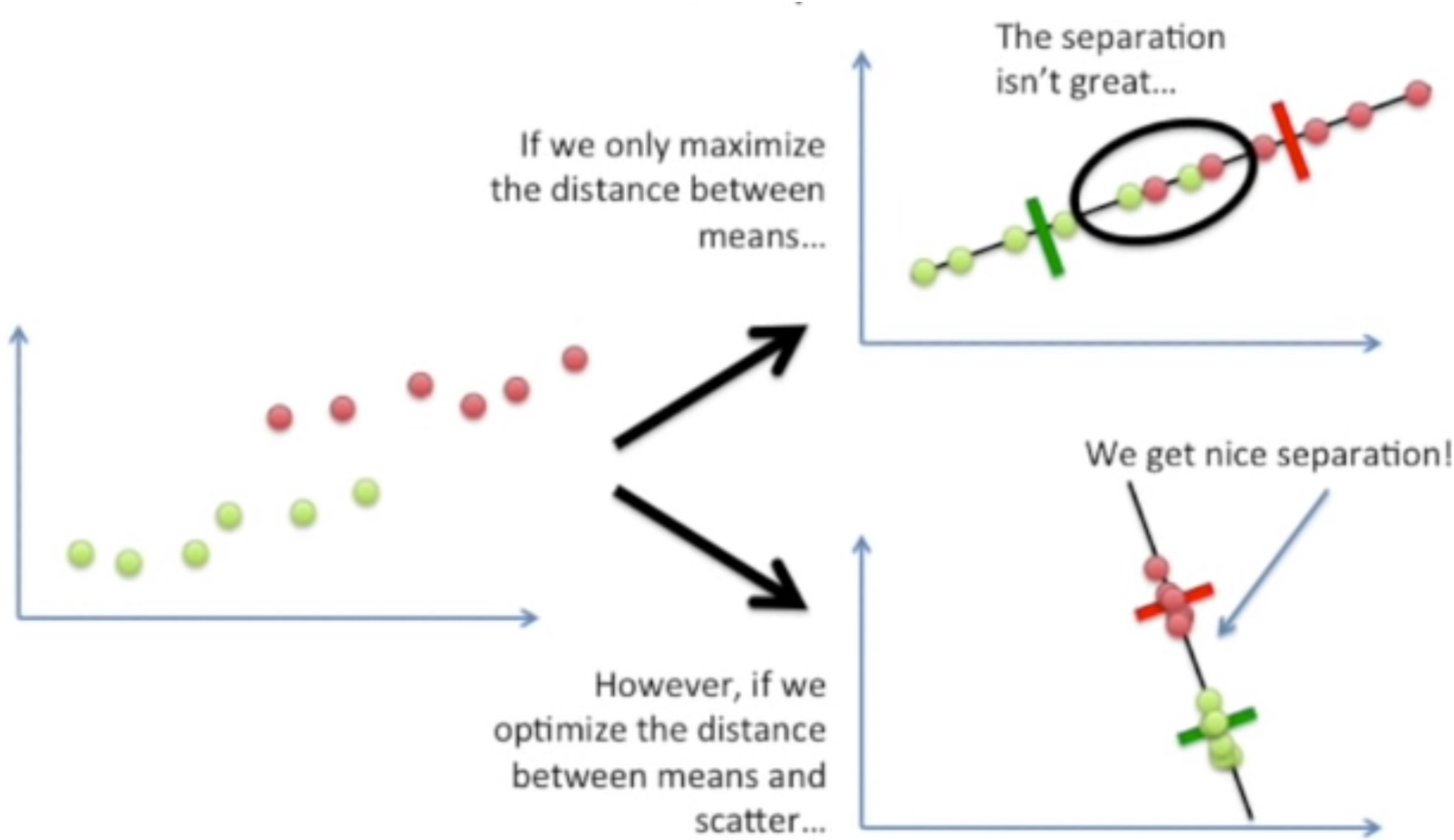


Let's call $(\mu - \mu)$ ***d*** for ***d***istance.

$$\frac{d^2}{s^2 + s^2} \rightarrow \frac{\text{Ideally large}}{\text{Ideally small}}$$

$$\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{\sigma^2}$$

2) Minimize the variation (which LDA calls "scatter" and is represented by s^2) within each category.


If we only maximize the distance between means...
The separation isn't great...
However, if we optimize the distance between means and scatter...
We get nice separation!

LDA classifier: $p = 1$ case

$$\hat{\pi}_k = \frac{n_k}{n}$$

n_k: number of training observations in the kth class

n: total number of training observations

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

The LDA classifier assigns the observation $X = x$ to class for which

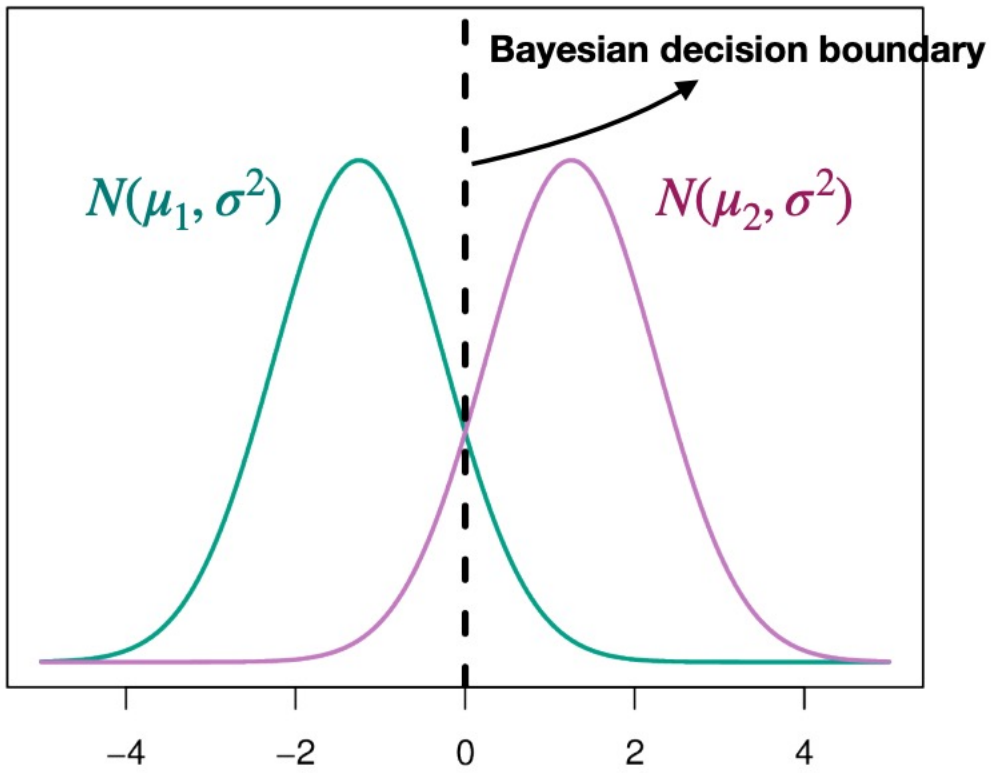
$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{\hat{\sigma}^2} + \log(\hat{\pi}_k)$$

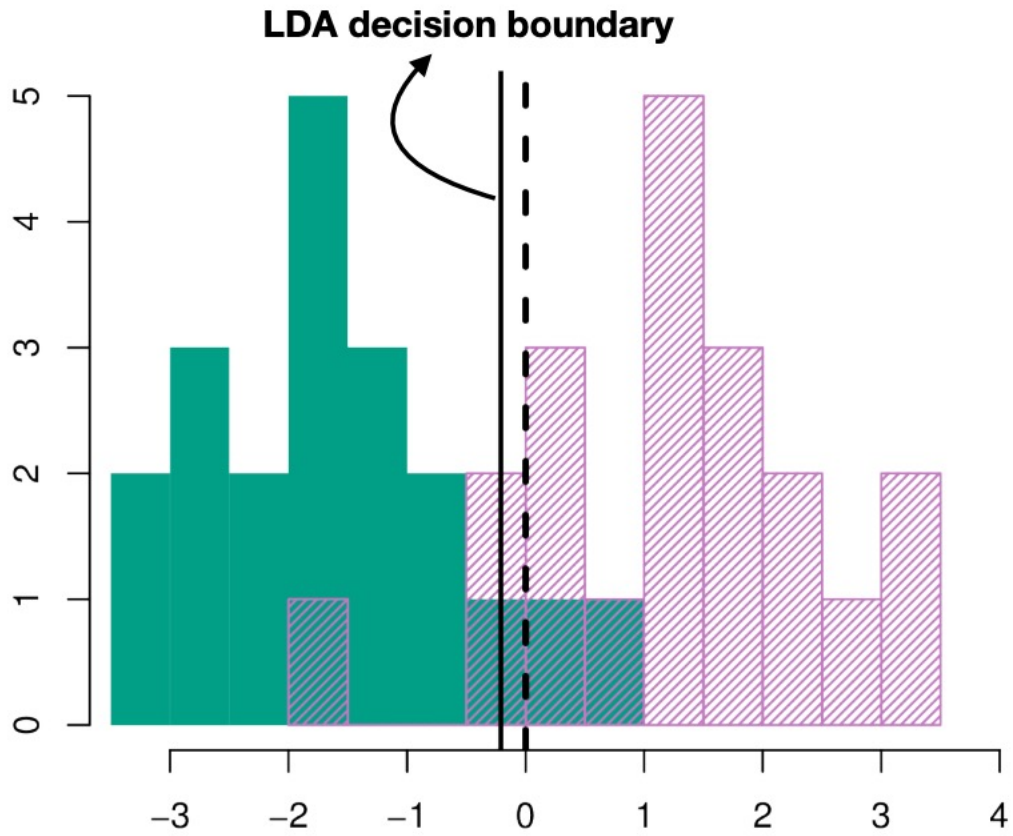
is largest

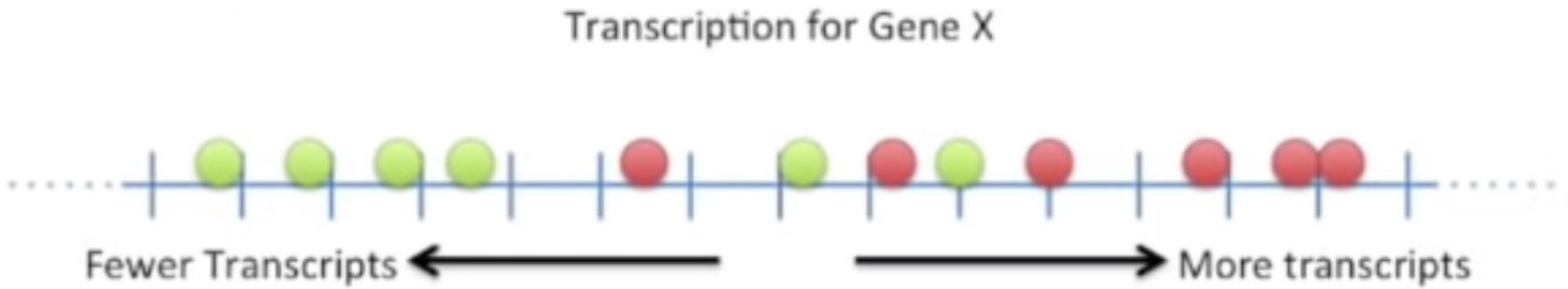
Discriminant function

linear in x

Decision boundary: $p = 1$ case







LDA: $p > 1$ case

We have multiple predictors $X_1, \ldots, X_p$, which we assume

$$(X_1, \ldots, X_p) \,|\, Y = k \sim N(\mu_k, \Sigma)$$

different mean, same covariance

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_k)^T \Sigma^{-1} (\mathbf{x} - \mu_k)\right)$$

with a lot of algebra…

The Bayes classifier assigns the observation $\mathbf{x}$ to class for which

$$\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

is largest

LDA classifier: $p > 1$ case

The Bayes classifier assigns the observation $\mathbf{x}$ to class for which

$$\delta_k(\mathbf{x}) = \mathbf{x}^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log \pi_k$$

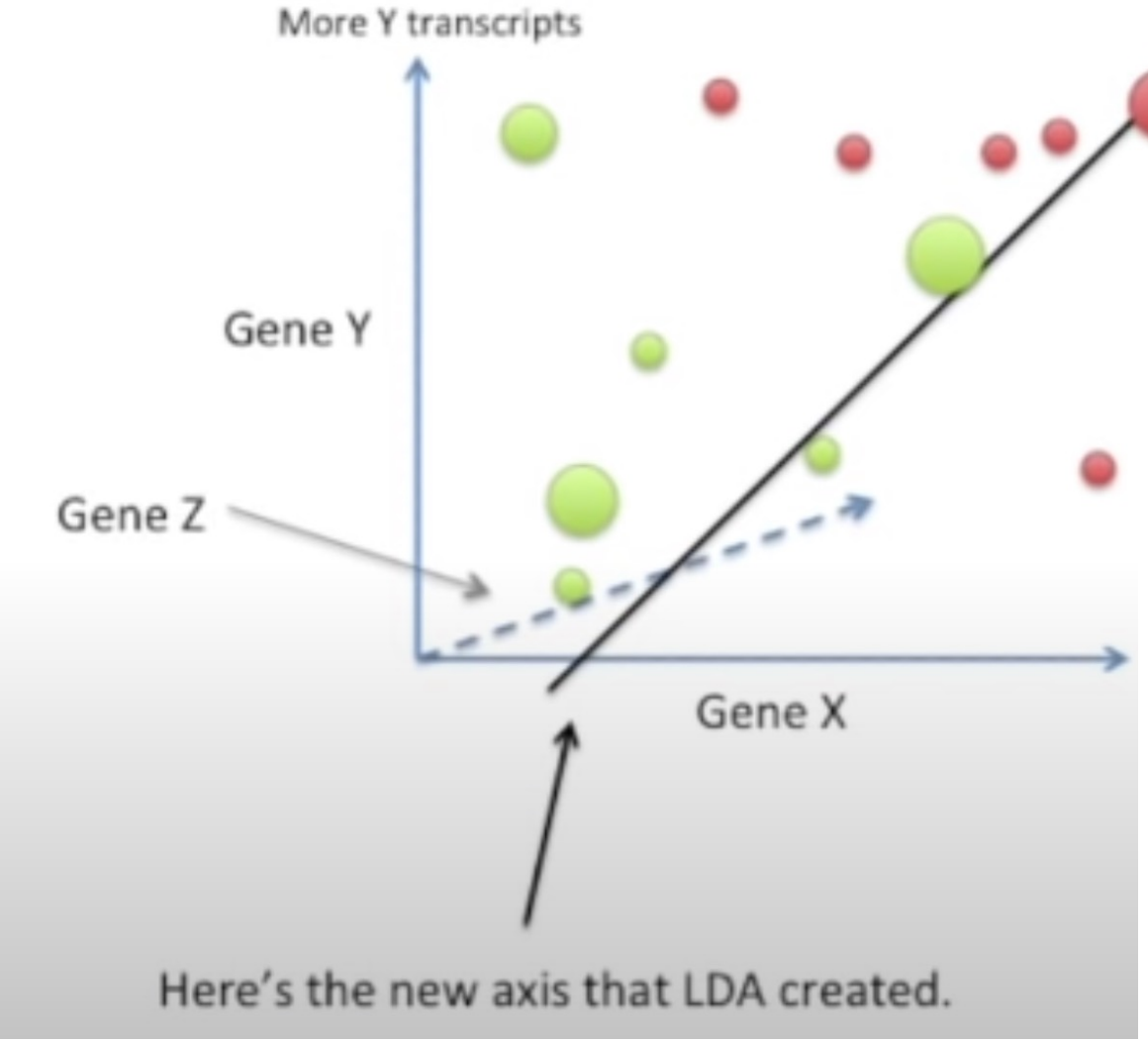
is largest

The LDA classifier assigns the observation $\mathbf{x}$ to class for which

$$\hat{\delta}_k(\mathbf{x}) = \boxed{\mathbf{x}^T\hat{\Sigma}^{-1}\hat{\mu}_k} - \frac{1}{2}\hat{\mu}_k^T\hat{\Sigma}^{-1}\hat{\mu}_k + \log \hat{\pi}_k$$

is largest

linear in x



Here's the new axis that LDA created.

The data are projected onto the new axis.



Decision boundary: $p > 1$ case

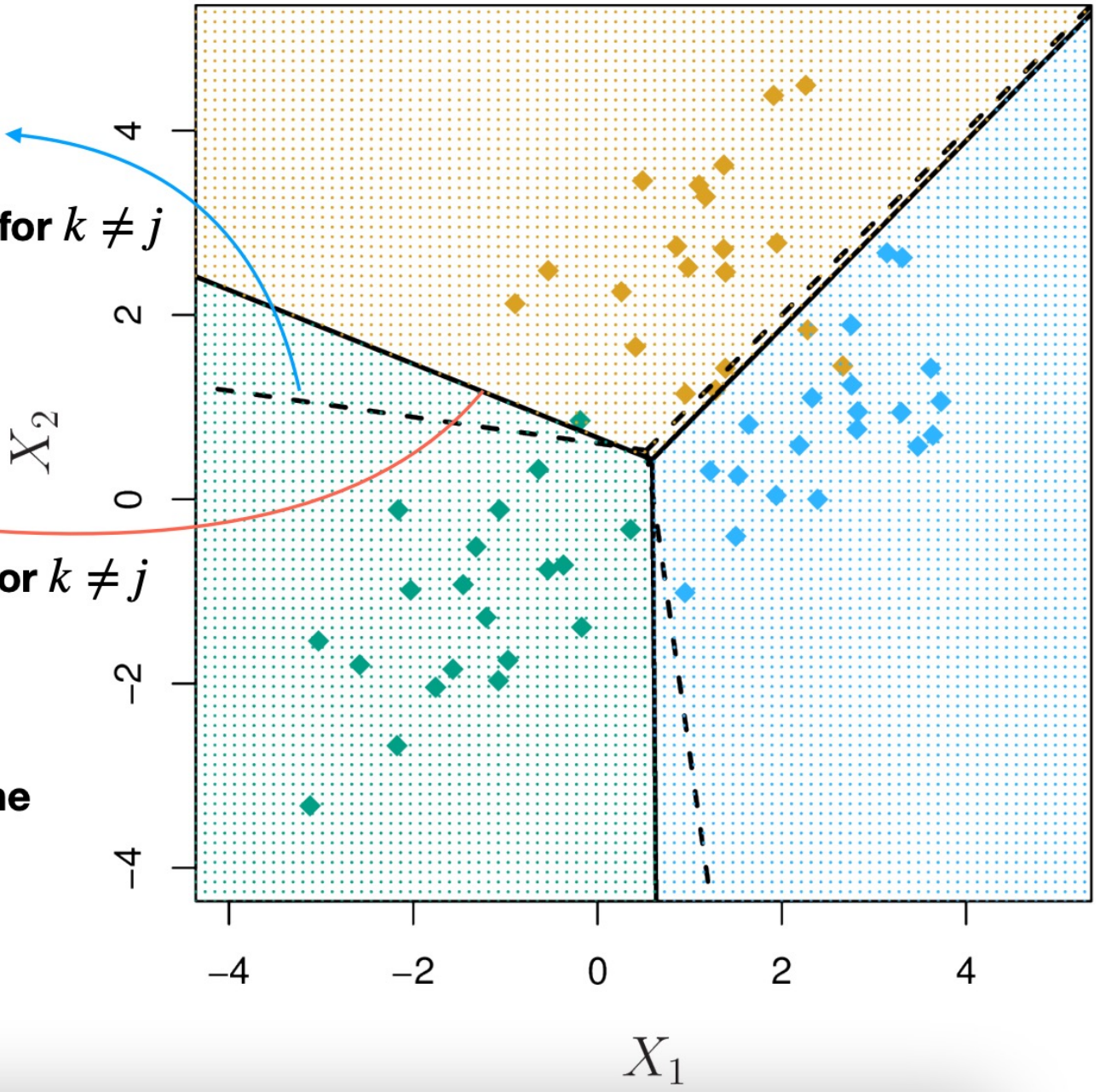
Bayesian decision boundary

set of x such that $\delta_k(\mathbf{x}) = \delta_j(\mathbf{x})$ **for** $k \neq j$

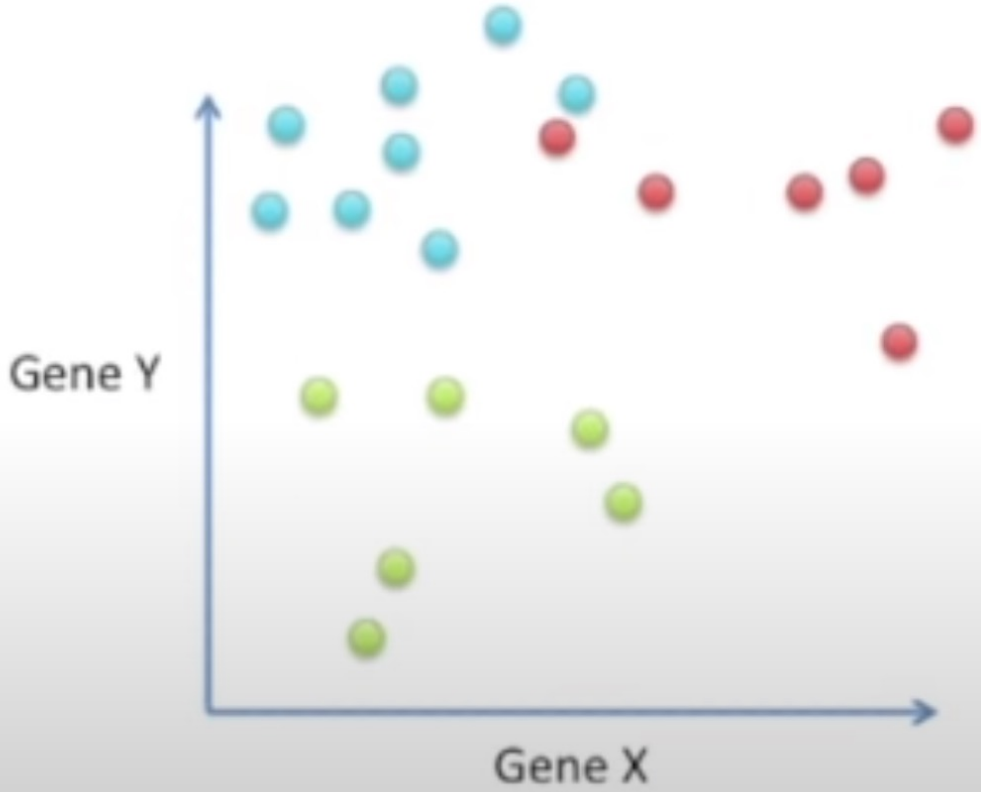
LDA decision boundary

set of x such that $\hat{\delta}_k(\mathbf{x}) = \hat{\delta}_j(\mathbf{x})$ **for** $k \neq j$

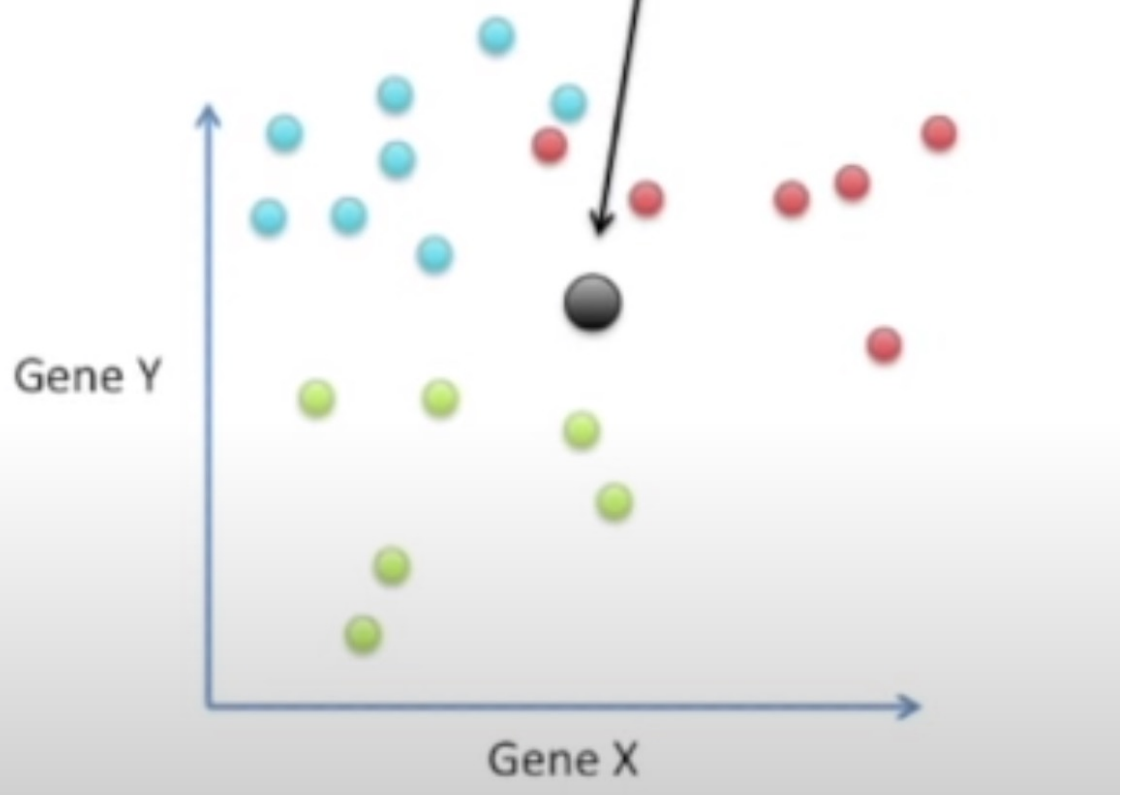
linear boundary, i.e., hyperplane



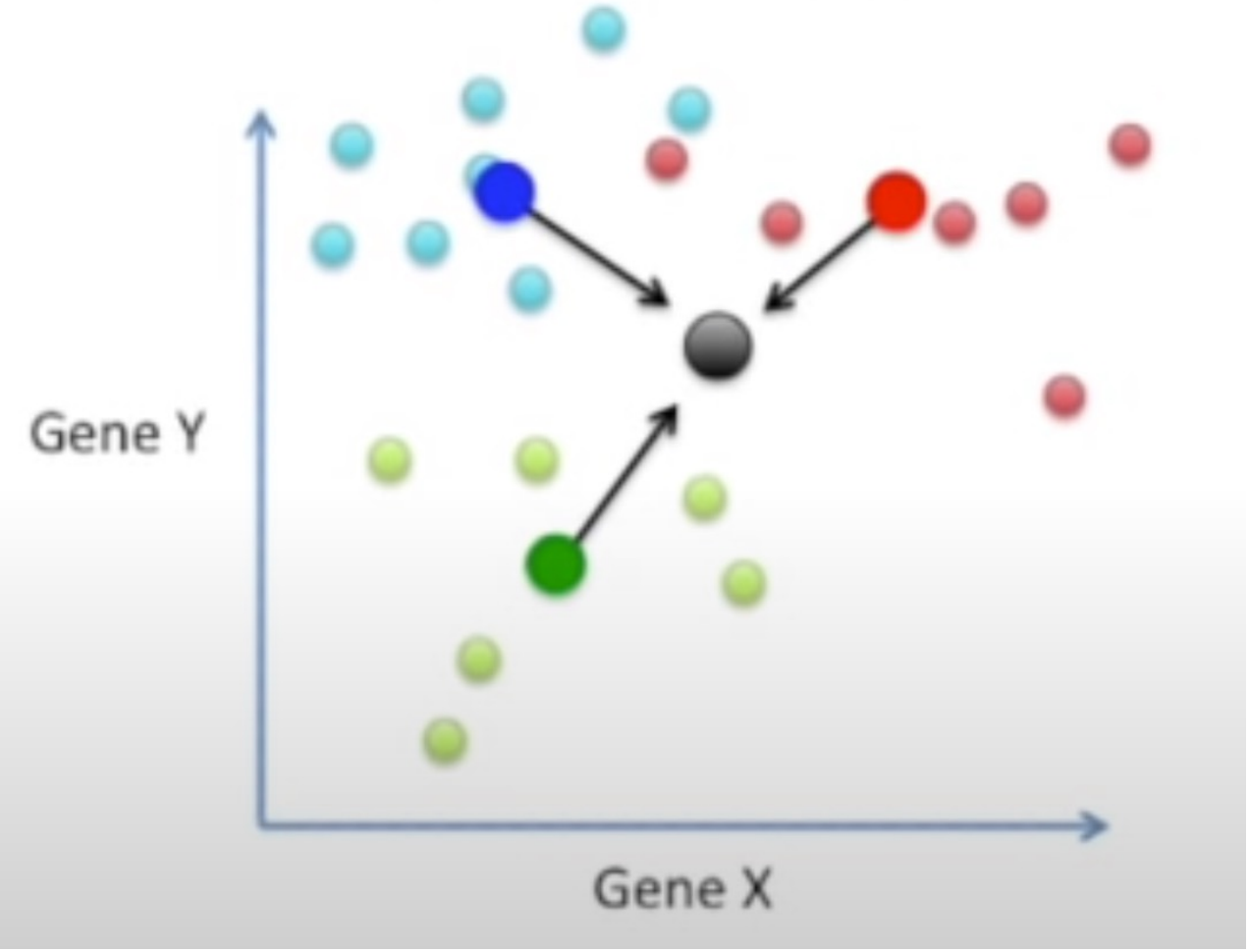
The first difference is how you measure the distances among the means.

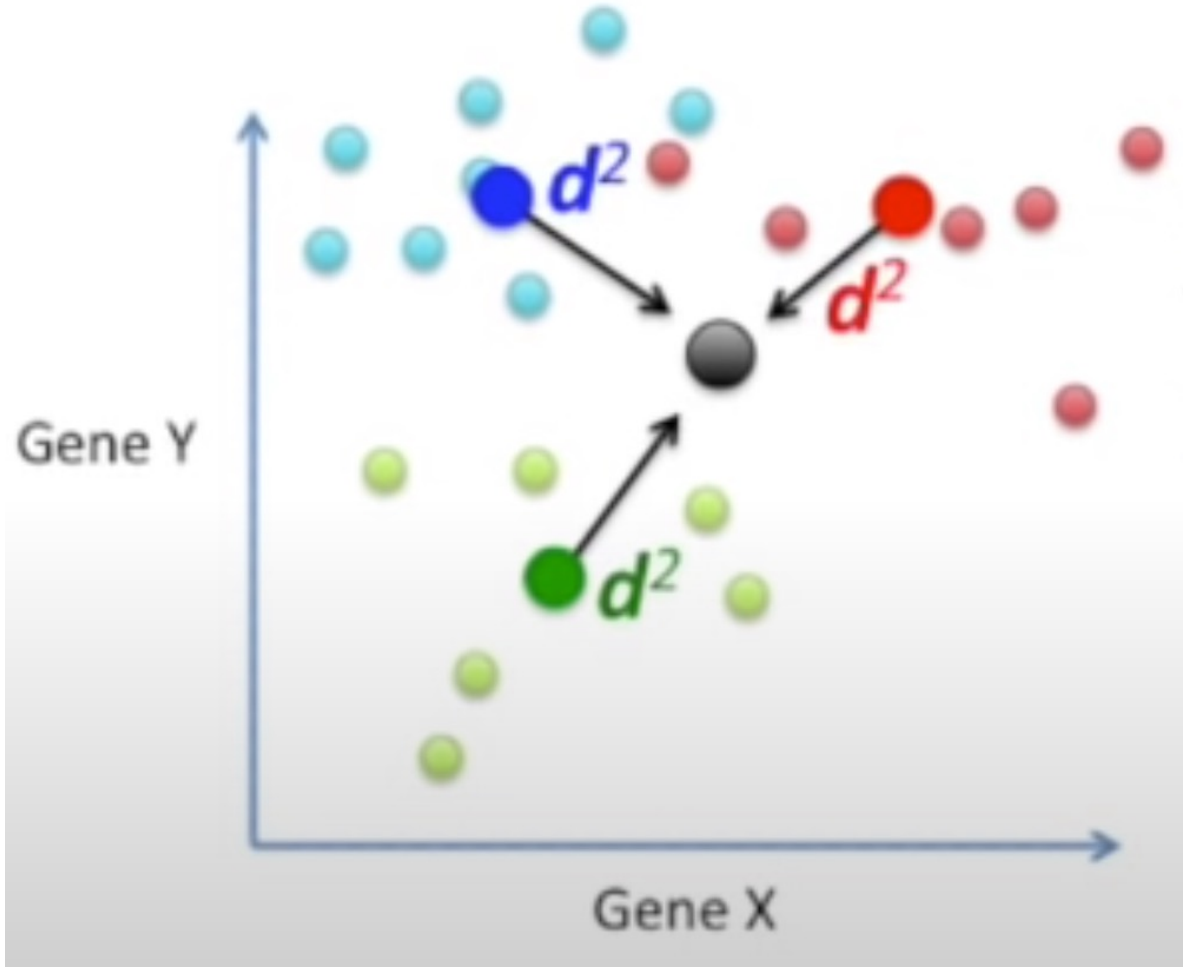


Find the point that is central to all of the data.



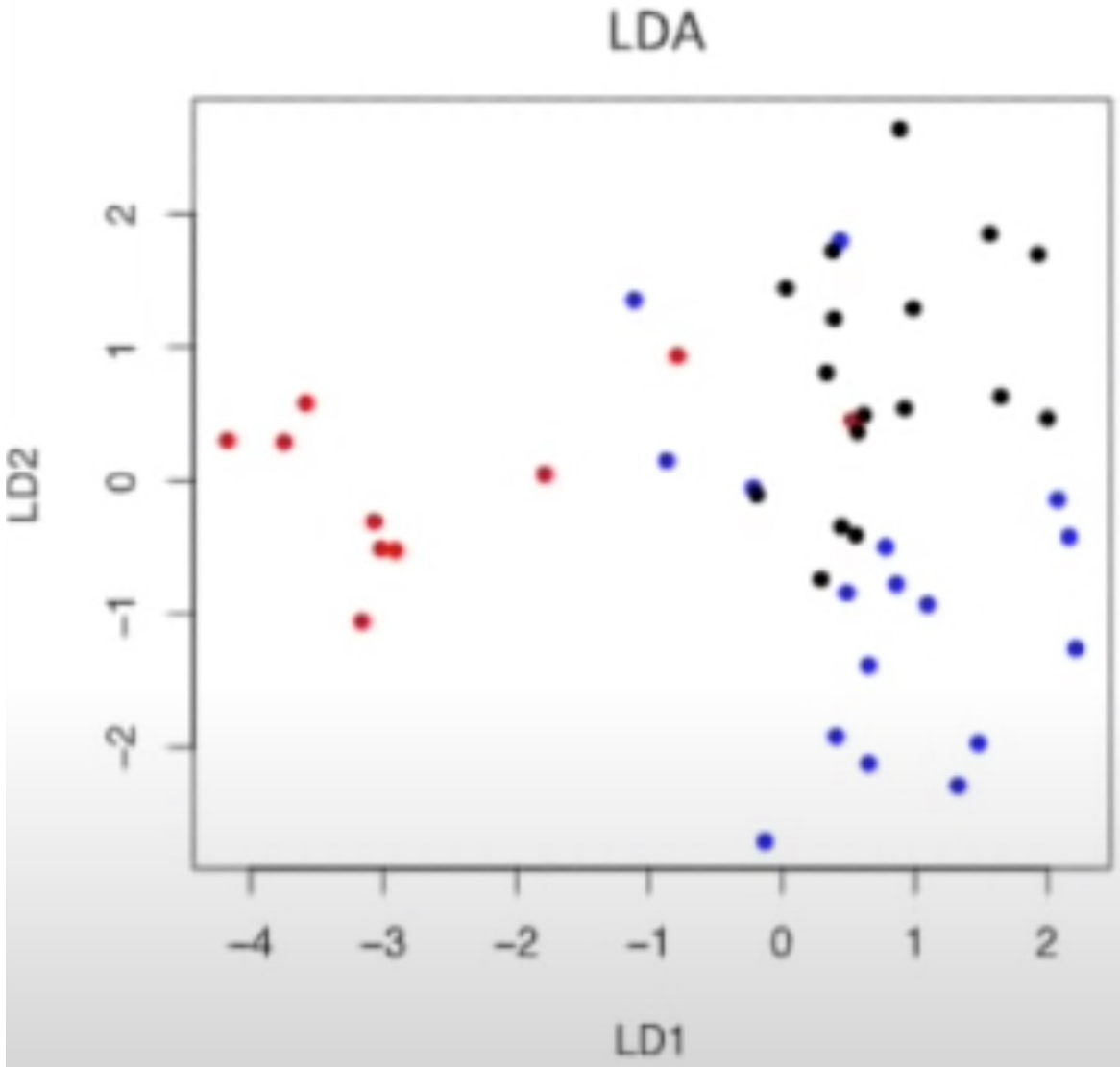
Then measure the distances between a point that is central in each category and the main central point.

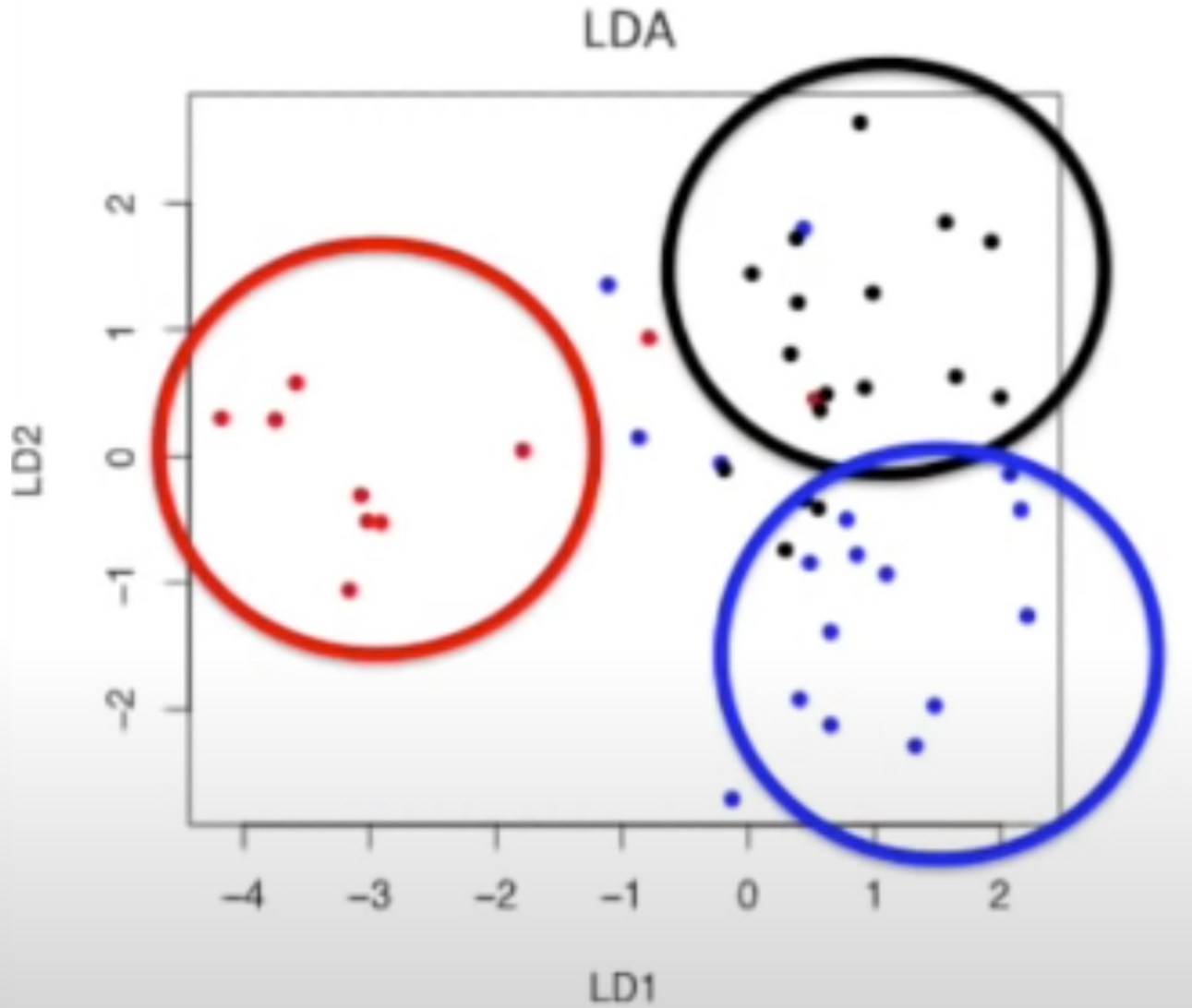




$$\frac{d^2 + d^2 + d^2}{s^2 + s^2 + s^2}$$


Gene Y
Gene X


LDA
LD2
LD1
2
1
0
−1
−2
−4
−3
−2
−1
0
1
2


LDA
LD2
LD1
2
1
0
−1
−2
−4
−3
−2
−1
0
1
2

Quadratic Discriminant Analysis (QDA)

Key in Discriminant Analysis

$$(X_1, \ldots, X_p) \mid Y \text{ is normally distributed}$$

In LDA, we assume

$$(X_1, \ldots, X_p) \mid Y = k \sim N(\mu_k, \Sigma)$$

different mean, same covariance

In QDA, we assume

$$(X_1, \ldots, X_p) \mid Y = k \sim N(\mu_k, \Sigma_k)$$

different mean, different covariance

QDA classifier

In QDA…

The Bayes classifier assigns the observation x to class for which

$$\delta_k(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T\Sigma_k^{-1}\mathbf{x} + \mathbf{x}^T\Sigma_k^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| + \log\pi_k$$

is largest

The QDA classifier assigns the observation x to class for which

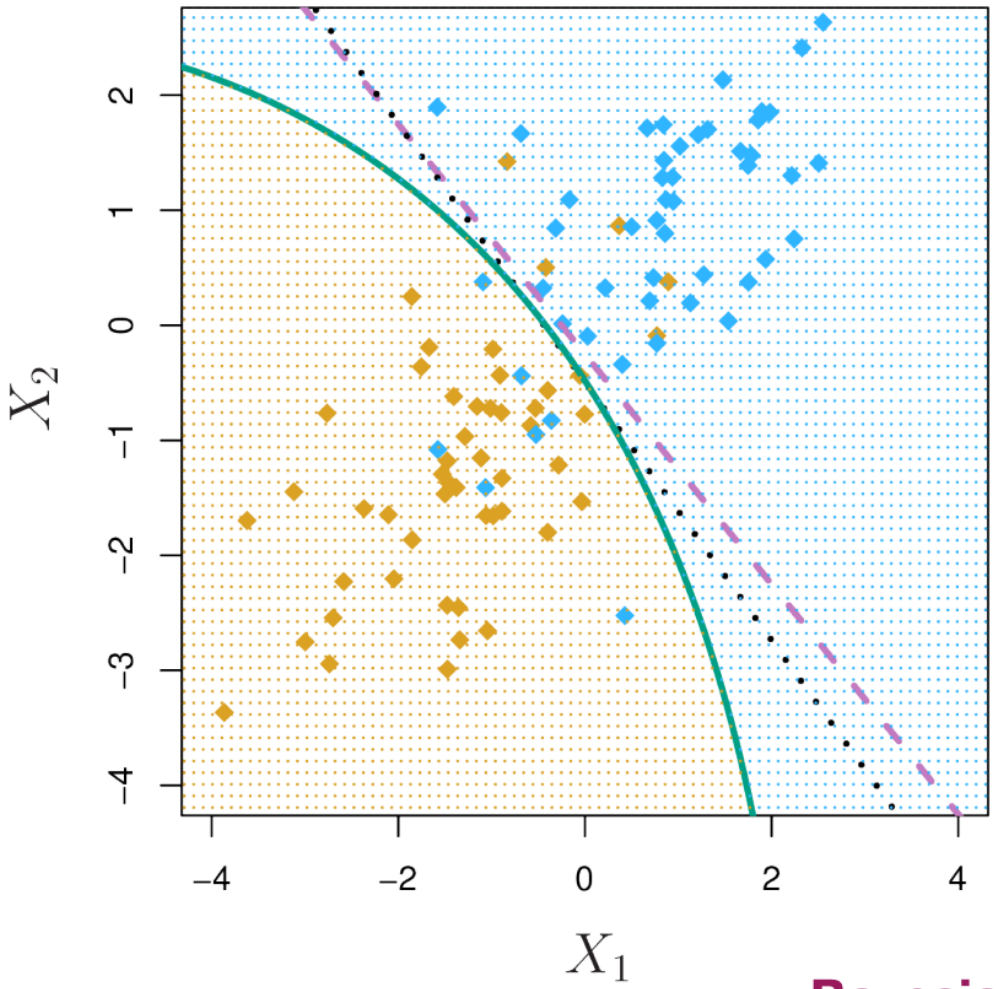
$$\hat{\delta}_k(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T\hat{\Sigma}_k^{-1}\mathbf{x} + \mathbf{x}^T\hat{\Sigma}_k^{-1}\hat{\mu}_k - \frac{1}{2}\hat{\mu}_k^T\hat{\Sigma}_k^{-1}\hat{\mu}_k - \frac{1}{2}\log|\hat{\Sigma}_k| + \log\hat{\pi}_k$$

quadratic in x

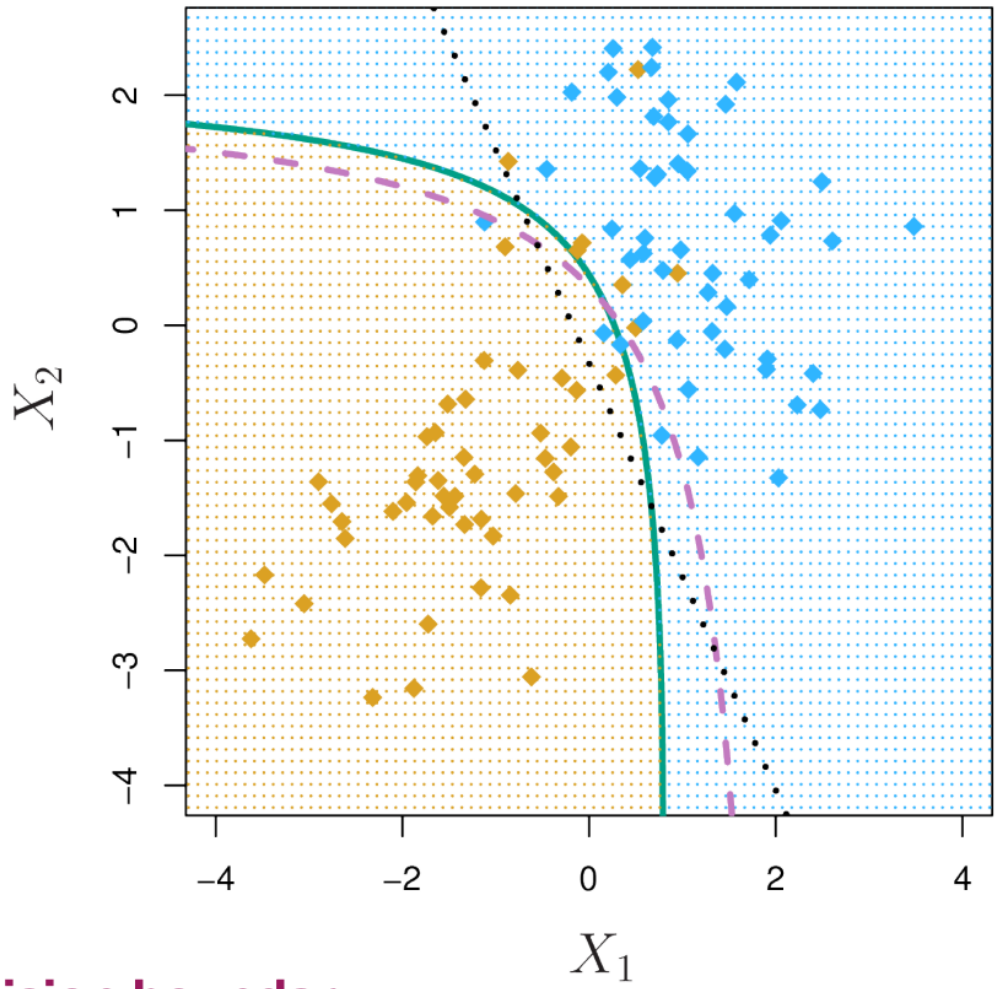
is largest

Decision boundary

$\Sigma_1 = \Sigma_2$



$\Sigma_1 \neq \Sigma_2$

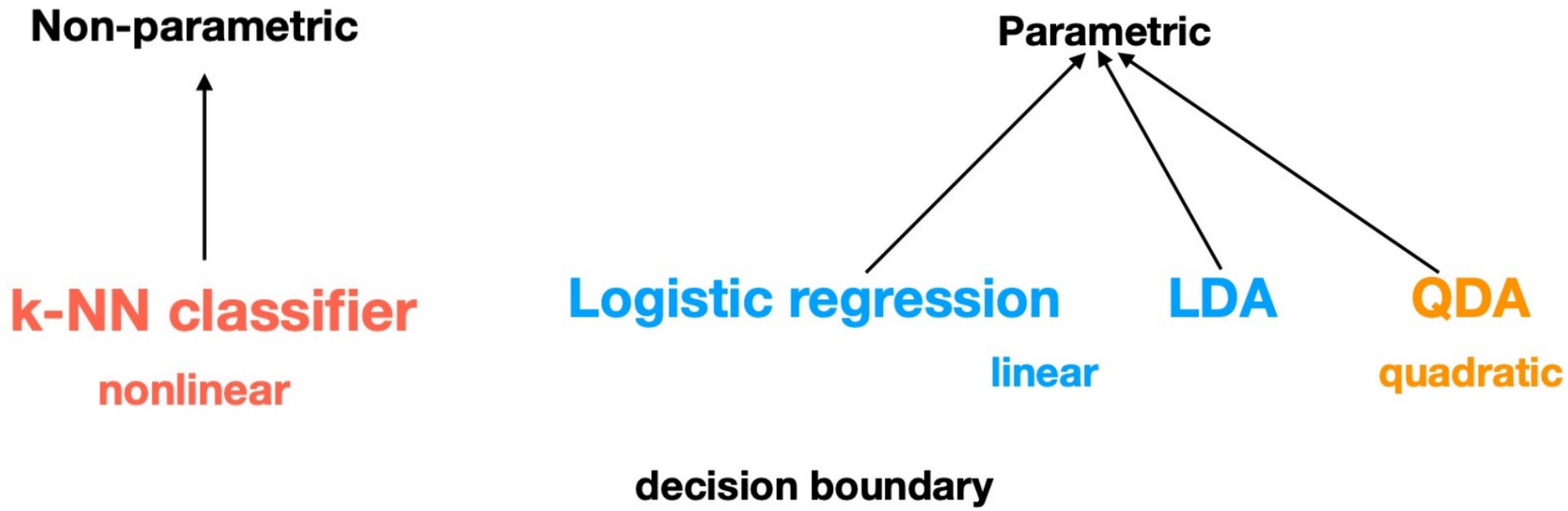


Bayesian decision boundary

LDA decision boundary

QDA decision boundary

Comparisons



model flexibility:

k-NN classifier > QDA > Logistic regression / LDA

via Bias-Variance tradeoff **via interpretability**

Logistic regression vs LDA

Both produce linear decision boundaries

Coefficient estimate procedures are different

LDA assumes normal distribution of predictors given response class

Logistic regression learns $\Pr(Y = k \mid X)$ **directly: discriminative learning**

LDA learns $\Pr(Y = k \mid X)$ **by modeling** $\Pr(Y = k, X)$**: generative learning**

LDA vs QDA

LDA can be thought as a special case of QDA

QDA is more flexible than LDA

QDA has more parameters to estimate (those Σ_k's)

via Bias-Variance tradeoff

In summary

Discriminant Analysis

- **LDA**
- **QDA**

Comparison of classifiers

- **Bias-variance tradeoff**
- **Interpretability**

Next…

In practice, we often do not know what assumptions or methods are more reasonable to use…

We need data to tell us…

How? Cross-validation